

Report LR-556

Stability and Control Derivatives of the De Havilland DHC-2 "BEAVER" aircraft

August 1988

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SUMMARY.

This memorandum presents a database of stability derivatives of the De Havilland DHC-2 "BEAVER" experimental aircraft of the Delft University of Technology at various stationary nominal flight conditions.

To compute these stability derivatives the nonlinear aerodynamic model selected in Ref. 1. is used. The flight test data of this model is based on data derived from the DHC-2 "BEAVER" test flights of the 1977/1978 period. These flights were performed in cooperation with DFVLR and NLR.

The stability derivatives are presented at several nominal flight conditions, at the c.g.-position used during the flight tests, throughout the normally used flight envelope of the aircraft.

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0. SYMBOLS AND DEFINITIONS.

0.1. Symbols.

a	engine model parameter
A	system matrix
b	wingspan
B	engine model parameter
B	input matrix
$\bar{c}$	mean aerodynamic chord (m.a.c.)
c.g.	center of gravity
C	vertical speed
C_l, C_m, C_n	dimensionless aerodynamic rolling, pitching and yawing moment
C_{l0}, C_{m0}, C_{n0}	constant part of C_l, C_m, C_n
C_{lp}, C_{np}	$\frac{\partial C_l}{\partial \left(\frac{pb}{2V}\right)}, \frac{\partial C_n}{\partial \left(\frac{pb}{2V}\right)}$
C_{lr}, C_{nr}	$\frac{\partial C_l}{\partial \left(\frac{rb}{2V}\right)}, \frac{\partial C_n}{\partial \left(\frac{rb}{2V}\right)}$
$C_{l\beta}, C_{n\beta}$	$\frac{\partial C_l}{\partial \beta}, \frac{\partial C_n}{\partial \beta}$
$C_{l\dot{\beta}}, C_{n\dot{\beta}}$	$\frac{\partial C_l}{\partial \left(\frac{\dot{\beta}b}{V}\right)}, \frac{\partial C_n}{\partial \left(\frac{\dot{\beta}b}{V}\right)}$
$C_{l\delta_a}, C_{n\delta_a}$	$\frac{\partial C_l}{\partial \delta_a}, \frac{\partial C_n}{\partial \delta_a}$
$C_{l\delta_r}, C_{n\delta_r}$	$\frac{\partial C_l}{\partial \delta_r}, \frac{\partial C_n}{\partial \delta_r}$
C_L	dimensionless lift coefficient
C_{m_u}	$\frac{1}{\frac{1}{2}\rho V S \bar{c}} \cdot \frac{\partial M}{\partial u} = \frac{\partial C_m}{\partial u}$
C_{m_q}	$\frac{\partial C_m}{\partial \left(\frac{q\bar{c}}{V}\right)}$
$C_{m\alpha}$	$\frac{\partial C_m}{\partial \alpha}$
$C_{m\dot{\alpha}}$	$\frac{\partial C_m}{\partial \left(\frac{\dot{\alpha}\bar{c}}{V}\right)}$
$C_{m\delta}$	$\frac{\partial C_m}{\partial \delta_e}$
$C_{m\delta_f}$	$\frac{\partial C_m}{\partial \delta_f}$

$C_{m\Delta P_t}$	$\frac{\partial C_m}{\partial (\Delta P_t / \frac{1}{2} \rho V^2)}$
C_X, C_Y, C_Z	dimensionless aerodynamic forces along the X_B -axis, the Y_B -axis and the Z_B -axis
$C_{X_0}, C_{Y_0}, C_{Z_0}$	constant part of C_X, C_Y, C_Z
C_{X_q}, C_{Z_q}	$\frac{\partial C_X}{\partial \left(\frac{q\bar{c}}{V} \right)}, \frac{\partial C_Z}{\partial \left(\frac{q\bar{c}}{V} \right)}$
C_{X_u}	$\frac{1}{\frac{1}{2} \rho V S} \cdot \frac{\partial X}{\partial u} \neq \frac{\partial C_X}{\partial u}$
$C_{X_\alpha}, C_{Z_\alpha}$	$\frac{\partial C_X}{\partial \alpha}, \frac{\partial C_Z}{\partial \alpha}$
$C_{X_{\dot{\alpha}}}, C_{Z_{\dot{\alpha}}}$	$\frac{\partial C_X}{\partial \left(\frac{\dot{\alpha}\bar{c}}{V} \right)}, \frac{\partial C_Z}{\partial \left(\frac{\dot{\alpha}\bar{c}}{V} \right)}$
$C_{X_\delta}, C_{Z_\delta}$	$\frac{\partial C_X}{\partial \delta_e}, \frac{\partial C_Z}{\partial \delta_e}$
$C_{X_{\delta_f}}, C_{Z_{\delta_f}}$	$\frac{\partial C_X}{\partial \delta_f}, \frac{\partial C_Z}{\partial \delta_f}$
C_{Y_p}	$\frac{\partial C_Y}{\partial \left(\frac{pb}{2V} \right)}$
C_{Y_r}	$\frac{\partial C_Y}{\partial \left(\frac{rb}{2V} \right)}$
C_{Y_β}	$\frac{\partial C_Y}{\partial \beta}$
$C_{Y_{\dot{\beta}}}$	$\frac{\partial C_Y}{\partial \left(\frac{\dot{\beta}\bar{c}}{V} \right)}$
$C_{Y_{\delta_a}}$	$\frac{\partial C_Y}{\partial \delta_a}$
$C_{Y_{\delta_r}}$	$\frac{\partial C_Y}{\partial \delta_r}$
C_{Z_u}	$\frac{1}{\frac{1}{2} \rho V S} \cdot \frac{\partial Z}{\partial u} \neq \frac{\partial C_Z}{\partial u}$
g	gravitational acceleration
h	pressure altitude
H	geopotential altitude
I_x, I_y, I_z	moment of inertia about the X_B -axis, Y_B -axis and Z_B -axis
J_{xz}	product of inertia
K_X^2	$\frac{I_x}{mb^2}$

K_{XZ}	$\frac{J_{xz}}{mb^2}$
K_Y^2	$\frac{I_y}{m\bar{c}^2}$
K_Z^2	$\frac{I_z}{mb^2}$
L	orthogonal matrix
L, M, N	aerodynamic rolling, pitching and yawing moment
m	mass of the aircraft
M	molecular weight
n	engine speed
O	origin of reference frame
p	pressure
p, q, r	angular rate of roll, pitch and yaw
P_Z	manifold pressure
P	engine power
P, Q, R	matrices
R_a	molar gas constant
S	wing area
t	time
T	temperature
T_M	molecular temperature
u, v, w	components of vectorial velocity along X-axis, Y-axis and Z-axis
$\bar{u}$	$\frac{\underline{u}}{V}$
$\underline{u}(t)$	input vector
V	true airspeed
W	aircraft weight
	constant atmospheric wind
$\underline{x}(t)$	state vector
$x_{c.g.}, z_{c.g.}$	position of center of gravity ($y_{c.g.} = 0$)
X, Y, Z	aerodynamic force along X-axis, Y-axis, Z-axis
$\underline{y}(t)$	output vector
α	angle of attack
β	angle of sideslip
γ	flight path angle
$\delta_a, \delta_e, \delta_f, \delta_r$	angle of deflection of ailerons ($\delta_a = \delta_{a_r} - \delta_{a_l}$) elevator, flaps and rudder
Δ	increment
ΔP_t	the increase of total pressure over the propeller
θ	pitch angle
λ	$\frac{\partial T}{\partial h}$
μ_b	$\frac{m}{\rho S b}$
μ_c	$\frac{m}{\rho S \bar{c}}$
ρ	air density
ϕ	roll angle
ψ	yaw angle

0.2. Index notations.

Superscripts

-1	matrix inverse
.	derivative with respect to time
-	mean value

Subscripts.

o	nominal value
	sea level
a	aileron
B	body-fixed reference frame F_B
e	elevator
E	earth-fixed reference frame F_E
f	flaps
M	measurement reference frame F_M
S	stability reference frame F_S
r	rudder

0.3. Abbreviations.

c.g.	center of gravity
DFVLR	Deutsche Forschungs- und Versuchsanstalt fuer Luft- und Raumfahrt
DHC	De Havilland of Canada Ltd.
DUT	Delft University of Technology, Faculty of Aerospace Engineering
m.a.c.	mean aerodynamic chord ($\bar{c}$)
NLR	National Aerospace Laboratory, Amsterdam
STOL	Short Take Off and Landing
TAS	True Air Speed, V
T.O.	Take Off

0.4. Reference frames.

The measurement reference frame F_M .

The measurement reference frame F_M is a left-handed orthogonal system $O_M X_M Y_M Z_M$ (See fig. 2). The origin O_M lies in a point resulting from the perpendicular projection of the foremost point of the wingchord, parallel to the aircraft's plane of symmetry, situated in the starboard wing 1.400 m from the aircraft's plane of symmetry, on this plane of symmetry. The $X_M O_M Z_M$ -plane coincides with this plane of symmetry. The positive X_M -axis points to the left. And so, the positive Z_M -axis points upwards.

The aircraft's body-fixed reference frame F_B .

The aircraft's body-fixed reference frame F_B is a right-handed orthogonal system $O_B X_B Y_B Z_B$ (See fig. 2). The origin O_B is placed in the aircraft's center of gravity. The $X_B O_B Z_B$ -plane coincides with the aircraft's plane of symmetry. The positive X_B -axis points forwards, the positive Y_B -axis points to starboard and so, the positive Z_B -axis points downwards.

The aircraft's body-fixed reference frame F_R .

The aircraft's body-fixed reference frame F_R is identical to F_B with only one exception. Its origin O_R is placed in an aircraft's body-fixed reference point. An actual flown c.g.-position is selected to be this point; it has the following coordinates in F_M :

x	=	0.5996	m
y	=	0.0	m
z	=	-0.8815	m

The stability reference frame F_S .

The stability reference frame F_S is a special body-fixed frame used in the study of small deviations from a given nominal flight condition. The difference between F_B and F_S is the orientation of the X_S -axis. In case of a symmetrical nominal flight condition, the X_S -axis is chosen parallel to true airspeed V . In case of a non-symmetrical flight condition the X_S -axis is chosen parallel to the projection of V on the aircraft's plane of symmetry.

The earth-fixed reference frame F_E .

The earth-fixed reference frame F_E is a right-handed orthogonal system $O_E X_E Y_E Z_E$. The origin can be placed in principle at an arbitrary position, often the aircraft's center of gravity at the start of a flight test manoeuvre. The Z_E -axis points downwards parallel to the local direction of gravitation. The X_E -axis points to the North, the Y_E -axis points to the East.

1. INTRODUCTION.

The nonlinear aerodynamic model selected in Ref. 1. is the "Beaver"-model on which this memorandum is based. The aerodynamic coefficients are determined from a dataset containing a DUT longitudinal + lateral manoeuvre at 35, 45 and 55 m/s nominal true airspeed.

In Chapter 2 the equations of motion are presented for aircraft like the "Beaver"-aircraft.

The atmosphere is briefly discussed in Chapter 3.

In Chapter 4 a description of the aerodynamic model used will be given together with an engine model determined of the aircraft.

Chapter 5 describes the computation of the nominal flight conditions, while Chapter 6 discusses the stability derivatives and their corrections for different c.g.-positions.

The correction of the moments and products of inertia in case of translation and rotation of axes is discussed in Chapter 7.

In Chapter 8 a flight envelope of the "Beaver"-aircraft is described.

Conclusions and comments with respect to the preceding Chapters are given in Chapter 9.

2. THE EQUATIONS OF MOTION.

The aircraft is considered as a rigid body moving through an airmass as a result of external forces and moments. Defining a suitable reference frame, the aircraft motions can be described using the Newtonian laws. The resulting set of equations are simplified by assuming the aircraft moves relative to a flat non-rotating earth.

2.1. Nonlinear kinematic models.

For an aircraft having a geometrical plane of symmetry, the equations of motion can be written in terms of a set of relations between the components of the total aerodynamic force (i.e. including the aerodynamic effects of engines) X , Y , and Z and the components of the kinematic acceleration $\dot{u}$, $\dot{v}$ and $\dot{w}$ along the axes X_B , Y_B and Z_B of the body-fixed reference frame F_B :

$$\begin{aligned} -W \sin \theta + X &= m (\dot{u} + qw - rv) , \\ W \cos \theta \sin \varphi + Y &= m (\dot{v} + ru - pw) , \\ W \cos \theta \cos \varphi + Z &= m (\dot{w} + pv - qu) , \end{aligned} \quad (2.1-a)$$

where: u , v and w denote the components of the true airspeed,
 p , q and r the rates of rotation about the axes of F_B .
 θ and φ pitch and roll angle respectively,
 m denotes the aircraft mass and
 W the aircraft weight. ($W = m.g$).

Furthermore a set of relations between the total aerodynamic moments L , M and N and the rotational accelerations $\dot{p}$, $\dot{q}$ and $\dot{r}$, about the axes of F_B , can be written as

$$\begin{aligned} L &= I_x \dot{p} + (I_z - I_x) qr - J_{xz} (\dot{r} + pq) , \\ M &= I_y \dot{q} + (I_x - I_z) rp + J_{xz} (p^2 - r^2) , \\ N &= I_z \dot{r} + (I_y - I_x) pq - J_{xz} (\dot{p} - qr) , \end{aligned} \quad (2.1-b)$$

where: I_x , I_y and I_z denote the moments of inertia and
 J_{xz} denotes the only (due to symmetry)
non-zero product of inertia in F_B .

The orientation of the body-fixed reference frame F_B with respect to the earth-fixed reference frame F_E is governed by a set of kinematic relations for the Euler angles ψ , θ and φ :

$$\begin{aligned} \dot{\psi} &= q \sin \varphi / \cos \theta + r \cos \varphi / \cos \theta , \\ \dot{\theta} &= q \cos \varphi - r \sin \varphi , \\ \dot{\varphi} &= p + q \sin \varphi \tan \theta + r \cos \varphi \tan \theta . \end{aligned} \quad (2.1-c)$$

The geographic position coordinates of the origin of F_B (i.e. the center of gravity) with respect to F_E are given by

$$\begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = L_{EB} \begin{bmatrix} u \\ v \\ w \end{bmatrix} + \begin{bmatrix} W_{xE} \\ W_{yE} \\ W_{zE} \end{bmatrix} \quad (2.2)$$

in which L_{EB} denotes an orthogonal matrix of the form:

$$L_{EB} = \begin{bmatrix} \cos \theta \cos \psi & \sin \varphi \sin \theta \cos \psi & \cos \varphi \sin \theta \cos \psi \\ & -\cos \varphi \sin \psi & +\sin \varphi \sin \psi \\ \cos \theta \sin \psi & \sin \varphi \sin \theta \sin \psi & \cos \varphi \sin \theta \sin \psi \\ & +\cos \varphi \cos \psi & -\sin \varphi \cos \psi \\ -\sin \theta & \sin \varphi \cos \theta & \cos \varphi \cos \theta \end{bmatrix} \quad (2.3)$$

where W_{xE} , W_{yE} and W_{zE} denote the components of the constant atmospheric wind along the axes of F_E .

2.2. Linearized kinematic models.

The nonlinear equations of motion (2.1-a) and (2.1-b) in the body-fixed reference frame F_B may readily be transferred into an equivalent set of relations in the stability reference system F_S .

These equations are linearized for small deviations from a nominal flight condition of steady, rectilinear flight with side slip angle equal to zero.

In the stability reference frame this implies

$$u_0 = V_0, \quad v_0 = 0, \quad w_0 = 0,$$

$$p_0 = 0, \quad q_0 = \dot{\theta}, \quad r_0 = 0,$$

while the nominal pitch angle is equal to the nominal flight path angle:

$$\theta_0 = \gamma_0, \quad \text{as } \alpha_0 = 0,$$

where the subscript 0 refers to the nominal flight condition.

According to Ref. 7 the symmetric and the nonsymmetric part of the equations of motion can be considered separately under the following conditions:

1. the aircraft is symmetric in the XOZ-plane,
2. only motions in steady, rectilinear, symmetric flight are considered,
3. the disturbances and the resulting deviations from the stationary initial condition are small enough to allow linearization.

When writing $u = du$, $v = dv$, $w = dw$, $\theta = d\theta$, $\delta_a = d\delta_a$, $\delta_e = d\delta_e$, etc. the equations of motion become in F_S :

Symmetric:

$$\begin{aligned} -W \cos \theta_0 \cdot \theta + X_u \cdot u + X_w \cdot w + X_{\delta_e} \cdot \delta_e &= m \cdot \dot{u} \\ -W \sin \theta_0 \cdot \theta + Z_u \cdot u + Z_w \cdot w + Z_{\dot{w}} \cdot \dot{w} + Z_{\delta_e} \cdot \delta_e &= m \cdot (\dot{w} - q \cdot V) \\ M_u \cdot u + M_w \cdot w + M_{\dot{w}} \cdot \dot{w} + M_{\delta_e} \cdot \delta_e &= I_y \cdot \dot{q} \\ \dot{\theta} &= q \end{aligned} \quad (2.4-a)$$

Nonsymmetric:

$$\begin{aligned} W \cos \theta_0 \cdot \varphi + Y_v \cdot v + Y_{\dot{v}} \cdot \dot{v} + Y_p \cdot p + Y_r \cdot r + Y_{\delta_a} \cdot \delta_a + Y_{\delta_r} \cdot \delta_r &= m(\dot{v} + rV) \\ L_v \cdot v + L_p \cdot p + L_r \cdot r + L_{\delta_a} \cdot \delta_a + L_{\delta_r} \cdot \delta_r &= I_x \dot{p} - J_{xz} \dot{r} \\ N_v \cdot v + N_{\dot{v}} \cdot \dot{v} + N_p \cdot p + N_r \cdot r + N_{\delta_a} \cdot \delta_a + N_{\delta_r} \cdot \delta_r &= I_z \dot{r} - J_{xz} \dot{p} \\ \dot{\psi} &= \frac{r}{\cos \theta_0} \\ \dot{\phi} &= p + r \operatorname{tg} \theta_0 \end{aligned} \quad (2.4-b)$$

2.3. The dimensionless equations.

It is customary in aircraft engineering to relate the aerodynamic forces X , Y and Z , and the aerodynamic moments L , M and N to the corresponding dimensionless coefficients C_X , C_Y , C_Z , C_l , C_m and C_n , writing:

$$\begin{aligned} X &= C_X \cdot \frac{1}{2} \rho V^2 S \\ Y &= C_Y \cdot \frac{1}{2} \rho V^2 S \\ Z &= C_Z \cdot \frac{1}{2} \rho V^2 S \\ L &= C_l \cdot \frac{1}{2} \rho V^2 S b \\ M &= C_m \cdot \frac{1}{2} \rho V^2 S \bar{c} \\ N &= C_n \cdot \frac{1}{2} \rho V^2 S b \end{aligned} \quad (2.5)$$

where the symmetric moment is related to the mean aerodynamic chord $\bar{c}$ and the nonsymmetric moments are related to the wing span b .

The equations of motion (2.4-a) and (2.4-b) can be made dimensionless by dividing each term of the equations of the forces X, Y and Z by $\frac{1}{2}\rho V^2 S$, the moment M by $\frac{1}{2}\rho V^2 S \bar{c}$ and the moments L and N by $\frac{1}{2}\rho V^2 S b$.

Using stability derivatives and adding the flap coefficients the equations can be written as

$$\begin{bmatrix} C_{X_u} - 2\mu_c D_c & C_{X_\alpha} & C_{Z_0} & C_{X_q} \\ C_{Z_u} & C_{Z_\alpha} + (C_{Z_{\dot{\alpha}}} - 2\mu_c) D_c & -C_{X_0} & C_{Z_q} + 2\mu_c \\ 0 & 0 & -D_c & 1 \\ C_{m_u} & C_{m_\alpha} + C_{m_{\dot{\alpha}}} D_c & 0 & C_{m_q} - 2\mu_c K_Y^2 D_c \end{bmatrix} \cdot \begin{bmatrix} \hat{u} \\ \alpha \\ \theta \\ \frac{q\bar{c}}{V} \end{bmatrix} = \begin{bmatrix} -C_{X_\delta} & -C_{X_{\delta_f}} \\ -C_{Z_\delta} & -C_{Z_{\delta_f}} \\ 0 & 0 \\ -C_{m_\delta} & -C_{m_{\delta_f}} \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_f \end{bmatrix} \quad (2.6-a)$$

with the operator $D_c = \frac{\bar{c}}{V} \frac{d}{dt}$.

$$\begin{bmatrix} C_{Y_\beta} + (C_{Y_{\dot{\beta}}} + 2\mu_b) D_b & C_L & C_{Y_p} & C_{Y_r} - 4\mu_b \\ 0 & -\frac{1}{2} D_b & 1 & 0 \\ C_{l_\beta} & 0 & C_{l_p} - 4\mu_b K_X^2 D_b & C_{l_r} + 4\mu_b K_X Z D_b \\ C_{n_\beta} + C_{n_{\dot{\beta}}} D_b & 0 & C_{n_p} + 4\mu_b K_X Z D_b & C_{n_r} - 4\mu_b K_Z^2 D_b \end{bmatrix} \cdot \begin{bmatrix} \beta \\ \varphi \\ \frac{pb}{2V} \\ \frac{rb}{2V} \end{bmatrix} = \begin{bmatrix} -C_{Y_{\delta_a}} \\ 0 \\ -C_{l_{\delta_a}} \\ -C_{n_{\delta_a}} \end{bmatrix} \cdot \delta_a + \begin{bmatrix} -C_{Y_{\delta_r}} \\ 0 \\ -C_{l_{\delta_r}} \\ -C_{n_{\delta_r}} \end{bmatrix} \cdot \delta_r \quad (2.6-b)$$

(in horizontal flight)

with the operator $D_b = \frac{b}{V} \frac{d}{dt}$.

2.4. The state-space description.

The above-mentioned sets of linear differential equations can be written in the state-space description using matrix-vector form as

$$\dot{\underline{x}}(t) = A \underline{x}(t) + B \underline{u}(t) \quad (2.7)$$

$$\underline{y}(t) = C_1 \underline{x}(t) + C_2 \underline{u}(t) \quad (2.8)$$

where the vectors $\underline{x}(t)$, $\underline{y}(t)$, and $\underline{u}(t)$ denote the system variables, the outputs and the inputs, respectively. The matrix A is called the system matrix and the matrix B the input matrix.

Writing the linearized equations of motion in this form, the dependent terms of D_c and D_b have to be separated from the independent terms of D_c and D_b (Ref. 7).

With $D_c = \frac{\bar{c}}{V} \frac{d}{dt}$ and $D_b = \frac{b}{V} \frac{d}{dt}$ it follows that

$$\begin{bmatrix} 2\mu_c \frac{\bar{c}}{V} & 0 & 0 & 0 \\ 0 & (2\mu_c - CZ_{\dot{\alpha}}) \frac{\bar{c}}{V} & 0 & 0 \\ 0 & 0 & \frac{\bar{c}}{V} & 0 \\ 0 & -C_{m\dot{\alpha}} \frac{\bar{c}}{V} & 0 & 2\mu_c K_Y \frac{2\bar{c}}{V} \end{bmatrix} \begin{bmatrix} \dot{\hat{u}}(t) \\ \dot{\alpha}(t) \\ \dot{\theta}(t) \\ \frac{\dot{g}\bar{c}}{V}(t) \end{bmatrix} = \begin{bmatrix} C_{X_u} & C_{X_{\alpha}} & C_{Z_0} & C_{X_q} \\ C_{Z_u} & C_{Z_{\alpha}} & -C_{X_0} & C_{Z_q} + 2\mu_c \\ 0 & 0 & 0 & 1 \\ C_{m_u} & C_{m_{\alpha}} & 0 & C_{m_q} \end{bmatrix} \begin{bmatrix} \hat{u}(t) \\ \alpha(t) \\ \theta(t) \\ \frac{g\bar{c}}{V}(t) \end{bmatrix} \\ + \begin{bmatrix} C_{X_{\delta}} & C_{X_{\delta f}} \\ C_{Z_{\delta}} & C_{Z_{\delta f}} \\ 0 & 0 \\ C_{m_{\delta}} & C_{m_{\delta f}} \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_f \end{bmatrix} \quad (2.9-a)$$

and

$$\begin{bmatrix} (2\mu_b - C_{Y_{\dot{\beta}}}) \frac{b}{V} & 0 & 0 & 0 \\ 0 & \frac{b}{V} & 0 & 0 \\ 0 & 0 & 4\mu_b K_X \frac{2b}{V} & -4\mu_b K_{XZ} \frac{b}{V} \\ -C_{n_{\dot{\beta}}} \frac{b}{V} & 0 & -4\mu_b K_{XZ} \frac{b}{V} & 4\mu_b K_Z \frac{2b}{V} \end{bmatrix} \begin{bmatrix} \dot{\beta}(t) \\ \dot{\phi}(t) \\ \frac{\dot{p}b}{2V}(t) \\ \frac{\dot{r}b}{2V}(t) \end{bmatrix} = \begin{bmatrix} C_{Y_{\beta}} & C_L & C_{Y_p} & C_{Y_r} - 4\mu_b \\ 0 & 0 & 1 & 0 \\ C_{l_{\beta}} & 0 & C_{l_p} & C_{l_r} \\ C_{n_{\beta}} & 0 & C_{n_p} & C_{n_r} \end{bmatrix} \begin{bmatrix} \beta(t) \\ \phi(t) \\ \frac{pb}{2V}(t) \\ \frac{rb}{2V}(t) \end{bmatrix} + \begin{bmatrix} C_{Y_{\delta a}} & C_{Y_{\delta r}} \\ 0 & 0 \\ C_{l_{\delta a}} & C_{l_{\delta r}} \\ C_{n_{\delta a}} & C_{n_{\delta r}} \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix} \quad (2.9-b)$$

Calling the three matrices in these equations P, Q and R, respectively, then (2.9-a) can be written as

$$P \begin{bmatrix} \dot{\hat{u}}(t) \\ \dot{\alpha}(t) \\ \dot{\theta}(t) \\ \frac{\dot{g}\bar{c}}{V}(t) \end{bmatrix} = Q \begin{bmatrix} \hat{u}(t) \\ \alpha(t) \\ \theta(t) \\ \frac{g\bar{c}}{V}(t) \end{bmatrix} + R \begin{bmatrix} \delta_e \\ \delta_f \end{bmatrix} \quad (2.10)$$

Because P is a square nonsingular matrix, the equation (2.10) can be written as

$$\begin{bmatrix} \dot{\hat{u}}(t) \\ \dot{\hat{\alpha}}(t) \\ \dot{\hat{\theta}}(t) \\ \dot{\frac{\hat{q}\bar{c}}{V}}(t) \end{bmatrix} = P^{-1} Q \begin{bmatrix} \hat{u}(t) \\ \alpha(t) \\ \theta(t) \\ \frac{q\bar{c}}{V}(t) \end{bmatrix} + P^{-1} R \begin{bmatrix} \delta_e \\ \delta_f \end{bmatrix} \quad (2.11)$$

In exactly the same manner the nonsymmetric equation (2.9-b) can be written in the form of (2.11).

Thus the equations are presented in the state-space form

$$\dot{\underline{x}}(t) = A \underline{x}(t) + B \underline{u}(t) \quad (2.7)$$

where $A = P^{-1} Q$ and $B = P^{-1} R$.

3. THE ATMOSPHERE.

A model describing the atmosphere at rest is the "U.S. Standard Atmosphere 1962" (Ref. 8). This model describes an idealized atmosphere, based on the ideal gas law and the basic hydrostatic equation. The described atmosphere is valid for flight altitudes upward to 11,000 m, which is sufficient for the aircraft concerned.

3.1. The air temperature T .

Starting from the assumption of dry air having a constant molecular weight M and a constant gravitational acceleration g , in the troposphere the air temperature can be considered as

$$T = T_0 + \lambda \cdot h, \text{ with } h \text{ in m.} \quad (3.1)$$

where $T_0 = 288.15 \text{ K}$ and $\lambda = -0.0065 \text{ K/m}$.

The altitude h is taken relatively to sea level.

3.2. The air pressure p .

The basic hydrostatic equation

$$dp = - \rho g dh \quad (3.2)$$

combined with the ideal gas law

$$\frac{p}{\rho} = \frac{R_a}{M} T \quad (3.3)$$

yields:

$$\frac{dp}{p} = - \frac{M g}{R_a T} dh \quad (3.4)$$

where: M = molecular weight
g = gravitational acceleration
T = temperature

Integration of (3.4) with respect to p and introduction of the geopotential altitude H yields:

$$\ln \frac{p_2}{p_1} = - \frac{M_0 g_0}{R_a} \int_{H_1}^{H_2} \frac{dH}{T_M} \quad (3.5)$$

where T_M : molecular temperature.
the indices 0 : denotes sea level
1, 2: denote top and bottom level of the layer considered.

The slight distinction between the geometrical altitude h and the geopotential altitude H, in view of the flight altitudes considered, will be neglected.

So, $h = H$ and $T = T_M$.

For the troposphere (0 to 11,000 m), this resolves into:

$$p = p_0 \left[\frac{T_0}{T} \right]^{\frac{M_0 g_0}{R_a \lambda}}, \quad p \text{ in [mb]} \quad (3.6)$$

with $R_a = 8314.32 \text{ J/K.kmol}$, $p_0 = 1013.12 \text{ mb}$, $M_0 = 28.9644 \text{ kg/kmol}$ and $g_0 = 9.80665 \text{ m/s}^2$.

3.3. The air density ρ .

The air density ρ can be found by

$$\rho = \rho_0 \left[\frac{T_0}{T} \right]^{\frac{M_0 g_0}{R_a \lambda} + 1}, \quad \rho \text{ in [kg/m}^3] \quad (3.7)$$

with $\rho_0 = 1.225 \text{ kg/m}^3$.

4. THE AIRCRAFT MODEL.

4.1. The aerodynamic model.

The selected aerodynamic model is presented in Table 3. In this model the effect of compressibility is neglected where airspeed is assumed to be low. Also scale effects have not been taken into account since, at relatively high Reynolds number, the effect of Reynolds number variations as occurring in flight is considered to be negligible.

Writing the nonlinear model in terms of the dimensionless components of the total aerodynamic forces C_X , C_Z and C_Y , and the dimensionless components of the aerodynamic moments C_m , C_l and C_n , the model becomes

Symmetric:

$$\begin{aligned}
 C_X &= C_{X0} + C_{X_{\Delta P t}} \frac{\Delta P t}{\frac{1}{2} \rho V^2} + C_{X_\alpha} \cdot \alpha + C_{X_{\alpha^2}} \cdot \alpha^2 + C_{X_{\alpha^3}} \cdot \alpha^3 + \\
 &C_{X_q} \cdot \frac{q \bar{c}}{V} + C_{X_{\alpha \Delta P t}} \cdot \alpha \cdot \left[\frac{\Delta P t}{\frac{1}{2} \rho V^2} \right]^2 + C_{X_{\delta_r}} \cdot \delta_r + C_{X_{\delta_f}} \cdot \delta_f + C_{X_{\alpha \delta_f}} \cdot \alpha \cdot \delta_f \\
 C_Z &= C_{Z0} + C_{Z_{\Delta P t}} \frac{\Delta P t}{\frac{1}{2} \rho V^2} + C_{Z_\alpha} \cdot \alpha + C_{Z_{\alpha^3}} \cdot \alpha^3 + C_{Z_q} \cdot \frac{q \bar{c}}{V} + \\
 &C_{Z_{\delta_e}} \cdot \delta_e + C_{Z_{\delta_e^2}} \cdot \delta_e \cdot \delta_e^2 + C_{Z_{\delta_f}} \cdot \delta_f + C_{Z_{\alpha \delta_f}} \cdot \alpha \cdot \delta_f \\
 C_m &= C_{m0} + C_{m_{\Delta P t}} \frac{\Delta P t}{\frac{1}{2} \rho V^2} + C_{m_\alpha} \cdot \alpha + C_{m_{\alpha^2}} \cdot \alpha^2 + C_{m_q} \cdot \frac{q \bar{c}}{V} + \\
 &C_{m_{\delta_e}} \cdot \delta_e + C_{m_{\delta^2}} \cdot \delta^2 + C_{m_r} \cdot \frac{r b}{2V} + C_{m_{\delta_f}} \cdot \delta_f
 \end{aligned} \tag{4.1-a}$$

Nonsymmetric:

$$\begin{aligned}
 C_Y &= C_{Y0} + C_{Y_\beta} \cdot \beta + C_{Y_p} \cdot \frac{p b}{2V} + C_{Y_r} \cdot \frac{r b}{2V} + C_{Y_{\delta_a}} \cdot \delta_a + C_{Y_{\delta_r}} \cdot \delta_r + \\
 &C_{Y_{\delta_r \alpha}} \cdot \delta_r \cdot \alpha + C_{Y_{\dot{\beta}}} \cdot \frac{\dot{\beta} b}{2V} \quad (\text{NOTE: } \frac{\dot{\beta} b}{2V} \text{ as used in Ref. 1 and} \\
 &\text{not according to the def. in Ref. 6)} \\
 C_l &= C_{l0} + C_{l_\beta} \cdot \beta + C_{l_p} \cdot \frac{p b}{2V} + C_{l_r} \cdot \frac{r b}{2V} + C_{l_{\delta_a}} \cdot \delta_a + C_{l_{\delta_r}} \cdot \delta_r + \\
 &C_{l_{\alpha^2 \Delta P t}} \cdot \alpha^2 \cdot \frac{\Delta P t}{\frac{1}{2} \rho V^2} + C_{l_{\delta_a \alpha}} \cdot \delta_a \cdot \alpha
 \end{aligned}$$

$$C_n = C_{n0} + C_{n\beta} \cdot \beta + C_{n_p} \cdot \frac{pb}{2V} + C_{n_r} \cdot \frac{rb}{2V} + C_{n\delta_a} \cdot \delta_a + C_{n\delta_r} \cdot \delta_r + C_{n\Delta p_t} \cdot \left[\frac{\Delta p_t}{\frac{1}{2}\rho V^2} \right]^3 + C_{n_q} \cdot \frac{q\bar{c}}{V} + C_{n\beta^3} \cdot \beta^3 \quad (4.1-b)$$

Apart from the aerodynamic coefficients found in earlier flight tests, it is interesting that the above-mentioned model also includes effects like longitudinal-lateral cross-coupling and unsteady aerodynamics. For instance the gyroscopical effect of the propeller is clearly visible in the coefficients C_{m_r} and C_{n_q} .

The dimensionless aerodynamic forces and moments are shown in Fig. 4. In Figs. 5-a and 5-b the same aerodynamic force and moment coefficients are given. These coefficients are measured on a 1:11 scale model of the 'Beaver'-aircraft in windtunnel tests at the Low Speed Windtunnel of DUT.

Comparison of the model with these windtunnel tests shows that there exists a reasonable correspondence between Figs. 4 and 5.

In this comparison it should be kept in mind that the windtunnel experiments were performed at a relatively low Reynolds number. Because of this scale effect the correspondence can only be seen in a qualitative way (Ref. 9).

An exception to this is that the coefficient $C_{m\Delta p_t}$ in comparison with the windtunnel tests has the wrong sign. As all earlier model predictions produce the same sign and the cause of the difference cannot explicitly be put down to the aerodynamic model, this model will yet be used.

4.2. The engine model.

If the propeller is represented as an ideal pulling disc, the following relation between $\Delta p_t / \frac{1}{2}\rho V^2$ and the airspeed V and the engine power P holds

$$\frac{\Delta p_t}{\frac{1}{2}\rho V^2} = a + b \cdot \frac{P}{\frac{1}{2}\rho V^3} \quad (4.2)$$

with $\frac{P}{\frac{1}{2}\rho V^3}$ in $\left[\frac{\text{kW}}{\text{kg/s}^3} \right]$, and $a = 0.08696$ and $b = 191.18$

Whereas Δp_t denotes the increase of total air pressure in the propeller slipstream. In case of piston engined aircraft, variations of airspeed V and engine power settings (engine speed n and manifold pressure p_z) can be replaced by the variable $\Delta p_t / \frac{1}{2}\rho V^2$ (Ref. 5).

The engine power P can be found by the equation

$$P = -326.5 + [0.00412(p_z + 7.4)(n + 2010) + (408.0 - 0.0965n) \cdot (1.0 - \rho/\rho_0)] \quad \text{in [BHP]} \quad (4.3)$$

with the manifold pressure p_z in inch Hg and the engine speed n in RPM.

5. THE INITIAL CONDITION COMPUTATION.

To compute a stationary nominal flight condition the equations (4.1-a) and (4.1-b) are used. The nominal flight condition can be determined by providing the wanted values of the true airspeed V , the engine speed n , the manifold pressure p_z , the pressure altitude h and the deflection of the flaps δ_f . It is also possible to make a choice for a horizontal flight. In this case the manifold pressure is changed in order to obtain a flight path angle γ which equals to zero.

By means of iterations of α , β , δ_a , δ_e and δ_r a nominal flight condition is found for,

$$\begin{aligned} C_X &= \frac{W \sin(\theta)}{\frac{1}{2} \rho V^2 S} ; & C_Z &= - \frac{W \cos(\theta)}{\frac{1}{2} \rho V^2 S} ; \\ C_Y &= C_l = C_m = C_n = 0 \end{aligned} \quad (5.1)$$

If the flight path angle γ does not equals to zero and a horizontal flight is desired, the manifold pressure is changed accordingly after which the iterations start all over again.

When the variations of all computed variables are within specified limits the iterations are ended.

The resulting nominal flight condition is used to compute the stability derivatives of the aircraft and can be used as an initial flight condition to compute the response of the aircraft for small deviations around this condition.

6. THE LINEAR SYSTEM DESCRIPTION.

For studying of small deviations from a given condition of steady flight the aerodynamic model is linearized and transformed into the stability reference system by a rotation through the angle of attack α_0 and the sideslip angle β_0 found in the nominal flight condition.

6.1. The stability derivatives.

Off this point the aircraft is assumed to be symmetric. Thus in steady flight condition the following can be written

$$C_Y = C_l = C_m = C_n = 0 \quad (6.1)$$

Fig. 3 shows the rotation from the body-fixed reference frame to the stability reference frame.

Using the mentioned angles of attack and sideslip, the stability derivatives are computed as follows:

$$\begin{aligned} C_{X_{0S}} &= C_X \cos(\alpha_0) \cos(\beta_0) + C_Z \sin(\alpha_0) \cos(\beta_0) \\ C_{Z_{0S}} &= -C_X \sin(\alpha_0) + C_Z \cos(\alpha_0) \end{aligned} \quad (6.2-a)$$

and

$$\begin{aligned} C_{X_{US}} &= \left[2C_X + \frac{\partial C_X}{\partial V} \cdot V \right] \cos(\alpha_0) \cos(\beta_0) + \left[2C_Z + \frac{\partial C_Z}{\partial V} \cdot V \right] \sin(\alpha_0) \cos(\beta_0) \\ C_{Z_{US}} &= - \left[2C_X + \frac{\partial C_X}{\partial V} \cdot V \right] \sin(\alpha_0) + \left[2C_Z + \frac{\partial C_Z}{\partial V} \cdot V \right] \cos(\alpha_0) \\ C_{m_{US}} &= \left[\frac{\partial C_m}{\partial V} \cdot V \right] \cos(\beta_0) - \left[\frac{\partial C_l}{\partial V} \cdot V \right] \cos(\alpha_0) \sin(\beta_0) - \left[\frac{\partial C_n}{\partial V} \cdot V \right] \sin(\alpha_0) \sin(\beta_0) \end{aligned} \quad (6.2-b)$$

where $\frac{\partial C_X}{\partial V} = C_{X_{\Delta P_t}} \cdot \frac{\partial}{\partial V} \left(\frac{\Delta P_t}{\frac{1}{2} \rho V^2} \right) + C_{X_{\alpha \Delta t^2}} \cdot \alpha \cdot \frac{\partial}{\partial V} \left(\frac{\Delta P_t}{\frac{1}{2} \rho V^2} \right)^2$

and $\frac{\partial C_Z}{\partial V} = C_{Z_{\Delta P_t}} \cdot \frac{\partial}{\partial V} \left(\frac{\Delta P_t}{\frac{1}{2} \rho V^2} \right) ; \frac{\partial C_m}{\partial V} = \dots \text{etc.}$

with $\frac{\partial}{\partial V} \left(\frac{\Delta P_t}{\frac{1}{2} \rho V^2} \right) = -3b \left(\frac{P}{\frac{1}{2} \rho} \right) \frac{1}{V^4} , \frac{\partial}{\partial V} \left(\frac{\Delta P_t}{\frac{1}{2} \rho V^2} \right)^2 = \dots \text{etc.}$

using equation (4.2).

The stability derivatives with respect to the angle of attack can be written as

$$\begin{aligned} C_{X_{\alpha S}} &= \frac{\partial C_X}{\partial \alpha} \cos(\alpha_0) \cos(\beta_0) + \frac{\partial C_Z}{\partial \alpha} \sin(\alpha_0) \cos(\beta_0) - \frac{\partial C_Y}{\partial \alpha} \sin(\beta_0) \\ C_{Z_{\alpha S}} &= - \frac{\partial C_X}{\partial \alpha} \sin(\alpha_0) + \frac{\partial C_Z}{\partial \alpha} \cos(\alpha_0) \end{aligned}$$

$$C_{m\alpha_S} = \frac{\partial C_m}{\partial \alpha} \cos(\beta_0) - \frac{\partial C_l}{\partial \alpha} \cos(\alpha_0) \sin(\beta_0) - \frac{\partial C_n}{\partial \alpha} \sin(\alpha_0) \sin(\beta_0) \quad (6.2-c)$$

The derivatives $C_{x_{qS}}$, $C_{z_{qS}}$, $C_{m_{qS}}$, $C_{x_{\delta S}}$, $C_{z_{\delta S}}$, $C_{m_{\delta S}}$,

$C_{x_{\delta fS}}$, $C_{z_{\delta fS}}$ and $C_{m_{\delta fS}}$ are computed similarly.

The derivatives with respect to $\dot{\alpha}$ are taken into account by the derivatives with respect to q (See Chapter 9).

The nonsymmetric part of the stability derivatives can be written as:

$$C_{Y_{0S}} = -C_X \cos(\alpha_0) \sin(\beta_0) - C_Z \sin(\alpha_0) \sin(\beta_0) \quad (6.3-a)$$

and

$$\begin{aligned} C_{Y_{\beta S}} &= \frac{\partial C_Y}{\partial \beta} \cos(\beta_0) - \frac{\partial C_X}{\partial \beta} \cos(\alpha_0) \sin(\beta_0) - \frac{\partial C_Z}{\partial \beta} \sin(\alpha_0) \sin(\beta_0) \\ C_{l_{\beta S}} &= \frac{\partial C_l}{\partial \beta} \cos(\alpha_0) \cos(\beta_0) + \frac{\partial C_n}{\partial \beta} \sin(\alpha_0) \cos(\beta_0) - \frac{\partial C_m}{\partial \beta} \sin(\beta_0) \\ C_{n_{\beta S}} &= \frac{\partial C_n}{\partial \beta} \cos(\alpha_0) - \frac{\partial C_l}{\partial \beta} \sin(\alpha_0) \end{aligned} \quad (6.3-b)$$

The other nonsymmetric derivatives e.g. $C_{Y_{pS}}$, $C_{l_{pS}}$, $C_{n_{pS}}$, $C_{Y_{rS}}$,

$C_{l_{rS}}$, $C_{n_{rS}}$, $C_{Y_{\delta aS}}$, $C_{l_{\delta aS}}$, $C_{n_{\delta aS}}$, $C_{Y_{\delta rS}}$, $C_{l_{\delta rS}}$, $C_{n_{\delta rS}}$ and $C_{Y_{\beta S}}$

are computed in the same manner.

6.2. The correction for the c.g.-position.

As the aerodynamic derivatives are measured in a certain c.g.-position of the aircraft, the computed stability derivatives are valid for this center of gravity. If the stability derivatives are desired for an other c.g.-position a correction is necessarily.

In Table 2 the c.g.-position, as used with the flight tests, is given in F_M .

The desired c.g.-position related to the mean aerodynamic chord c is converted to the measurement reference frame F_M as follows:

$$x_{c.g.M} = x_{c.g.} + 0.006 \text{ m}, \quad x \text{ in [m]}$$

$$z_{c.g.M} = z_{c.g.} + 0.007 \text{ m}, z \text{ in [m]} \quad (6.4)$$

where $x_{c.g.}$ and $z_{c.g.}$ in meters of m.a.c.

Now, the c.g.-positions are known within the measurement reference frame F_M . In order to correct the dimensionless derivatives for the c.g.-position the displacement of the c.g. is related to the mean aerodynamic chord $\bar{c}$ (symmetric). Thus the corrections of the stability derivatives become:

$$C_{m_{\Delta Pt_2}} = C_{m_{\Delta Pt_1}} - C_{Z_{\Delta Pt}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} + C_{X_{\Delta Pt}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.5)$$

Using this equation and (6.2-b) the correction of C_{m_u} can be computed.

The correction for the derivatives with respect to α can be found by using the equilibrium of moments.

$$C_{m_{\alpha_2}} = C_{m_{\alpha_1}} - C_{Z_{\alpha}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} + C_{X_{\alpha}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.6)$$

When considering the coefficients related to q a displacement of the c.g. parallel to the X_M -axis induces a change of angle of attack, while a displacement parallel to the Z_M -axis induces a change in airspeed in every point of the aircraft (See Ref. 6).

This implies:

$$C_{X_{q_2}} = C_{X_{q_1}} + C_{X_u} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.7-a)$$

$$C_{Z_{q_2}} = C_{Z_{q_1}} - C_{Z_{\alpha}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} \quad (6.7-b)$$

$$\begin{aligned} C_{m_{q_2}} = C_{m_{q_1}} - C_{Z_{q_2}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} - C_{m_{\alpha_1}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} \\ + C_{X_{q_2}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} + C_{m_{u_1}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \end{aligned} \quad (6.7-c)$$

Furthermore, the corrections for the derivatives with respect to the elevator angle and flap angle are:

$$C_{m_{\delta_2}} = C_{m_{\delta_1}} - C_{Z_{\delta}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} + C_{X_{\delta}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.8)$$

$$C_{m_{\delta f_2}} = C_{m_{\delta f_1}} - C_{Z_{\delta f}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} + C_{X_{\delta f}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.9)$$

For the sake of completeness the correction of $C_{m\dot{\alpha}}$ is also given here:

$$C_{m\dot{\alpha}2} = C_{m\dot{\alpha}1} - C_{Z\dot{\alpha}} \frac{x_{cg2} - x_{cg1}}{\bar{c}} + C_{X\dot{\alpha}} \frac{z_{cg2} - z_{cg1}}{\bar{c}} \quad (6.10)$$

The impact of the displacement of the c.g.-position on the stability derivatives for the nonsymmetric motions is in general very small.

Nonsymmetric stability derivatives on which the displacement of the c.g. could have some influence are $C_{n\beta}$, C_{Yr} and C_{nr} . So, the corrections for these derivatives are given here:

$$C_{n\beta 2} = C_{n\beta 1} + C_{Y\beta} \frac{x_{cg2} - x_{cg1}}{b/2} \quad (6.11)$$

In exactly the same way as in the case of the corrections with respect to the rate of pitch q , the corrections with respect to the rate of yaw r in consequence of a displacement of the c.g. parallel to the X_M -axis have to be entered into details. (see Ref. 6).

In case of the corrections with respect to the rate of yaw this induces a change in sideslip angle β in every point of the aircraft.
It follows that:

$$C_{Yr2} = C_{Yr1} - C_{Y\beta} \frac{x_{cg2} - x_{cg1}}{b/2} \quad (6.12-a)$$

$$C_{nr2} = C_{nr1} + C_{Yr2} \frac{x_{cg2} - x_{cg1}}{b/2} - C_{n\beta 1} \frac{x_{cg2} - x_{cg1}}{b/2} \quad (6.12-b)$$

If necessary, the corrections with respect to the remaining nonsymmetric stability derivatives can be computed in the same manner as described earlier.

7. THE CONVERSION OF MOMENTS AND PRODUCTS OF INERTIA.

When using a different c.g.-position , the moments and products of inertia have to be corrected towards the new center of gravity, e.g. translation of axes. Correction also has to take place when converting to stability reference system, e.g. rotation of axes.

7.1. Translation of axes.

Conversion formulae must take the position of the center of gravity into account. In this case the correction is towards a new center of gravity, thus simplifying to:

$$\begin{aligned}I_{x_2} &= I_x - m (\Delta z_{c.g.})^2 \\I_{y_2} &= I_y \\I_{z_2} &= I_z - m (\Delta x_{c.g.})^2 \\J_{xz_2} &= J_{xz} - m \Delta x_{c.g.} \Delta z_{c.g.}\end{aligned}\tag{7.1}$$

where m denotes the mass of the aircraft.
And $\Delta x_{c.g.}$ and $\Delta z_{c.g.}$ in [m].

7.2. Rotation of axes.

The moments and products of inertia are given in the body-fixed reference frame F_B . When converting to the stability reference frame F_S a rotation has to take place over the angles α_0 and β_0 . As the angle of side slip β_0 in stationary flight is small and the aircraft is considered symmetric, β_0 is neglected. So, the rotation only takes place in the aircraft's plane of symmetry. The conversion formulae can be written as:

$$\begin{aligned}I_{x_2} &= \frac{I_x + I_z}{2} + \frac{I_x - I_z}{2} \cos(2\alpha) - J_{xz} \sin(2\alpha) \\I_{y_2} &= I_y \\I_{z_2} &= \frac{I_x + I_z}{2} - \frac{I_x - I_z}{2} \cos(2\alpha) + J_{xz} \sin(2\alpha) \\J_{xz_2} &= \frac{I_x - I_z}{2} \sin(2\alpha) + J_{xz} \cos(2\alpha)\end{aligned}\tag{7.2}$$

8. THE FLIGHT ENVELOPE.

The aircraft condition is determined by the mass, the mass distribution and the external shape of the aircraft, e.g. weight, c.g.-position, deflection of flaps, position of the under-carriage, engine settings, etc.

In case of the DHC-2 'Beaver' the under-carriage is fixed. The flap settings, Climb (15°) and T.O. (35°) are normally used in flight. Due to the STOL characteristics of the 'Beaver'-aircraft, Landing (50°) and Full (58°) are rarely applied in actual practice.

If the aircraft condition is defined, a number of different flight conditions can be distinguished:

8.1. Take Off.

Flaps : T.O.
Power : $n = 2350$ RPM
 $p_z = 36.5$ inch Hg
TAS : 70 kts (Rotation at about 50 kts)
C : 800/900 ft/min.

8.2. Climb.

Flaps : Climb/Cruise
Power : $n = 2000$ RPM
 $p_z = 30.0$ inch Hg
TAS : 70 kts
C : 700/800 ft/min.

8.3. Cruise.

Flaps : Up (0°)
Power : $n = 1800$ RPM
 $p_z = 28.0$ inch Hg
TAS : 100 kts
C : 0 ft/min. (horizontal flight)
h : 2000-10000 ft

8.4. Descend.

Flaps : Up
Power : $n = 1800$ RPM
 $p_z = 15.0$ inch Hg
TAS : 120 kts
C : -1000 ft/min.

8.5. Approach.

Flaps : Climb/Cruise
Power : $n = 2000$ RPM
 $p_z = 15.0$ inch Hg
TAS : 80 kts
C : -400/500 ft/min.

The given values are an average of the normally applied flap deflections, engine settings and airspeeds. For instance, in cruise it is possible to vary the airspeed between 70 kts and 120 kts.

In Table 4 the stability derivatives are presented for the above-mentioned flight conditions and various aircraft conditions.

9. CONCLUSIONS.

The aerodynamic model presented in Chapter 4 can be used in nonlinear simulation. This model describes the aircraft over a wide range of flight conditions.

Except for C_{x_α} , the accuracy of all parameters is so excellent that this model is more accurate than the parameters of the former aerodynamic model (Ref. 11).

The role of C_{x_α} is completely taken into account by other variables in the C_x -model. Therefore C_{x_α} statistically equals zero.

As the aerodynamic coefficients are obtained from flight tests the effects of the coefficients with respect to $\dot{\alpha}$ are taken into account by coefficients with respect to the rate of pitch q . Consequently, the variable $\dot{\alpha}$ is omitted from the aerodynamic model (Ref 1).

In linear simulation the aircraft's response to small deviations from usual stationary flight conditions can be studied. The stability derivatives in Table 4 are computed for this purpose.

If required, the derivatives with respect to $\dot{\alpha}$ and q can be computed with the formulae given in Ref. 6. It holds that in this case $C_{m_{\dot{\alpha}}}$ added to the new C_{m_q} has to be equal to the C_{m_q} given in Table 4.

This will be unlikely as the formulae in Ref. 6. are approximations. Thus an acceptable ratio between $\dot{\alpha}$ and q has to be searched for.

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APPENDIX A

The flight test program and the development of a nonlinear aerodynamic model of the DHC-2 "Beaver" aircraft.

The flight tests on which the aerodynamic model is based were performed in the 1977/1978 period on the DHC-2 "Beaver", owned and operated by the Delft University of Technology.

For this purpose the aircraft was equipped with a high-accuracy measurement system with high registration speed and a hydraulic control system consisting of electro-hydraulic servo's, a side-stick controller and a taperecorder to implement the input signals.

The flight tests were carried out under the following conditions:

- Several input signals were implemented by the electro-hydraulic system.
- The longitudinal (δ_e) and lateral (δ_a , δ_r) excitations were applied separately.
- Each input signal was implemented at the following flight conditions: 35 m/s TAS, 45 m/s TAS, 55 m/s TAS at 6000 ft pressure altitude.

Furthermore flight tests were carried out at several altitudes and flap settings.

An extensive description of the preparation and the performing of the flight tests can be found in Ref. 2.

In Ref. 1 the DUT input signal is chosen to measure an aerodynamic model of the 'Beaver'-aircraft. The results of these measurements are used to compute the aerodynamic coefficients. This is done by means of a model identification procedure based on a Stepwise Regression procedure.

The aerodynamic coefficients selected by this procedure are given in Table 3. and are valid within the 35-55 TAS-range.

Manufacturer	The De Havilland Aircraft of Canada Ltd.
Serial no.	1244
Type	Seven-seat, single engine, high-wing all-metal aircraft.
Wing span (b)	14.63 m
Wing area (S)	23.23 m ²
Mean aerodynamic chord ($\bar{c}$)	1.5875 m
Wing sweep	0°
Wing dihedral	1°
Wing profile	NACA 64 A 416
Fuselage length	9.22 m
Max. Take-off weight	2315 kgf = 22800 N
Empty weight	1520 kgf = 14970 N
Engine	Pratt and Whitney Wasp Jr.
Type	R-985
Max. power	450 hp at n=2300 RPM, p_z = 26 inch Hg, h = 750 m.
Aircrew	Hamilton Standard.
Type	Two-bladed metal regulator propeller.
Diameter	2.59 m
Total contents of fuel tanks	115 imp. gallon = 521 l
Fuselage front tank	131 l = 0.1318 m ³
Fuselage center tank	131 l = 0.1318 m ³
Fuselage rear tank	95 l = 0.1318 m ³
Wing tip tanks	2 x 82 l = 2 x 0.0818 m ³
Most forward admissible c.g.-position	17.36% m.a.c. at 1725 kgf = 16989 N
	29.92% m.a.c. at 2315 kgf = 22800 N
Most backward admissible c.g.-position	40.24% m.a.c.

Table 1. General aircraft data of the DHC-2 "Beaver", PH-VTH.

$x_{c.g.}$	=	0.5996 m	in F_M
$y_{c.g.}$	=	0.0 m	in F_M
$z_{c.g.}$	=	-0.8851 m	in F_M
I_x	=	5368.39 kgm^2	in F_R
I_y	=	6928.93 kgm^2	in F_R
I_z	=	11158.75 kgm^2	in F_R
J_{xz}	=	117.64 kgm^2	in F_R
m	=	2288.231 kg	
h	=	1828.8 m	
ρ	=	1.024 kg/m^3	

Table 2. Aircraft data on which the aerodynamic model is based.

C _x			C _z			C _m			
parameter	value	rel. stand. dev.	parameter	value	rel. stand. dev.	parameter	value	rel. stand. dev.	
1	1	-0.03554	4.9%	1	-0.05504	4.2%	1	0.09448	1.9%
2	$\Delta p_t / \rho v^2$	0.1161	2.6%	$\Delta p_t / \rho v^2$	-0.1563	5.3%	$\Delta p_t / \rho v^2$	-0.07695	4.7%
3	α	0.002920	897 %	α	-5.578	0.52%	α	-0.6029	3.2%
4	α^2	5.459	3.5%	-	-	-	α^2	-2.140	2.6%
5	$q\bar{c}/V$	-0.6748	6.2%	$q\bar{c}/V$	-2.988	10 %	$q\bar{c}/V$	-15.56	0.86%
6	-	-	-	δ_a	-0.3980	4.9%	δ_e	-1.921	0.44%
7	δ_r	0.03412	5.7%	$\delta_e \delta^2$	-15.93	3.1%	δ^2	0.6921	3.1%
8	$\alpha \Delta p_t^2$	0.1453	7.6%	α^3	3.442	8.3%	$rb/2v$	-0.3118	5.4%
9	α^3	-5.162	8.1%	-	-	-	-	-	

C_Y				C_l			C_n		
	parameter	value	rel. stand. dev.	parameter	value	rel. stand. dev.	parameter	value	rel. stand. dev.
1	1	-0.002226	10 %	1	0.0005910	6.8%	1	-0.003117	1.3%
2	β	-0.7678	0.61%	β	-0.06180	1.3%	β	0.006719	11 %
3	$pb/2V$	-0.1240	27 %	$pb/2V$	-0.5045	1.1%	$pb/2V$	-0.1585	3.2%
4	$rb/2V$	0.3666	4.7%	$rb/2V$	0.1695	1.4%	$rb/2V$	-0.1112	1.9%
5	δ_a	-0.02956	23 %	δ_a	-0.09917	2.2%	δ_a	-0.003672	26 %
6	δ_r	0.1159	6.2%	δ_r	0.006934	8.7%	δ_r	-0.05265	0.66%
7	$\delta_r \alpha$	0.5238	6.0%	$\alpha^2 \Delta p_t$	-0.01406	8.9%	Δp_t^2	-0.003026	3.0%
8	$\delta b/2V$	-0.1600	9.9%	$\delta_a \alpha$	-0.08269	13 %	$q \bar{c}/V$	0.1595	5.2%
9	-	-	-	-	-	-	β^3	0.1373	7.9%

	C_x	C_z	C_m
δ_r -parameter	-0.02447	-1.377	0.4072
$\alpha \delta_r$ -parameter	1.106	-1.261	
rel. stand. dev.	0.6%	0.2%	0.3%
" " "	1.2%	4.7%	

Table 3. The nonlinear aerodynamic model of the 'Beaver'-aircraft (Ref. 1).
Valid within the 35-55 m/s TAS-range.

Index.

Flight condition	Airspeed [m/s]	Altitude [ft]
TAKE OFF	35,40,45	0
CLIMB	35, 45	2000
	35, 45	4000
	35, 45	6000
	45	8000
CRUISE	30,35,40,45,50,55,60	2000
	30,35,40,45,50,55,60	4000
	35,40,45,50,55,60	6000
	40,45,50,55,60	8000
	50,55,60	10000
DESCEND	50,55,60	2000
	50,55,60	6000
	50,55,60	10000
APPROACH	35,40	0
	35,40	2000
	35,40	4000

Table 4. The Stability and Control Derivatives of the De Havilland DHC-2 "BEAVER" aircraft for several nominal flight conditions.

TAKE OFF

Table 4. (continued)

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.017	RAD
α	=	.005	RAD	θ	=	.104	RAD	δ_e	=	.096	RAD
β	=	-.055	RAD	ψ	=	.000	RAD	δ_r	=	-.335	RAD
γ	=	.099	RAD	P	=	.000	RAD/S	δ_f	=	.611	RAD
C	=	3.46	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				
Pz	=	32.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	2200.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	262.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.998		μ_b	=	5.50		K _{XZ}	=	.00024	
				μ_c	=	50.65					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.18775	C _{Z0}	=	-.43569						
C _{Xu}	=	-.30526	C _{Zu}	=	.02469	C _{m_u}	=	.44588			
C _{Xα}	=	1.30811	C _{Zα}	=	-6.34826	C _{mα}	=	-.58125			
C _{Xq}	=	-.67540	C _{Zq}	=	-2.94941	C _{m_q}	=	-15.53872			
C _{Xδ}	=	.00000	C _{Zδ}	=	-.44646	C _{mδ}	=	-1.91829			
C _{Xδ_f}	=	-.08943	C _{Zδ_f}	=	-1.38274	C _{mδ_f}	=	.41543			
C _{Y0}	=	.00000									
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00696			
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000			
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850			
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11073			
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.09955	C _{nδ_a}	=	-.00387			
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265			

TABLE 4. (CONTINUED). TAKE OFF.
V = 35.0 M/S, H = 0.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.017	RAD
α	=	.005	RAD	θ	=	.104	RAD	δ_e	=	.096	RAD
β	=	-.055	RAD	ψ	=	.000	RAD	δ_r	=	-.335	RAD
γ	=	.099	RAD	P	=	.000	RAD/S	δ_f	=	.611	RAD
C	=	3.46	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

Pz	=	32.00	"HG	Xcg	=	.380	M.A.C.	Kx2	=	.01096
n	=	2200.0	RPM	Ycg	=	.000	M.A.C.	Ky2	=	1.20150
P	=	262.5	KW	Zcg	=	-.560	M.A.C.	Kz2	=	.02279
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	1.998		μ_b	=	5.50		Kxz	=	.00019
				μ_c	=	50.65				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

Cx0	=	.18548	Cz0	=	-.43654		
Cxu	=	-.30468	Czu	=	.02608	Cmu	= .44525
Cx α	=	1.26756	Cz α	=	-6.35415	Cm α	= -.58044
Cxq	=	-.68795	Czq	=	-2.94627	Cmq	= -15.51507
Cx δ	=	-.00203	Cz δ	=	-.44573	Cm δ	= -1.91537
Cx δ_f	=	-.09558	Cz δ_f	=	-1.38232	Cm δ_f	= .41480
Cy0	=	.01024					
Cy β	=	-.76659	Cl β	=	-.06588	Cn β	= .00724
Cy $\dot{\beta}$	=	-.07988	Cl $\dot{\beta}$	=	.00000	Cn $\dot{\beta}$	= .00000
Cyp	=	-.12381	Clp	=	-.50445	Cnp	= -.15620
Cyr	=	.36705	Clr	=	.15155	Cnr	= -.11150
Cy δ_a	=	-.02952	Cl δ_a	=	-.09941	Cn δ_a	= -.00342
Cy δ_r	=	.11750	Cl δ_r	=	.00655	Cn δ_r	= -.08268

TABLE 4. (CONTINUED). TAKE OFF.
V = 35.0 M/S, H = 0.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00 M/S	φ	=	.000 RAD	δ_a	=	.010 RAD
α	=	-.032 RAD	θ	=	.054 RAD	δ_e	=	.124 RAD
β	=	-.026 RAD	ψ	=	.000 RAD	δ_r	=	-.177 RAD
γ	=	.087 RAD	P	=	.000 RAD/S	δ_f	=	.611 RAD
C	=	3.47 M/S	q	=	.000 RAD/S			
h	=	0.0 FT	r	=	.000 RAD/S			

Pz	=	34.00 "HG	Xcg	=	.380 M.A.C.	Kx2	=	.01096
n	=	2300.0 RPM	Ycg	=	.000 M.A.C.	Ky2	=	1.20150
P	=	300.6 KW	Zcg	=	-.560 M.A.C.	Kz2	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.553	μ_b	=	5.50	Kxz	=	.00024
			μ_c	=	50.65			

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

CX ₀	=	.13315	CZ ₀	=	-.16787			
CX _u	=	-.17996	CZ _u	=	.35160	Cm _u	=	.34201
CX _{α}	=	.65899	CZ _{α}	=	-6.33763	Cm _{α}	=	-.42440
CX _q	=	-.67515	CZ _q	=	-2.94948	Cm _q	=	-15.53989
CX _{δ}	=	.00000	CZ _{δ}	=	-.40886	Cm _{δ}	=	-1.91851
CX _{δ_f}	=	-.13031	CZ _{δ_f}	=	-1.33614	Cm _{δ_f}	=	.41507
CY ₀	=	.00000						
CY _{β}	=	-.76780	Cl _{β}	=	-.06180	Cn _{β}	=	.00599
CY _{$\dot{\beta}$}	=	-.08000	Cl _{$\dot{\beta}$}	=	.00000	Cn _{$\dot{\beta}$}	=	.00000
CY _p	=	-.12400	Cl _p	=	-.50450	Cn _p	=	-.15850
CY _r	=	.36761	Cl _r	=	.16950	Cn _r	=	-.11072
CY _{δ_a}	=	-.02956	Cl _{δ_a}	=	-.09649	Cn _{δ_a}	=	-.00387
CY _{δ_r}	=	.11580	Cl _{δ_r}	=	.00693	Cn _{δ_r}	=	-.08265

TABLE 4. (CONTINUED). TAKE OFF.
V = 40.0 M/S, H = 0.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	-.032	RAD	θ	=	.054	RAD	δ_e	=	.124	RAD
β	=	-.026	RAD	ψ	=	.000	RAD	δ_r	=	-.177	RAD
γ	=	.087	RAD	P	=	.000	RAD/S	δ_f	=	.611	RAD
C	=	3.47	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

P _Z	=	34.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01099
n	=	2300.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	300.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02276
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.553		μ_b	=	5.50		K _{XZ}	=	.00062
				μ_c	=	50.65				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.13847	C_{Z_0}	=	-.16346		
C_{X_u}	=	-.19119	C_{Z_u}	=	.34559	C_{m_u}	= .34181
C_{X_α}	=	.86125	C_{Z_α}	=	-6.31295	C_{m_α}	= -.42398
C_{X_q}	=	-.57782	C_{Z_q}	=	-2.96992	$C_{m_{\dot{q}}}$	= -15.53441
C_{X_δ}	=	.01324	C_{Z_δ}	=	-.40872	C_{m_δ}	= -1.91783
$C_{X_{\delta_f}}$	=	-.08692	$C_{Z_{\delta_f}}$	=	-1.33966	$C_{m_{\delta_f}}$	= .41503
C_{Y_0}	=	.00362					
C_{Y_β}	=	-.76763	C_{l_β}	=	-.06292	C_{n_β}	= .00398
$C_{Y_{\dot{\beta}}}$	=	-.07997	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12396	C_{l_p}	=	-.49893	C_{n_p}	= -.17476
C_{Y_r}	=	.36749	C_{l_r}	=	.16481	C_{n_r}	= -.10517
$C_{Y_{\delta_a}}$	=	-.02955	$C_{l_{\delta_a}}$	=	-.09628	$C_{n_{\delta_a}}$	= -.00700
$C_{Y_{\delta_r}}$	=	.11665	$C_{l_{\delta_r}}$	=	.00960	$C_{n_{\delta_r}}$	= -.08238

TABLE 4. (CONTINUED). TAKE OFF.
V = 40.0 M/S, H = 0.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	-.058	RAD	θ	=	.017	RAD	δ_e	=	.143	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.107	RAD
γ	=	.075	RAD	P	=	.000	RAD/S	δ_f	=	.611	RAD
C	=	3.36	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

P _Z	=	36.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	2350.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	333.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.228		μ_b	=	5.50		K _{XZ}	=	.00024
				μ_c	=	50.65				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.10987	C_{Z_0}	=	.01801		
C_{X_u}	=	-.10704	C_{Z_u}	=	.57127	C_{m_u}	= .26634
C_{X_α}	=	.21370	C_{Z_α}	=	-6.31384	C_{m_α}	= -.31622
C_{X_q}	=	-.67501	C_{Z_q}	=	-2.94962	C_{m_q}	= -15.54070
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40164	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	-.15852	$C_{Z_{\delta f}}$	=	-1.30397	$C_{m_{\delta f}}$	= .41482
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00580
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.09438	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). TAKE OFF.
V = 45.0 M/S, H = 0.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	-.058	RAD	θ	=	.017	RAD	δ_e	=	.143	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.107	RAD
γ	=	.075	RAD	P	=	.000	RAD/S	δ_f	=	.611	RAD
C	=	3.36	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

Pz	=	36.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01103
n	=	2350.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	333.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02272
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.228		μ_b	=	5.50		K _{XZ}	=	.00092
				μ_c	=	50.65				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	.10863	C _{Z0}	=	.02434			
C _{Xu}	=	-.13991	C _{Zu}	=	.56411	C _{m_u}	=	.26627
C _{Xα}	=	.57789	C _{Zα}	=	-6.29088	C _{mα}	=	-.31559
C _{Xq}	=	-.50093	C _{Zq}	=	-2.98381	C _{m_q}	=	-15.53851
C _{Xδ}	=	.02325	C _{Zδ}	=	-.40160	C _{mδ}	=	-1.91829
C _{Xδ_f}	=	-.08277	C _{Zδ_f}	=	-1.31096	C _{mδ_f}	=	.41496
C _{Y0}	=	.00164						
C _{Yβ}	=	-.76777	C _{lβ}	=	-.06240	C _{nβ}	=	.00221
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.49442	C _{n_p}	=	-.18744
C _{Yr}	=	.36757	C _{l_r}	=	.17092	C _{n_r}	=	-.10072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.09399	C _{nδ_a}	=	-.00933
C _{Yδ_r}	=	.11630	C _{lδ_r}	=	.01171	C _{nδ_r}	=	-.08211

TABLE 4. (CONTINUED). TAKE OFF.
V = 45.0 M/S, H = 0.0 FT.
STABILITY REFERENCE SYSTEM.

CLIMB

Table 4. (continued)

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.015	RAD
α	=	.120	RAD	θ	=	.180	RAD	δ_e	=	-.008	RAD
β	=	-.039	RAD	ψ	=	.000	RAD	δ_r	=	-.154	RAD
γ	=	.060	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.11	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	176.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.453		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.23497	C_{Z_0}	=	-.94296		
C_{X_u}	=	-.21375	C_{Z_u}	=	-1.24540	C_{m_u}	= .31871
C_{X_α}	=	1.68777	C_{Z_α}	=	-5.75928	C_{m_α}	= -1.07890
C_{X_q}	=	-.67522	C_{Z_q}	=	-2.95299	C_{m_q}	= -15.53595
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.42265	C_{m_δ}	= -1.91843
$C_{X_{\delta_f}}$	=	.03843	$C_{Z_{\delta_f}}$	=	-1.52853	$C_{m_{\delta_f}}$	= .41657
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00634
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10911	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.015	RAD
α	=	.120	RAD	θ	=	.180	RAD	δ_e	=	-.008	RAD
β	=	-.039	RAD	ψ	=	.000	RAD	δ_r	=	-.154	RAD
γ	=	.060	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.11	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
P _Z	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01107	
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	176.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02267	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.453		μ_b	=	5.83		K _{XZ}	=	-.00117	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.12014	C_{Z_0}	=	-.96432		
C_{X_u}	=	-.36122	C_{Z_u}	=	-1.21080	C_{m_u}	= .31887
C_{X_α}	=	.98127	C_{Z_α}	=	-5.92007	C_{m_α}	= -1.07868
C_{X_q}	=	-1.02803	C_{Z_q}	=	-2.84958	C_{m_q}	= -15.52447
C_{X_δ}	=	-.05063	C_{Z_δ}	=	-.42233	C_{m_δ}	= -1.91705
$C_{X_{\delta f}}$	=	-.14497	$C_{Z_{\delta f}}$	=	-1.52211	$C_{m_{\delta f}}$	= .41585
C_{Y_0}	=	.00473					
C_{Y_β}	=	-.76725	C_{l_β}	=	-.06257	C_{n_β}	= .01370
$C_{Y_{\dot{\beta}}}$	=	-.07994	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12390	C_{l_p}	=	-.51946	C_{n_p}	= -.09688
C_{Y_r}	=	.36733	C_{l_r}	=	.14256	C_{n_r}	= -.13025
$C_{Y_{\delta a}}$	=	-.02954	$C_{l_{\delta a}}$	=	-.10870	$C_{n_{\delta a}}$	= .00924
$C_{Y_{\delta r}}$	=	.11704	$C_{l_{\delta r}}$	=	-.00302	$C_{n_{\delta r}}$	= -.08289

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.018	RAD
α	=	.130	RAD	θ	=	.200	RAD	δ_e	=	-.020	RAD
β	=	-.051	RAD	ψ	=	.000	RAD	δ_r	=	-.196	RAD
γ	=	.070	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.45	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	185.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.611		μ_b	=	6.19		K _{XZ}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.27554	C_{Z_0}	=	-1.01687		
C_{X_u}	=	-.25863	C_{Z_u}	=	-1.31907	C_{m_u}	= .35562
C_{X_α}	=	1.82922	C_{Z_α}	=	-5.73319	C_{m_α}	= -1.12196
C_{X_q}	=	-.67531	C_{Z_q}	=	-2.95315	C_{m_q}	= -15.53561
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.43873	C_{m_δ}	= -1.91833
$C_{X_{\delta f}}$	=	.04959	$C_{Z_{\delta f}}$	=	-1.54125	$C_{m_{\delta f}}$	= .41667
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00676
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11073
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10994	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.018	RAD
α	=	.130	RAD	θ	=	.200	RAD	δ_e	=	-.020	RAD
β	=	-.051	RAD	ψ	=	.000	RAD	δ_r	=	-.196	RAD
γ	=	.070	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.45	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01110
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	185.9	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02265
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.611		μ_b	=	6.19		K _{xz}	=	-.00129
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.14095	C_{Z_0}	=	-1.04405		
C_{X_u}	=	-.42722	C_{Z_u}	=	-1.27430	C_{m_u}	= .35593
C_{X_α}	=	1.06252	C_{Z_α}	=	-5.92222	C_{m_α}	= -1.12120
C_{X_q}	=	-1.05666	C_{Z_q}	=	-2.83904	C_{m_q}	= -15.51612
C_{X_δ}	=	-.05691	C_{Z_δ}	=	-.43817	C_{m_δ}	= -1.91599
$C_{X_{\delta_f}}$	=	-.15082	$C_{Z_{\delta_f}}$	=	-1.53463	$C_{m_{\delta_f}}$	= .41571
C_{Y_0}	=	.00713					
C_{Y_β}	=	-.76703	C_{l_β}	=	-.06373	C_{n_β}	= .01472
$C_{Y_{\dot{\beta}}}$	=	-.07990	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12384	C_{l_p}	=	-.52015	C_{n_p}	= -.09163
C_{Y_r}	=	.36714	C_{l_r}	=	.13766	C_{n_r}	= -.13181
$C_{Y_{\delta_a}}$	=	-.02952	$C_{l_{\delta_a}}$	=	-.10937	$C_{n_{\delta_a}}$	= .01044
$C_{Y_{\delta_r}}$	=	.11736	$C_{l_{\delta_r}}$	=	-.00385	$C_{n_{\delta_r}}$	= -.08285

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.022	RAD
α	=	.141	RAD	θ	=	.222	RAD	δ_e	=	-.033	RAD
β	=	-.066	RAD	ψ	=	.000	RAD	δ_r	=	-.252	RAD
γ	=	.080	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.81	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	194.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.782		μ_b	=	6.58		K _{XZ}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.32236	C_{Z_0}	=	-1.09522		
C_{X_u}	=	-.31683	C_{Z_u}	=	-1.39587	C_{m_u}	= .39537
C_{X_α}	=	1.98600	C_{Z_α}	=	-5.70283	C_{m_α}	= -1.16822
C_{X_q}	=	-.67542	C_{Z_q}	=	-2.95333	C_{m_q}	= -15.53525
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.46718	C_{m_δ}	= -1.91816
$C_{X_{\delta_f}}$	=	.06158	$C_{Z_{\delta_f}}$	=	-1.55491	$C_{m_{\delta_f}}$	= .41677
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00749
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11073
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11084	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.022	RAD
α	=	.141	RAD	θ	=	.222	RAD	δ_e	=	-.033	RAD
β	=	-.066	RAD	ψ	=	.000	RAD	δ_r	=	-.252	RAD
γ	=	.080	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.81	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	27.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01113
n	=	1900.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	194.6	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02262
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.782		μ_b	=	6.58		K _{xz}	=	-.00142
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.16478	C_{Z_0}	=	-1.12967		
C_{X_u}	=	-.50887	C_{Z_u}	=	-1.33745	C_{m_u}	= .39596
C_{X_α}	=	1.15311	C_{Z_α}	=	-5.92544	C_{m_α}	= -1.16645
C_{X_q}	=	-1.08691	C_{Z_q}	=	-2.82740	C_{m_q}	= -15.50158
C_{X_δ}	=	-.06555	C_{Z_δ}	=	-.46617	C_{m_δ}	= -1.91412
$C_{X_{\delta_f}}$	=	-.15735	$C_{Z_{\delta_f}}$	=	-1.54812	$C_{m_{\delta_f}}$	= .41542
C_{Y_0}	=	.01087					
C_{Y_β}	=	-.76678	C_{l_β}	=	-.06587	C_{n_β}	= .01610
$C_{Y_{\dot{\beta}}}$	=	-.07983	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12373	C_{l_p}	=	-.52064	C_{n_p}	= -.08598
C_{Y_r}	=	.36682	C_{l_r}	=	.13132	C_{n_r}	= -.13347
$C_{Y_{\delta_a}}$	=	-.02950	$C_{l_{\delta_a}}$	=	-.11004	$C_{n_{\delta_a}}$	= .01175
$C_{Y_{\delta_r}}$	=	.11777	$C_{l_{\delta_r}}$	=	-.00475	$C_{n_{\delta_r}}$	= -.08280

TABLE 4. (CONTINUED). CLIMB.
V = 35.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.044	RAD	θ	=	.069	RAD	δ_e	=	.056	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.057	RAD
γ	=	.025	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.11	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

P _Z	=	28.00	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	193.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.791		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.06888	C_{Z_0}	=	-.44868		
C_{X_u}	=	-.12921	C_{Z_u}	=	-.56703	C_{m_u}	= .16437
C_{X_α}	=	.83807	C_{Z_α}	=	-5.88801	C_{m_α}	= -.75556
C_{X_q}	=	-.67505	C_{Z_q}	=	-2.95221	C_{m_q}	= -15.53823
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40098	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	-.04535	$C_{Z_{\delta f}}$	=	-1.43300	$C_{m_{\delta f}}$	= .41582
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00578
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10284	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.044	RAD	θ	=	.069	RAD	δ_e	=	.056	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.057	RAD
γ	=	.025	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.11	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
P _Z	=	28.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	193.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.791		μ_b	=	5.83		K _{XZ}	=	-.00029	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	.04888	C _{Z0}	=	-.45130			
C _{X_u}	=	-.15425	C _{Z_u}	=	-.56073	C _{m_u}	=	.16437
C _{X_α}	=	.57535	C _{Z_α}	=	-5.91942	C _{m_α}	=	-.75583
C _{X_q}	=	-.80702	C _{Z_q}	=	-2.91911	C _{m_q}	=	-15.53713
C _{X_δ}	=	-.01780	C _{Z_δ}	=	-.40095	C _{m_δ}	=	-1.91842
C _{X_{δ_f}}	=	-.10892	C _{Z_{δ_f}}	=	-1.42958	C _{m_{δ_f}}	=	.41564
C _{Y0}	=	.00067						
C _{Y_β}	=	-.76771	C _{l_β}	=	-.06169	C _{n_β}	=	.00852
C _{Y_{β̇}}	=	-.07999	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	=	.00000
C _{Y_p}	=	-.12399	C _{l_p}	=	-.51099	C _{n_p}	=	-.13594
C _{Y_r}	=	.36758	C _{l_r}	=	.16011	C _{n_r}	=	-.11814
C _{Y_{δ_a}}	=	-.02956	C _{l_{δ_a}}	=	-.10290	C _{n_{δ_a}}	=	.00070
C _{Y_{δ_r}}	=	.11626	C _{l_{δ_r}}	=	.00326	C _{n_{δ_r}}	=	-.08288

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.051	RAD	θ	=	.083	RAD	δ_e	=	.050	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.063	RAD
γ	=	.032	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.43	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	28.00	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	202.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.869		μ_b	=	6.19		K _{XZ}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.08258	C_{Z_0}	=	-.49626		
C_{X_u}	=	-.13755	C_{Z_u}	=	-.62584	C_{m_u}	= .18245
C_{X_α}	=	.92106	C_{Z_α}	=	-5.88128	C_{m_α}	= -.78460
C_{X_q}	=	-.67507	C_{Z_q}	=	-2.95225	C_{m_q}	= -15.53801
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40162	C_{m_δ}	= -1.91856
$C_{X_{\delta_f}}$	=	-.03781	$C_{Z_{\delta_f}}$	=	-1.44160	$C_{m_{\delta_f}}$	= .41589
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00580
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10341	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.051	RAD	θ	=	.083	RAD	δ_e	=	.050	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.063	RAD
γ	=	.032	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.43	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				
Pz	=	28.00	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01097	
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	202.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.869		μ_b	=	6.19		K _{XZ}	=	-.00037	
				μ_c	=	57.04					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.05706	C_{Z_0}	=	-.49983		
C_{X_u}	=	-.16940	C_{Z_u}	=	-.61797	C_{m_u}	= .18245
C_{X_α}	=	.61813	C_{Z_α}	=	-5.92073	C_{m_α}	= -.78489
C_{X_q}	=	-.82715	C_{Z_q}	=	-2.91354	C_{m_q}	= -15.53665
C_{X_δ}	=	-.02056	C_{Z_δ}	=	-.40158	C_{m_δ}	= -1.91838
$C_{X_{\delta f}}$	=	-.11157	$C_{Z_{\delta f}}$	=	-1.43777	$C_{m_{\delta f}}$	= .41568
C_{Y_0}	=	.00086					
C_{Y_β}	=	-.76769	C_{l_β}	=	-.06168	C_{n_β}	= .00896
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51190	C_{n_p}	= -.13246
C_{Y_r}	=	.36757	C_{l_r}	=	.15886	C_{n_r}	= -.11926
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10346	$C_{n_{\delta_a}}$	= .00143
$C_{Y_{\delta_r}}$	=	.11630	$C_{l_{\delta_r}}$	=	.00269	$C_{n_{\delta_r}}$	= -.08290

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.011	RAD
α	=	.059	RAD	θ	=	.097	RAD	δ_e	=	.043	RAD
β	=	-.017	RAD	ψ	=	.000	RAD	δ_r	=	-.071	RAD
γ	=	.039	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.74	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	28.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	211.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.952		μ_b	=	6.58		K _{XZ}	=	.00024	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.09816	C_{Z_0}	=	-.54726		
C_{X_u}	=	-.14718	C_{Z_u}	=	-.68871	C_{m_u}	= .20193
C_{X_α}	=	1.01057	C_{Z_α}	=	-5.87299	C_{m_α}	= -.81578
C_{X_q}	=	-.67509	C_{Z_q}	=	-2.95230	C_{m_q}	= -15.53778
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40254	C_{m_δ}	= -1.91855
$C_{X_{\delta f}}$	=	-.02972	$C_{Z_{\delta f}}$	=	-1.45082	$C_{m_{\delta f}}$	= .41596
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00582
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10401	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.011	RAD
α	=	.059	RAD	θ	=	.097	RAD	δ_e	=	.043	RAD
β	=	-.017	RAD	ψ	=	.000	RAD	δ_r	=	-.071	RAD
γ	=	.039	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	1.74	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	28.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01097	
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	211.2	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02277	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.952		μ_b	=	6.58		K _{xz}	=	-.00045	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.06597	C_{Z_0}	=	-.55206		
C_{X_u}	=	-.18719	C_{Z_u}	=	-.67891	C_{m_u}	= .20192
C_{X_α}	=	.66450	C_{Z_α}	=	-5.92206	C_{m_α}	= -.81608
C_{X_q}	=	-.84871	C_{Z_q}	=	-2.90740	C_{m_q}	= -15.53602
C_{X_δ}	=	-.02355	C_{Z_δ}	=	-.40249	C_{m_δ}	= -1.91833
$C_{X_{\delta_f}}$	=	-.11454	$C_{Z_{\delta_f}}$	=	-1.44660	$C_{m_{\delta_f}}$	= .41571
C_{Y_0}	=	.00111					
C_{Y_β}	=	-.76767	C_{l_β}	=	-.06168	C_{n_β}	= .00943
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12398	C_{l_p}	=	-.51284	C_{n_p}	= -.12871
C_{Y_r}	=	.36756	C_{l_r}	=	.15741	C_{n_r}	= -.12046
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10404	$C_{n_{\delta_a}}$	= .00222
$C_{Y_{\delta_r}}$	=	.11636	$C_{l_{\delta_r}}$	=	.00209	$C_{n_{\delta_r}}$	= -.08291

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.011	RAD
α	=	.066	RAD	θ	=	.112	RAD	δ_e	=	.036	RAD
β	=	-.019	RAD	ψ	=	.000	RAD	δ_r	=	-.081	RAD
γ	=	.045	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.04	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

P _Z	=	28.00	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	219.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.042		μ_b	=	6.99		K _{XZ}	=	.00024
				μ_c	=	64.44				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.11594	C_{Z_0}	=	-.60197		
C_{X_u}	=	-.15855	C_{Z_u}	=	-.75593	C_{m_u}	= .22292
C_{X_α}	=	1.10723	C_{Z_α}	=	-5.86286	C_{m_α}	= -.84928
C_{X_q}	=	-.67511	C_{Z_q}	=	-2.95236	C_{m_q}	= -15.53754
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40391	C_{m_δ}	= -1.91854
$C_{X_{\delta_f}}$	=	-.02104	$C_{Z_{\delta_f}}$	=	-1.46073	$C_{m_{\delta_f}}$	= .41604
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00586
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10466	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.011	RAD
α	=	.066	RAD	θ	=	.112	RAD	δ_e	=	.036	RAD
β	=	-.019	RAD	ψ	=	.000	RAD	δ_r	=	-.081	RAD
γ	=	.045	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	2.04	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

Pz	=	28.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01098
n	=	1950.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	219.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02276
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.042		μ_b	=	6.99		K _{XZ}	=	-.00054
				μ_c	=	64.44				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.07573	C_{Z_0}	=	-.60833		
C_{X_u}	=	-.20832	C_{Z_u}	=	-.74374	C_{m_u}	= .22292
C_{X_α}	=	.71484	C_{Z_α}	=	-5.92340	C_{m_α}	= -.84957
C_{X_q}	=	-.87181	C_{Z_q}	=	-2.90064	C_{m_q}	= -15.53515
C_{X_δ}	=	-.02679	C_{Z_δ}	=	-.40384	C_{m_δ}	= -1.91824
$C_{X_{\delta_f}}$	=	-.11788	$C_{Z_{\delta_f}}$	=	-1.45611	$C_{m_{\delta_f}}$	= .41575
C_{Y_0}	=	.00146					
C_{Y_β}	=	-.76763	C_{l_β}	=	-.06171	C_{n_β}	= .00994
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12398	C_{l_p}	=	-.51381	C_{n_p}	= -.12468
C_{Y_r}	=	.36754	C_{l_r}	=	.15571	C_{n_r}	= -.12173
$C_{Y_{\delta_a}}$	=	-.02955	$C_{l_{\delta_a}}$	=	-.10467	$C_{n_{\delta_a}}$	= .00308
$C_{Y_{\delta_r}}$	=	.11643	$C_{l_{\delta_r}}$	=	.00143	$C_{n_{\delta_r}}$	= -.08293

TABLE 4. (CONTINUED). CLIMB.
V = 45.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

CRUISE

Table 4. (continued)

NOMINAL FLIGHT CONDITION :

V	=	30.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.304	RAD	θ	=	.304	RAD	δ_e	=	-.203	RAD
β	=	-.050	RAD	ψ	=	.000	RAD	δ_r	=	-.132	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
Pz	=	21.44	" HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	102.6	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.346		μ_b	=	5.83		K _{xz}	=	.00024	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.55578	C_{Z_0}	=	-1.77369		
C_{X_u}	=	.22490	C_{Z_u}	=	-2.95723	C_{m_u}	= .29365
C_{X_α}	=	2.15338	C_{Z_α}	=	-4.62586	C_{m_α}	= -1.87023
C_{X_q}	=	-.67436	C_{Z_q}	=	-2.95988	C_{m_q}	= -15.53118
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.43836	C_{m_δ}	= -1.91834
$C_{X_{\delta f}}$	=	.24138	$C_{Z_{\delta f}}$	=	-1.75991	$C_{m_{\delta f}}$	= .41837
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00675
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11073
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.12428	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 30.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	30.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.304	RAD	θ	=	.304	RAD	δ_e	=	-.203	RAD
β	=	-.050	RAD	ψ	=	.000	RAD	δ_r	=	-.132	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
Pz	=	21.44	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01188	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	102.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02186	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.346		μ_b	=	5.83		K _{XZ}	=	-.00318	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-1.85873			
C _{X_u}	=	-.66879	C _{Z_u}	=	-2.38918	C _{m_u}	=	.29445
C _{X_α}	=	.66735	C _{Z_α}	=	-5.05811	C _{m_α}	=	-1.86868
C _{X_q}	=	-1.53674	C _{Z_q}	=	-2.61878	C _{m_q}	=	-15.51284
C _{X_δ}	=	-.13091	C _{Z_δ}	=	-.43780	C _{m_δ}	=	-1.91616
C _{X_{δ_f}}	=	-.29553	C _{Z_{δ_f}}	=	-1.75157	C _{m_{δ_f}}	=	.41675
C _{Y0}	=	0.00000						
C _{Y_β}	=	-.77173	C _{l_β}	=	-.06008	C _{n_β}	=	.02487
C _{Y_{β̇}}	=	-.07990	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	=	.00000
C _{Y_p}	=	-.12384	C _{l_p}	=	-.52814	C _{n_p}	=	-.00040
C _{Y_r}	=	.36715	C _{l_r}	=	.11265	C _{n_r}	=	-.15634
C _{Y_{δ_a}}	=	-.02952	C _{l_{δ_a}}	=	-.11960	C _{n_{δ_a}}	=	.03347
C _{Y_{δ_r}}	=	.11729	C _{l_{δ_r}}	=	-.01807	C _{n_{δ_r}}	=	-.08094

TABLE 4. (CONTINUED). CRUISE.
V = 30.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	30.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.324	RAD	θ	=	.324	RAD	δ_e	=	-.229	RAD
β	=	-.062	RAD	ψ	=	.000	RAD	δ_r	=	-.158	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	20.88	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	105.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.463		μ_b	=	6.19		K _{XZ}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.62878	C_{Z_0}	=	-1.87032		
C_{X_u}	=	.20930	C_{Z_u}	=	-3.09550	C_{m_u}	= .32101
C_{X_α}	=	2.22589	C_{Z_α}	=	-4.49189	C_{m_α}	= -1.95933
C_{X_q}	=	-.67439	C_{Z_q}	=	-2.96069	C_{m_q}	= -15.53058
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.45947	C_{m_δ}	= -1.91821
$C_{X_{\delta f}}$	=	.26423	$C_{Z_{\delta f}}$	=	-1.78597	$C_{m_{\delta f}}$	= .41857
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00730
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11073
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.12599	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 30.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	30.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.324	RAD	θ	=	.324	RAD	δ_e	=	-.229	RAD
β	=	-.062	RAD	ψ	=	.000	RAD	δ_r	=	-.158	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	20.88	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01202
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	105.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02173
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.463		μ_b	=	6.19		K _{XZ}	=	-.00338
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-1.97318			
C _{X_u}	=	-.78651	C _{Z_u}	=	-3.00082	C _{m_u}	=	.32234
C _{X_α}	=	.67200	C _{Z_α}	=	-4.96703	C _{m_α}	=	-1.95647
C _{X_q}	=	-1.59025	C _{Z_q}	=	-2.58700	C _{m_q}	=	-15.50149
C _{X_δ}	=	-.14613	C _{Z_δ}	=	-.45858	C _{m_δ}	=	-1.91479
C _{X_{δ_f}}	=	-.31805	C _{Z_{δ_f}}	=	-1.77706	C _{m_{δ_f}}	=	.41660
C _{Y0}	=	0.00000						
C _{Y_β}	=	-.77527	C _{l_β}	=	-.06116	C _{n_β}	=	.02655
C _{Y_{β̇}}	=	-.07985	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	=	.00000
C _{Y_p}	=	-.12376	C _{l_p}	=	-.52769	C _{n_p}	=	.01053
C _{Y_r}	=	.36692	C _{l_r}	=	.10563	C _{n_r}	=	-.15897
C _{Y_{δ_a}}	=	-.02950	C _{l_{δ_a}}	=	-.12042	C _{n_{δ_a}}	=	.03648
C _{Y_{δ_r}}	=	.11758	C _{l_{δ_r}}	=	-.01973	C _{n_{δ_r}}	=	-.08055

TABLE 4. (CONTINUED). CRUISE.
V = 30.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.219	RAD	θ	=	.219	RAD	δ_e	=	-.109	RAD
β	=	-.023	RAD	ψ	=	.000	RAD	δ_r	=	-.065	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

P _Z	=	21.40	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	102.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.877		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.29660	C_{Z_0}	=	-1.33300		
C_{X_u}	=	.18611	C_{Z_u}	=	-2.29577	C_{m_u}	= .18422
C_{X_α}	=	1.76263	C_{Z_α}	=	-5.08302	C_{m_α}	= -1.50563
C_{X_q}	=	-.67444	C_{Z_q}	=	-2.95710	C_{m_q}	= -15.53362
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40614	C_{m_δ}	= -1.91853
$C_{X_{\delta f}}$	=	.14768	$C_{Z_{\delta f}}$	=	-1.65309	$C_{m_{\delta f}}$	= .41754
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00592
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.11727	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, h = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.219	RAD	θ	=	.219	RAD	δ_e	=	-.109	RAD
β	=	-.023	RAD	ψ	=	.000	RAD	δ_r	=	-.065	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	21.40	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01142
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	102.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02233
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.877		μ_b	=	5.83		K _{XZ}	=	-.00229
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-1.36559		
C_{X_u}	=	-.31689	C_{Z_u}	=	-2.28139	C_{m_u}	= .18429
C_{X_α}	=	.61561	C_{Z_α}	=	-5.34451	C_{m_α}	= -1.50602
C_{X_q}	=	-1.30796	C_{Z_q}	=	-2.73762	C_{m_q}	= -15.53143
C_{X_δ}	=	-.08819	C_{Z_δ}	=	-.40604	C_{m_δ}	= -1.91821
$C_{X_{\delta f}}$	=	-.21484	$C_{Z_{\delta f}}$	=	-1.64570	$C_{m_{\delta f}}$	= .41668
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76799	C_{l_β}	=	-.05951	C_{n_β}	= .01917
$C_{Y_{\dot{\beta}}}$	=	-.07998	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12397	C_{l_p}	=	-.52675	C_{n_p}	= -.04514
C_{Y_r}	=	.36752	C_{l_r}	=	.13422	C_{n_r}	= -.14490
$C_{Y_{\delta_a}}$	=	-.02955	$C_{l_{\delta_a}}$	=	-.11529	$C_{n_{\delta_a}}$	= .02169
$C_{Y_{\delta_r}}$	=	.11652	$C_{l_{\delta_r}}$	=	-.01118	$C_{n_{\delta_r}}$	= -.08218

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.233	RAD	θ	=	.233	RAD	δ_e	=	-.123	RAD
β	=	-.025	RAD	ψ	=	.000	RAD	δ_r	=	-.072	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				
Pz	=	20.85	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	105.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.950		μ_b	=	6.19		K _{XZ}	=	.00024	
				μ_c	=	57.04					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.33505	C_{Z_0}	=	-1.41044		
C_{X_u}	=	.20274	C_{Z_u}	=	-2.41615	C_{m_u}	= .20138
C_{X_α}	=	1.83810	C_{Z_α}	=	-5.01630	C_{m_α}	= -1.56704
C_{X_q}	=	-.67440	C_{Z_q}	=	-2.95751	C_{m_q}	= -15.53321
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40827	C_{m_δ}	= -1.91852
$C_{X_{\delta f}}$	=	.16348	$C_{Z_{\delta f}}$	=	-1.67110	$C_{m_{\delta f}}$	= .41768
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00597
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.11846	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.233	RAD	θ	=	.233	RAD	δ_e	=	-.123	RAD
β	=	-.025	RAD	ψ	=	.000	RAD	δ_r	=	-.072	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	20.85	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01148
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	105.3	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02226
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.950		μ_b	=	6.19		K _{xz}	=	-.00244
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-1.44969		
C_{X_u}	=	-.36106	C_{Z_u}	=	-2.39759	C_{m_u}	= .20149
C_{X_α}	=	.62781	C_{Z_α}	=	-5.30531	C_{m_α}	= -1.56730
C_{X_q}	=	-1.34739	C_{Z_q}	=	-2.71889	C_{m_q}	= -15.53002
C_{X_δ}	=	-.09433	C_{Z_δ}	=	-.40814	C_{m_δ}	= -1.91808
$C_{X_{\delta_f}}$	=	-.22710	$C_{Z_{\delta_f}}$	=	-1.66364	$C_{m_{\delta_f}}$	= .41674
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76814	C_{l_β}	=	-.05939	C_{n_β}	= .02007
$C_{Y_{\dot{\beta}}}$	=	-.07997	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12396	C_{l_p}	=	-.52730	C_{n_p}	= -.03761
C_{Y_r}	=	.36750	C_{l_r}	=	.13125	C_{n_r}	= -.14691
$C_{Y_{\delta_a}}$	=	-.02955	$C_{l_{\delta_a}}$	=	-.11611	$C_{n_{\delta_a}}$	= .02361
$C_{Y_{\delta_r}}$	=	.11661	$C_{l_{\delta_r}}$	=	-.01235	$C_{n_{\delta_r}}$	= -.08201

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.249	RAD	θ	=	.249	RAD	δ_e	=	-.140	RAD
β	=	-.029	RAD	ψ	=	.000	RAD	δ_r	=	-.081	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	20.36	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	108.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.033		μ_b	=	6.58		K _{XZ}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.37902	C _{Z0}	=	-1.49290			
C _{Xu}	=	.21675	C _{Zu}	=	-2.54222	C _{m_u}	=	.22072
C _{Xα}	=	1.91521	C _{Zα}	=	-4.93967	C _{mα}	=	-1.63328
C _{Xq}	=	-.67438	C _{Zq}	=	-2.95797	C _{m_q}	=	-15.53277
C _{Xδ}	=	.00000	C _{Zδ}	=	-.41149	C _{mδ}	=	-1.91850
C _{Xδ_f}	=	.18052	C _{Zδ_f}	=	-1.69052	C _{mδ_f}	=	.41783
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00605
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11973	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.249	RAD	θ	=	.249	RAD	δ_e	=	-.140	RAD
β	=	-.029	RAD	ψ	=	.000	RAD	δ_r	=	-.081	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	20.36	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01156
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	108.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02218
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	1.033		μ_b	=	6.58		K _{XZ}	=	-.00261
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-1.54027		
C_{X_u}	=	-.41533	C_{Z_u}	=	-2.51739	C_{m_u}	= .22089
C_{X_α}	=	.63927	C_{Z_α}	=	-5.25907	C_{m_α}	= -1.63334
C_{X_q}	=	-1.38955	C_{Z_q}	=	-2.69810	C_{m_q}	= -15.52801
C_{X_δ}	=	-.10122	C_{Z_δ}	=	-.41132	C_{m_δ}	= -1.91788
$C_{X_{\delta f}}$	=	-.24093	$C_{Z_{\delta f}}$	=	-1.68296	$C_{m_{\delta f}}$	= .41679
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76840	C_{l_β}	=	-.05931	C_{n_β}	= .02104
$C_{Y_{\dot{\beta}}}$	=	-.07997	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12395	C_{l_p}	=	-.52777	C_{n_p}	= -.02948
C_{Y_r}	=	.36746	C_{l_r}	=	.12779	C_{n_r}	= -.14904
$C_{Y_{\delta_a}}$	=	-.02955	$C_{l_{\delta_a}}$	=	-.11695	$C_{n_{\delta_a}}$	= .02571
$C_{Y_{\delta_r}}$	=	.11671	$C_{l_{\delta_r}}$	=	-.01361	$C_{n_{\delta_r}}$	= -.08181

TABLE 4. (CONTINUED). CRUISE.
V = 35.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.164	RAD	θ	=	.164	RAD	δ_e	=	-.058	RAD
β	=	-.016	RAD	ψ	=	.000	RAD	δ_r	=	-.049	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	21.72	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	105.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.635		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.17091	C _{Z0}	=	-1.03147			
C _{Xu}	=	.10128	C _{Zu}	=	-1.80611	C _{m_u}	=	.12780
C _{Xα}	=	1.43670	C _{Zα}	=	-5.29957	C _{mα}	=	-1.27068
C _{Xq}	=	-.67460	C _{Zq}	=	-2.95579	C _{m_q}	=	-15.53516
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40211	C _{mδ}	=	-1.91856
C _{Xδ_f}	=	.08714	C _{Zδ_f}	=	-1.58406	C _{mδ_f}	=	.41700
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00581
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11275	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	ϕ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.164	RAD	θ	=	.164	RAD	δ_e	=	-.058	RAD
β	=	-.016	RAD	ψ	=	.000	RAD	δ_r	=	-.049	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
Pz	=	21.72	" HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01120	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	105.9	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02254	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.635		μ_b	=	5.83		K _{xz}	=	-.00168	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-1.04553			
C _{Xu}	=	-.19530	C _{Zu}	=	-1.79837	C _{m_u}	=	.12781
C _{Xα}	=	.55057	C _{Zα}	=	-5.46314	C _{mα}	=	-1.27131
C _{Xq}	=	-1.15440	C _{Zq}	=	-2.80429	C _{m_q}	=	-15.53459
C _{Xδ}	=	-.06572	C _{Zδ}	=	-.40206	C _{mδ}	=	-1.91844
C _{Xδ_f}	=	-.17295	C _{Zδ_f}	=	-1.57700	C _{mδ_f}	=	.41640
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76778	C _{lβ}	=	-.06020	C _{nβ}	=	.01582
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12398	C _{l_p}	=	-.52356	C _{n_p}	=	-.07390
C _{Yr}	=	.36757	C _{l_r}	=	.14401	C _{n_r}	=	-.13695
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11185	C _{nδ_a}	=	.01461
C _{Yδ_r}	=	.11633	C _{lδ_r}	=	-.00667	C _{nδ_r}	=	-.08267

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.175	RAD	θ	=	.175	RAD	δ_e	=	-.068	RAD
β	=	-.017	RAD	ψ	=	.000	RAD	δ_r	=	-.051	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				
Pz	=	21.05	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	107.6	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.678		μ_b	=	6.19		K _{xz}	=	.00024	
				μ_c	=	57.04					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.19356	C _{Z0}	=	-1.09291				
C _{Xu}	=	.12013	C _{Zu}	=	-1.90877	C _{m_u}	=	.13786	
C _{Xα}	=	1.50766	C _{Zα}	=	-5.26072	C _{mα}	=	-1.31822	
C _{Xq}	=	-.67457	C _{Zq}	=	-2.95602	C _{m_q}	=	-15.53485	
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40261	C _{mδ}	=	-1.91855	
C _{Xδ_f}	=	.09940	C _{Zδ_f}	=	-1.59804	C _{mδ_f}	=	.41711	
C _{Y0}	=	.00000							
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00583	
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000	
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850	
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072	
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11366	C _{nδ_a}	=	-.00387	
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265	

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.175	RAD	θ	=	.175	RAD	δ_e	=	-.068	RAD
β	=	-.017	RAD	ψ	=	.000	RAD	δ_r	=	-.051	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S	.			
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	21.05	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01124
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	107.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02251
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.678		μ_b	=	6.19		K _{XZ}	=	-.00180
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-1.10992				
C _{Xu}	=	-.21455	C _{Zu}	=	-1.90047	C _{m_u}	=	.13787	
C _{Xα}	=	.56658	C _{Zα}	=	-5.44303	C _{mα}	=	-1.31882	
C _{Xq}	=	-1.18580	C _{Zq}	=	-2.79144	C _{m_q}	=	-15.53413	
C _{Xδ}	=	-.07020	C _{Zδ}	=	-.40255	C _{mδ}	=	-1.91841	
C _{Xδ_f}	=	-.18078	C _{Zδ_f}	=	-1.59089	C _{mδ_f}	=	.41646	
C _{Y0}	=	0.00000							
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06005	C _{nβ}	=	.01650	
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000	
C _{Yp}	=	-.12398	C _{l_p}	=	-.52433	C _{n_p}	=	-.06809	
C _{Yr}	=	.36756	C _{l_r}	=	.14219	C _{n_r}	=	-.13859	
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11258	C _{nδ_a}	=	.01601	
C _{Yδ_r}	=	.11635	C _{lδ_r}	=	-.00758	C _{nδ_r}	=	-.08259	

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.187	RAD	θ	=	.187	RAD	δ_e	=	-.078	RAD
β	=	-.018	RAD	ψ	=	.000	RAD	δ_r	=	-.054	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	20.46	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	109.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.728		μ_b	=	6.58		K _{XZ}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.21945	C_{Z_0}	=	-1.15867		
C_{X_u}	=	.13973	C_{Z_u}	=	-2.01697	C_{m_u}	= .14946
C_{X_α}	=	1.58089	C_{Z_α}	=	-5.21621	C_{m_α}	= -1.36924
C_{X_q}	=	-.67453	C_{Z_q}	=	-2.95629	C_{m_q}	= -15.53452
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40327	C_{m_δ}	= -1.91855
$C_{X_{\delta f}}$	=	.11255	$C_{Z_{\delta f}}$	=	-1.61303	$C_{m_{\delta f}}$	= .41723
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00584
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.11465	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.187	RAD	θ	=	.187	RAD	δ_e	=	-.078	RAD
β	=	-.018	RAD	ψ	=	.000	RAD	δ_r	=	-.054	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	20.46	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01128
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	109.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02246
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.728		μ_b	=	6.58		K _{XZ}	=	-.00194
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-1.17927			
C _{Xu}	=	-.23801	C _{Zu}	=	-2.00774	C _{m_u}	=	.14948
C _{Xα}	=	.58199	C _{Zα}	=	-5.41929	C _{mα}	=	-1.36980
C _{Xq}	=	-1.21931	C _{Zq}	=	-2.77729	C _{m_q}	=	-15.53356
C _{Xδ}	=	-.07503	C _{Zδ}	=	-.40320	C _{mδ}	=	-1.91838
C _{Xδ_f}	=	-.18955	C _{Zδ_f}	=	-1.60580	C _{mδ_f}	=	.41652
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76783	C _{lβ}	=	-.05989	C _{nβ}	=	.01722
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12398	C _{l_p}	=	-.52510	C _{n_p}	=	-.06185
C _{Yr}	=	.36755	C _{l_r}	=	.14015	C _{n_r}	=	-.14034
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11335	C _{nδ_a}	=	.01753
C _{Yδ_r}	=	.11639	C _{lδ_r}	=	-.00857	C _{nδ_r}	=	-.08250

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.200	RAD	θ	=	.200	RAD	δ_e	=	-.090	RAD
β	=	-.020	RAD	ψ	=	.000	RAD	δ_r	=	-.057	RAD
γ	=	.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				
Pz	=	19.95	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	112.5	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.785		μ_b	=	6.99		K _{xz}	=	.00024	
				μ_c	=	64.44					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.24908	C _{Z0}	=	-1.22906			
C _{Xu}	=	.15965	C _{Zu}	=	-2.13091	C _{m_u}	=	.16282
C _{Xα}	=	1.65635	C _{Zα}	=	-5.16515	C _{mα}	=	-1.42407
C _{Xq}	=	-.67449	C _{Zq}	=	-2.95660	C _{m_q}	=	-15.53415
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40419	C _{mδ}	=	-1.91854
C _{Xδ_f}	=	.12668	C _{Zδ_f}	=	-1.62914	C _{mδ_f}	=	.41735
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00587
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11570	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.008	RAD
α	=	.200	RAD	θ	=	.200	RAD	δ_e	=	-.090	RAD
β	=	-.020	RAD	ψ	=	.000	RAD	δ_r	=	-.057	RAD
γ	=	.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

P _Z	=	19.95	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01133
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	112.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02241
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.785		μ_b	=	6.99		K _{XZ}	=	-.00208
				μ_c	=	64.44				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.00000	C_{Z_0}	=	-1.25405		
C_{X_u}	=	-.26673	C_{Z_u}	=	-2.12016	C_{m_u}	= .16285
C_{X_α}	=	.59672	C_{Z_α}	=	-5.39123	C_{m_α}	= -1.42458
C_{X_q}	=	-1.25512	C_{Z_q}	=	-2.76166	C_{m_q}	= -15.53283
C_{X_δ}	=	-.08027	C_{Z_δ}	=	-.40411	C_{m_δ}	= -1.91833
$C_{X_{\delta f}}$	=	-.19939	$C_{Z_{\delta f}}$	=	-1.62184	$C_{m_{\delta f}}$	= .41659
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76787	C_{l_β}	=	-.05973	C_{n_β}	= .01800
$C_{Y_{\dot{\beta}}}$	=	-.07998	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12398	C_{l_p}	=	-.52583	C_{n_p}	= -.05514
C_{Y_r}	=	.36754	C_{l_r}	=	.13786	C_{n_r}	= -.14219
$C_{Y_{\delta a}}$	=	-.02955	$C_{l_{\delta a}}$	=	-.11415	$C_{n_{\delta a}}$	= .01919
$C_{Y_{\delta r}}$	=	.11644	$C_{l_{\delta r}}$	=	-.00962	$C_{n_{\delta r}}$	= -.08238

TABLE 4. (CONTINUED). CRUISE.
V = 40.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.126	RAD	θ	=	.126	RAD	δ_e	=	-.029	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.044	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

P _Z	=	22.66	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	116.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.511		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.10386	C_{Z_0}	=	-.81955		
C_{X_u}	=	.03613	C_{Z_u}	=	-1.44018	C_{m_u}	= .09898
C_{X_α}	=	1.17109	C_{Z_α}	=	-5.41392	C_{m_α}	= -1.10722
C_{X_q}	=	-.67473	C_{Z_q}	=	-2.95509	C_{m_q}	= -15.53621
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40100	C_{m_δ}	= -1.91856
$C_{X_{\delta_f}}$	=	.04495	$C_{Z_{\delta_f}}$	=	-1.53596	$C_{m_{\delta_f}}$	= .41662
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00578
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10959	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.126	RAD	θ	=	.126	RAD	δ_e	=	-.029	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.044	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	22.66	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01109
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	116.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02266
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.511		μ_b	=	5.83		K _{XZ}	=	-.00124
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.82610			
C _{X_u}	=	-.14520	C _{Z_u}	=	-1.43329	C _{m_u}	=	.09898
C _{X_α}	=	.48079	C _{Z_α}	=	-5.51819	C _{m_α}	=	-1.10787
C _{X_q}	=	-1.04529	C _{Z_q}	=	-2.84591	C _{m_q}	=	-15.53586
C _{X_δ}	=	-.05041	C _{Z_δ}	=	-.40096	C _{m_δ}	=	-1.91848
C _{X_{δ_f}}	=	-.14850	C _{Z_{δ_f}}	=	-1.52942	C _{m_{δ_f}}	=	.41617
C _{Y0}	=	0.00000						
C _{Y_β}	=	-.76775	C _{l_β}	=	-.06071	C _{n_β}	=	.01350
C _{Y_{β̇}}	=	-.07999	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	=	.00000
C _{Y_p}	=	-.12399	C _{l_p}	=	-.52038	C _{n_p}	=	-.09382
C _{Y_r}	=	.36758	C _{l_r}	=	.14988	C _{n_r}	=	-.13116
C _{Y_{δ_a}}	=	-.02956	C _{l_{δ_a}}	=	-.10920	C _{n_{δ_a}}	=	.00994
C _{Y_{δ_r}}	=	.11625	C _{l_{δ_r}}	=	-.00351	C _{n_{δ_r}}	=	-.08287

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.135	RAD	θ	=	.135	RAD	δ_e	=	-.035	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.045	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				
Pz	=	21.83	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	116.6	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.537		μ_b	=	6.19		K _{XZ}	=	.00024	
				μ_c	=	57.04					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.11799	C _{Z0}	=	-.86900			
C _{Xu}	=	.05102	C _{Zu}	=	-1.52719	C _{m_u}	=	.10489
C _{Xα}	=	1.23614	C _{Zα}	=	-5.38994	C _{mα}	=	-1.14533
C _{Xq}	=	-.67470	C _{Zq}	=	-2.95524	C _{m_q}	=	-15.53597
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40120	C _{mδ}	=	-1.91856
C _{Xδ_f}	=	.05479	C _{Zδ_f}	=	-1.54718	C _{mδ_f}	=	.41671
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00579
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11033	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.135	RAD	θ	=	.135	RAD	δ_e	=	-.035	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.045	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	21.83	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01111
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	116.6	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02263
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.537		μ_b	=	6.19		K _{xz}	=	-.00134
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.87697		
C_{X_u}	=	-.15491	C_{Z_u}	=	-1.52017	C_{m_u}	= .10490
C_{X_α}	=	.49932	C_{Z_α}	=	-5.50724	C_{m_α}	= -1.14597
C_{X_q}	=	-1.07088	C_{Z_q}	=	-2.83656	C_{m_q}	= -15.53560
C_{X_δ}	=	-.05397	C_{Z_δ}	=	-.40116	C_{m_δ}	= -1.91847
$C_{X_{\delta_f}}$	=	-.15386	$C_{Z_{\delta_f}}$	=	-1.54048	$C_{m_{\delta_f}}$	= .41622
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06059	C_{n_β}	= .01404
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52119	C_{n_p}	= -.08918
C_{Y_r}	=	.36758	C_{l_r}	=	.14856	C_{n_r}	= -.13253
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10984	$C_{n_{\delta_a}}$	= .01101
$C_{Y_{\delta_r}}$	=	.11627	$C_{l_{\delta_r}}$	=	-.00425	$C_{n_{\delta_r}}$	= -.08283

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.145	RAD	θ	=	.145	RAD	δ_e	=	-.043	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.046	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	21.08	"HG	x _{cg}	=	.380	M.A.C.	K _X ²	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _Y ²	=	1.20150	
P	=	116.9	KW	z _{cg}	=	-.560	M.A.C.	K _Z ²	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.566		μ_b	=	6.58		K _{XZ}	=	.00024	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.13417	C_{Z_0}	=	-.92205		
C_{X_u}	=	.06731	C_{Z_u}	=	-1.61946	C_{m_u}	= .11178
C_{X_α}	=	1.30379	C_{Z_α}	=	-5.36238	C_{m_α}	= -1.18623
C_{X_q}	=	-.67467	C_{Z_q}	=	-2.95540	C_{m_q}	= -15.53571
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40145	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	.06535	$C_{Z_{\delta f}}$	=	-1.55922	$C_{m_{\delta f}}$	= .41681
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00580
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11112	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.145	RAD	θ	=	.145	RAD	δ_e	=	-.043	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.046	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	21.08	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01114
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	116.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02261
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.566		μ_b	=	6.58		K _{XZ}	=	-.00145
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.93177		
C_{X_u}	=	-.16657	C_{Z_u}	=	-1.61227	C_{m_u}	= .11179
C_{X_α}	=	.51761	C_{Z_α}	=	-5.49424	C_{m_α}	= -1.18687
C_{X_q}	=	-1.09824	C_{Z_q}	=	-2.82629	C_{m_q}	= -15.53530
C_{X_δ}	=	-.05780	C_{Z_δ}	=	-.40141	C_{m_δ}	= -1.91846
$C_{X_{\delta_f}}$	=	-.15984	$C_{Z_{\delta_f}}$	=	-1.55238	$C_{m_{\delta_f}}$	= .41628
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76776	C_{l_β}	=	-.06047	C_{n_β}	= .01462
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52201	C_{n_p}	= -.08420
C_{Y_r}	=	.36757	C_{l_r}	=	.14711	C_{n_r}	= -.13398
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11051	$C_{n_{\delta_a}}$	= .01217
$C_{Y_{\delta_r}}$	=	.11628	$C_{l_{\delta_r}}$	=	-.00504	$C_{n_{\delta_r}}$	= -.08279

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.155	RAD	θ	=	.155	RAD	δ_e	=	-.051	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.047	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

Pz	=	20.41	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	117.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.600		μ_b	=	6.99		K _{XZ}	=	.00024
				μ_c	=	64.44				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.15273	C_{Z_0}	=	-.97901		
C_{X_u}	=	.08499	C_{Z_u}	=	-1.71723	C_{m_u}	= .11981
C_{X_α}	=	1.37405	C_{Z_α}	=	-5.33070	C_{m_α}	= -1.23017
C_{X_q}	=	-.67463	C_{Z_q}	=	-2.95560	C_{m_q}	= -15.53542
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40177	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	.07669	$C_{Z_{\delta f}}$	=	-1.57215	$C_{m_{\delta f}}$	= .41691
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00580
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11197	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	45.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.155	RAD	θ	=	.155	RAD	δ_e	=	-.051	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.047	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

P _Z	=	20.41	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01117
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	117.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02258
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.600		μ_b	=	6.99		K _{XZ}	=	-.00157
				μ_c	=	64.44				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.99085		
C_{X_u}	=	-.18069	C_{Z_u}	=	-1.70981	C_{m_u}	= .11982
C_{X_α}	=	.53552	C_{Z_α}	=	-5.47879	C_{m_α}	= -1.23080
C_{X_q}	=	-1.12752	C_{Z_q}	=	-2.81497	C_{m_q}	= -15.53494
C_{X_δ}	=	-.06192	C_{Z_δ}	=	-.40172	C_{m_δ}	= -1.91845
$C_{X_{\delta f}}$	=	-.16654	$C_{Z_{\delta f}}$	=	-1.56518	$C_{m_{\delta f}}$	= .41634
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76777	C_{l_β}	=	-.06033	C_{n_β}	= .01525
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52284	C_{n_p}	= -.07884
C_{Y_r}	=	.36757	C_{l_r}	=	.14552	C_{n_r}	= -.13553
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11121	$C_{n_{\delta_a}}$	= .01343
$C_{Y_{\delta_r}}$	=	.11630	$C_{l_{\delta_r}}$	=	-.00589	$C_{n_{\delta_r}}$	= -.08273

TABLE 4. (CONTINUED). CRUISE.
V = 45.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.098	RAD	θ	=	.098	RAD	δ_e	=	-.011	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	24.25	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	135.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.445		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.06574	C_{Z_0}	=	-.66590		
C_{X_u}	=	-.00690	C_{Z_u}	=	-1.16392	C_{m_u}	= .08354
C_{X_α}	=	.95609	C_{Z_α}	=	-5.47802	C_{m_α}	= -.98889
C_{X_q}	=	-.67481	C_{Z_q}	=	-2.95470	$C_{m_{\dot{q}}}$	= -15.53697
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40049	C_{m_δ}	= -1.91857
$C_{X_{\delta_f}}$	=	.01436	$C_{Z_{\delta_f}}$	=	-1.50108	$C_{m_{\delta_f}}$	= .41635
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10731	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.098	RAD	θ	=	.098	RAD	δ_e	=	-.011	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	24.25	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01103
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	135.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02272
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.445		μ_b	=	5.83		K _{XZ}	=	-.00092
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.66914		
C_{X_u}	=	-.12121	C_{Z_u}	=	-1.15761	C_{m_u}	= .08354
C_{X_α}	=	.41298	C_{Z_α}	=	-5.54544	C_{m_α}	= -.98949
C_{X_q}	=	-.96525	C_{Z_q}	=	-2.87354	C_{m_q}	= -15.53662
C_{X_δ}	=	-.03934	C_{Z_δ}	=	-.40046	C_{m_δ}	= -1.91849
$C_{X_{\delta f}}$	=	-.13317	$C_{Z_{\delta f}}$	=	-1.49523	$C_{m_{\delta f}}$	= .41600
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06105	C_{n_β}	= .01181
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51759	C_{n_p}	= -.10817
C_{Y_r}	=	.36758	C_{l_r}	=	.15385	C_{n_r}	= -.12685
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10716	$C_{n_{\delta a}}$	= .00669
$C_{Y_{\delta r}}$	=	.11622	$C_{l_{\delta r}}$	=	-.00122	$C_{n_{\delta r}}$	= -.08293

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.106	RAD	θ	=	.106	RAD	δ_e	=	-.015	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	23.24	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	132.9	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.461		μ_b	=	6.19		K _{XZ}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.07494	C_{Z_0}	=	-.70638		
C_{X_u}	=	.00385	C_{Z_u}	=	-1.23753	C_{m_u}	= .08719
C_{X_α}	=	1.01469	C_{Z_α}	=	-5.46266	C_{m_α}	= -1.02006
C_{X_q}	=	-.67479	C_{Z_q}	=	-2.95479	C_{m_q}	= -15.53677
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40061	C_{m_δ}	= -1.91856
$C_{X_{\delta_f}}$	=	.02242	$C_{Z_{\delta_f}}$	=	-1.51027	$C_{m_{\delta_f}}$	= .41642
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10791	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.106	RAD	θ	=	.106	RAD	δ_e	=	-.015	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _z	=	23.24	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01104
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	132.9	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02270
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.461		μ_b	=	6.19		K _{xz}	=	-.00101
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-0.71035		
C_{X_u}	=	-0.12672	C_{Z_u}	=	-1.23103	C_{m_u}	= 0.08719
C_{X_α}	=	0.43243	C_{Z_α}	=	-5.53922	C_{m_α}	= -1.02067
C_{X_q}	=	-0.98641	C_{Z_q}	=	-2.86647	C_{m_q}	= -15.53643
C_{X_δ}	=	-0.04226	C_{Z_δ}	=	-0.40058	C_{m_δ}	= -1.91849
$C_{X_{\delta f}}$	=	-0.13701	$C_{Z_{\delta f}}$	=	-1.50421	$C_{m_{\delta f}}$	= 0.41604
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-0.76775	C_{l_β}	=	-0.06096	C_{n_β}	= 0.01226
$C_{Y_{\dot{\beta}}}$	=	-0.07999	$C_{l_{\dot{\beta}}}$	=	0.00000	$C_{n_{\dot{\beta}}}$	= 0.00000
C_{Y_p}	=	-0.12399	C_{l_p}	=	-0.51836	C_{n_p}	= -0.10439
C_{Y_r}	=	0.36758	C_{l_r}	=	0.15282	C_{n_r}	= -0.12799
$C_{Y_{\delta_a}}$	=	-0.02956	$C_{l_{\delta_a}}$	=	-0.10771	$C_{n_{\delta_a}}$	= 0.00753
$C_{Y_{\delta_r}}$	=	0.11622	$C_{l_{\delta_r}}$	=	-0.00182	$C_{n_{\delta_r}}$	= -0.08292

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.114	RAD	θ	=	.114	RAD	δ_e	=	-.020	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	22.31	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	131.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.479		μ_b	=	6.58		K _{XZ}	=	.00024	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.08549	C _{Z0}	=	-.74987		
C _{X_u}	=	.01590	C _{Z_u}	=	-1.31597	C _{m_u}	= .09144
C _{X_α}	=	1.07607	C _{Z_α}	=	-5.44494	C _{m_α}	= -1.05356
C _{X_q}	=	-.67477	C _{Z_q}	=	-2.95490	C _{m_q}	= -15.53656
C _{X_δ}	=	.00000	C _{Z_δ}	=	-.40075	C _{m_δ}	= -1.91856
C _{X_{δ_f}}	=	.03108	C _{Z_{δ_f}}	=	-1.52015	C _{m_{δ_f}}	= .41650
C _{Y0}	=	.00000					
C _{Y_β}	=	-.76780	C _{l_β}	=	-.06180	C _{n_β}	= .00578
C _{Y_{β̇}}	=	-.08000	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	= .00000
C _{Y_p}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	= -.15850
C _{Y_r}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	= -.11072
C _{Y_{δ_a}}	=	-.02956	C _{l_{δ_a}}	=	-.10856	C _{n_{δ_a}}	= -.00387
C _{Y_{δ_r}}	=	.11580	C _{l_{δ_r}}	=	.00693	C _{n_{δ_r}}	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.114	RAD	θ	=	.114	RAD	δ_e	=	-.020	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	22.31	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01106
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	131.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02269
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.479		μ_b	=	6.58		K _{XZ}	=	-.00110
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.75473		
C_{X_u}	=	-.13325	C_{Z_u}	=	-1.30931	C_{m_u}	= .09144
C_{X_α}	=	.45204	C_{Z_α}	=	-5.53173	C_{m_α}	= -1.05418
C_{X_q}	=	-1.00909	C_{Z_q}	=	-2.85871	C_{m_q}	= -15.53622
C_{X_δ}	=	-.04539	C_{Z_δ}	=	-.40071	C_{m_δ}	= -1.91849
$C_{X_{\delta f}}$	=	-.14130	$C_{Z_{\delta f}}$	=	-1.51388	$C_{m_{\delta f}}$	= .41609
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06087	C_{n_β}	= .01273
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51916	C_{n_p}	= -.10033
C_{Y_r}	=	.36758	C_{l_r}	=	.15171	C_{n_r}	= -.12922
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10829	$C_{n_{\delta_a}}$	= .00845
$C_{Y_{\delta_r}}$	=	.11624	$C_{l_{\delta_r}}$	=	-.00247	$C_{n_{\delta_r}}$	= -.08290

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.122	RAD	θ	=	.122	RAD	δ_e	=	-.026	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.044	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				
Pz	=	21.47	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	130.1	KW	z _{cg}	=	-.560 [®]	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.500		μ_b	=	6.99		K _{XZ}	=	.00024	
				μ_c	=	64.44					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.09762	C _{Z0}	=	-.79663				
C _{Xu}	=	.02937	C _{Zu}	=	-1.39953	C _{m_u}	=	.09640	
C _{Xα}	=	1.14028	C _{Zα}	=	-5.42448	C _{mα}	=	-1.08957	
C _{Xq}	=	-.67474	C _{Zq}	=	-2.95503	C _{m_q}	=	-15.53633	
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40091	C _{mδ}	=	-1.91856	
C _{Xδ_f}	=	.04039	C _{Zδ_f}	=	-1.53076	C _{mδ_f}	=	.41658	
C _{Y0}	=	.00000							
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00578	
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000	
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850	
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072	
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10925	C _{nδ_a}	=	-.00387	
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265	

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

STATIONAIRY INITIAL CONDITION :

V	=	50.00	M/S	PHI	=	.000	RAD
ALFA	=	.122	RAD	THETA	=	.122	RAD
BETA	=	-.014	RAD	PSI	=	.000	RAD
GAMMA	=	0.000	RAD	P	=	.000	RAD/S
C	=	0.00	M/S	Q	=	.000	RAD/S
H	=	8000.0	FT	R	=	.000	RAD/S
PZ	=	21.47	INCH HG	DAIL	=	.009	RAD
N	=	1800.0	RPM	DELV	=	-.026	RAD
P	=	176.8	BHP	DRUC	=	-.044	RAD
DPT/.5RV2	=	.500		DFLAP	=	.000	RAD
X-CG	=	.374	M.A.C.	Y-CG	=	.000	M.A.C.
Z-CG	=	-.562	M.A.C.				
MUC	=	64.44		MUB	=	6.99	
KY2	=	1.20		KX2	=	.01108	
				KZ2	=	.02267	
				KXZ	=	-.00119	

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

CX0	=	0.00000	CZ0	=	-.80259	CMU	=	.09786
CXU	=	-.14106	CZU	=	-1.39271	CMA	=	-1.12470
CXA	=	.47168	CZA	=	-5.52290	CMFA	=	.00000
CXFA	=	.00000	CZFA	=	.00000	CMQ	=	-15.55832
CXQ	=	-1.03313	CZQ	=	-2.88374	CMDE	=	-1.92082
CXDE	=	-.04876	CZDE	=	-.40087	CMDF	=	.40716
CXDF	=	-.14609	CZDF	=	-1.52430			
CY0	=	0.00000	CLB	=	-.06076	CNB	=	.01426
CYB	=	-.76775	CLFB	=	.00000	CNFB	=	.00000
CYFB	=	-.07999	CLP	=	-.51999	CNP	=	-.09596
CYP	=	-.12399	CLR	=	.15049	CNR	=	-.13099
CYR	=	.36657	CLDA	=	-.10890	CNDA	=	.00949
CYDA	=	-.02956	CLDR	=	-.00317	CNDR	=	-.08288
CYDR	=	.11625						

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.131	RAD	θ	=	.131	RAD	δ_e	=	-.033	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.045	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	20.72	"HG	Xcg	=	.380	M.A.C.	Kx2	=	.01096
n	=	1800.0	RPM	Ycg	=	.000	M.A.C.	Ky2	=	1.20150
P	=	129.5	KW	Zcg	=	-.560	M.A.C.	Kz2	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.525		μ_b	=	7.44		Kxz	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.11157	C_{Z_0}	=	-.84694		
C_{X_u}	=	.04433	C_{Z_u}	=	-1.48849	C_{m_u}	= .10220
C_{X_α}	=	1.20736	C_{Z_α}	=	-5.40084	C_{m_α}	= -1.12833
C_{X_q}	=	-.67471	C_{Z_q}	=	-2.95517	C_{m_q}	= -15.53608
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40111	C_{m_δ}	= -1.91856
$C_{X_{\delta_f}}$	=	.05040	$C_{Z_{\delta_f}}$	=	-1.54217	$C_{m_{\delta_f}}$	= .41667
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00579
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11000	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.131	RAD	θ	=	.131	RAD	δ_e	=	-.033	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.045	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

P _z	=	20.72	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01110
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	129.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02264
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.525	μ_b	=	7.44	K _{XZ}	=	-.00130		
			μ_c	=	68.59					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.85425			
C _{Xu}	=	-.15044	C _{Zu}	=	-1.48153	C _{m_u}	=	.10220
C _{Xα}	=	.49124	C _{Zα}	=	-5.51227	C _{mα}	=	-1.12897
C _{Xq}	=	-1.05947	C _{Zq}	=	-2.84076	C _{m_q}	=	-15.53572
C _{Xδ}	=	-.05238	C _{Zδ}	=	-.40107	C _{mδ}	=	-1.91848
C _{Xδ_f}	=	-.15144	C _{Zδ_f}	=	-1.53554	C _{mδ_f}	=	.41620
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06065	C _{nβ}	=	.01380
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.52083	C _{n_p}	=	-.09125
C _{Yr}	=	.36758	C _{l_r}	=	.14916	C _{n_r}	=	-.13192
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10955	C _{nδ_a}	=	.01053
C _{Yδ_r}	=	.11626	C _{lδ_r}	=	-.00392	C _{nδ_r}	=	-.08285

TABLE 4. (CONTINUED). CRUISE.
V = 50.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.078	RAD	θ	=	.078	RAD	δ_e	=	.001	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	26.47	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	160.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.407		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.04296	C _{Z0}	=	-.55134		
C _{Xu}	=	-.03432	C _{Zu}	=	-.95267	C _{m_u}	= .07464
C _{Xα}	=	.78237	C _{Zα}	=	-5.51555	C _{mα}	= -.90068
C _{Xq}	=	-.67487	C _{Zq}	=	-2.95447	C _{m_q}	= -15.53752
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40019	C _{mδ}	= -1.91857
C _{Xδ_f}	=	-.00846	C _{Zδ_f}	=	-1.47506	C _{mδ_f}	= .41615
C _{Y0}	=	.00000					
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	= .00576
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	= .00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	= -.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	= -.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10560	C _{nδ_a}	= -.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.078	RAD	θ	=	.078	RAD	δ_e	=	.001	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	26.47	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01100
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	160.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02275
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.407		μ_b	=	5.83		K _{XZ}	=	-.00068
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.55301			
C_{X_u}	=	-.10822	C_{Z_u}	=	-.94713	C_{m_u}	=	.07464
C_{X_α}	=	.35123	C_{Z_α}	=	-5.55966	C_{m_α}	=	-.90121
C_{X_q}	=	-.90504	C_{Z_q}	=	-2.89275	C_{m_q}	=	-15.53713
C_{X_δ}	=	-.03109	C_{Z_δ}	=	-.40017	C_{m_δ}	=	-1.91850
$C_{X_{\delta_f}}$	=	-.12302	$C_{Z_{\delta_f}}$	=	-1.46995	$C_{m_{\delta_f}}$	=	.41587
C_{Y_0}	=	0.00000						
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06127	C_{n_β}	=	.01054
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	=	.00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51525	C_{n_p}	=	-.11883
C_{Y_r}	=	.36759	C_{l_r}	=	.15668	C_{n_r}	=	-.12356
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10557	$C_{n_{\delta_a}}$	=	.00434
$C_{Y_{\delta_r}}$	=	.11619	$C_{l_{\delta_r}}$	=	.00049	$C_{n_{\delta_r}}$	=	-.08294

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.084	RAD	θ	=	.084	RAD	δ_e	=	-.002	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

P _Z	=	25.27	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	156.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.417		μ_b	=	6.19		K _{XZ}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.04916	C_{Z_0}	=	-.58500		
C_{X_u}	=	-.02677	C_{Z_u}	=	-1.01516	C_{m_u}	= .07705
C_{X_α}	=	.83464	C_{Z_α}	=	-5.50543	C_{m_α}	= -.92660
C_{X_q}	=	-.67485	C_{Z_q}	=	-2.95453	C_{m_q}	= -15.53736
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40028	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	-.00175	$C_{Z_{\delta f}}$	=	-1.48271	$C_{m_{\delta f}}$	= .41621
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10610	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.084	RAD	θ	=	.084	RAD	δ_e	=	-.002	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	25.27	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01100
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	156.3	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02274
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.417		μ_b	=	6.19		K _{xz}	=	-.00075
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.58706		
C_{X_u}	=	-.11167	C_{Z_u}	=	-1.00935	C_{m_u}	= .07704
C_{X_α}	=	.37044	C_{Z_α}	=	-5.55599	C_{m_α}	= -.92715
C_{X_q}	=	-.92277	C_{Z_q}	=	-2.88723	C_{m_q}	= -15.53698
C_{X_δ}	=	-.03351	C_{Z_δ}	=	-.40025	C_{m_δ}	= -1.91850
$C_{X_{\delta f}}$	=	-.12589	$C_{Z_{\delta f}}$	=	-1.47736	$C_{m_{\delta f}}$	= .41591
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06121	C_{n_β}	= .01092
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51596	C_{n_p}	= -.11570
C_{Y_r}	=	.36759	C_{l_r}	=	.15586	C_{n_r}	= -.12453
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10605	$C_{n_{\delta a}}$	= .00503
$C_{Y_{\delta r}}$	=	.11620	$C_{l_{\delta r}}$	=	-.00001	$C_{n_{\delta r}}$	= -.08294

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.090	RAD	θ	=	.090	RAD	δ_e	=	-.006	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	24.15	" HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	152.4	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.429		μ_b	=	6.58		K _{xz}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.05628	C_{Z_0}	=	-.62120		
C_{X_u}	=	-.01817	C_{Z_u}	=	-1.08198	C_{m_u}	= .07982
C_{X_α}	=	.88970	C_{Z_α}	=	-5.49371	C_{m_α}	= -.95447
C_{X_q}	=	-.67484	C_{Z_q}	=	-2.95461	C_{m_q}	= -15.53718
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40037	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.00546	$C_{Z_{\delta f}}$	=	-1.49093	$C_{m_{\delta f}}$	= .41627
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10664	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.090	RAD	θ	=	.090	RAD	δ_e	=	-.006	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	24.15	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01101
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	152.4	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02273
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.429		μ_b	=	6.58		K _{XZ}	=	-.00083
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	.00000	C_{Z_0}	=	-.62374		
C_{X_u}	=	-.11571	C_{Z_u}	=	-1.07592	C_{m_u}	= .07982
C_{X_α}	=	.39009	C_{Z_α}	=	-5.55157	C_{m_α}	= -.95504
C_{X_q}	=	-.94180	C_{Z_q}	=	-2.88118	C_{m_q}	= -15.53682
C_{X_δ}	=	-.03612	C_{Z_δ}	=	-.40034	C_{m_δ}	= -1.91849
$C_{X_{\delta_f}}$	=	-.12908	$C_{Z_{\delta_f}}$	=	-1.48534	$C_{m_{\delta_f}}$	= .41595
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06114	C_{n_β}	= .01132
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51670	C_{n_p}	= -.11233
C_{Y_r}	=	.36759	C_{l_r}	=	.15496	C_{n_r}	= -.12557
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10655	$C_{n_{\delta_a}}$	= .00577
$C_{Y_{\delta_r}}$	=	.11621	$C_{l_{\delta_r}}$	=	-.00055	$C_{n_{\delta_r}}$	= -.08294

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.097	RAD	θ	=	.097	RAD	δ_e	=	-.010	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

Pz	=	23.12	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	149.1	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.443		μ_b	=	6.99		K _{XZ}	=	.00024
				μ_c	=	64.44				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.06448	C_{Z_0}	=	-.66016		
C_{X_u}	=	-.00839	C_{Z_u}	=	-1.15342	C_{m_u}	= .08304
C_{X_α}	=	.94765	C_{Z_α}	=	-5.48011	C_{m_α}	= -.98446
C_{X_q}	=	-.67482	C_{Z_q}	=	-2.95469	C_{m_q}	= -15.53699
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40047	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.01322	$C_{Z_{\delta f}}$	=	-1.49978	$C_{m_{\delta f}}$	= .41634
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10722	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.097	RAD	θ	=	.097	RAD	δ_e	=	-.010	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

P _Z	=	23.12	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01103
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	149.1	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02272
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.443		μ_D	=	6.99		K _{XZ}	=	-.00091
				μ_C	=	64.44				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-0.66330		
C_{X_u}	=	-0.12047	C_{Z_u}	=	-1.14714	C_{m_u}	= 0.08304
C_{X_α}	=	0.41012	C_{Z_α}	=	-5.54628	C_{m_α}	= -0.99506
C_{X_q}	=	-0.96223	C_{Z_q}	=	-2.87453	C_{m_q}	= -15.53664
C_{X_δ}	=	-0.03893	C_{Z_δ}	=	-0.40044	C_{m_δ}	= -1.91849
$C_{X_{\delta_f}}$	=	-0.13263	$C_{Z_{\delta_f}}$	=	-1.49396	$C_{m_{\delta_f}}$	= 0.41599
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-0.76775	C_{l_β}	=	-0.06106	C_{n_β}	= 0.01175
$C_{Y_{\dot{\beta}}}$	=	-0.07999	$C_{l_{\dot{\beta}}}$	=	0.00000	$C_{n_{\dot{\beta}}}$	= 0.00000
C_{Y_p}	=	-0.12399	C_{l_p}	=	-0.51748	C_{n_p}	= -0.10871
C_{Y_r}	=	0.36758	C_{l_r}	=	0.15399	C_{n_r}	= -0.12668
$C_{Y_{\delta_a}}$	=	-0.02956	$C_{l_{\delta_a}}$	=	-0.10708	$C_{n_{\delta_a}}$	= 0.00657
$C_{Y_{\delta_r}}$	=	0.11621	$C_{l_{\delta_r}}$	=	-0.00113	$C_{n_{\delta_r}}$	= -0.08293

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.105	RAD	θ	=	.105	RAD	δ_e	=	-.015	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

P _Z	=	22.18	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	146.4	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.459		μ_b	=	7.44		K _{XZ}	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.07394	C _{Z0}	=	-.70211		
C _{X_u}	=	.00269	C _{Z_u}	=	-1.22979	C _{m_u}	= .08680
C _{X_α}	=	1.00857	C _{Z_α}	=	-5.46433	C _{m_α}	= -1.01677
C _{X_q}	=	-.67479	C _{Z_q}	=	-2.95478	C _{m_q}	= -15.53679
C _{X_δ}	=	.00000	C _{Z_δ}	=	-.40060	C _{m_δ}	= -1.91856
C _{X_{δ_f}}	=	.02157	C _{Z_{δ_f}}	=	-1.50930	C _{m_{δ_f}}	= .41642
C _{Y0}	=	.00000					
C _{Y_β}	=	-.76780	C _{l_β}	=	-.06180	C _{n_β}	= .00577
C _{Y_{β̇}}	=	-.08000	C _{l_{β̇}}	=	.00000	C _{n_{β̇}}	= .00000
C _{Y_p}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	= -.15850
C _{Y_r}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	= -.11072
C _{Y_{δ_a}}	=	-.02956	C _{l_{δ_a}}	=	-.10785	C _{n_{δ_a}}	= -.00387
C _{Y_{δ_r}}	=	.11580	C _{l_{δ_r}}	=	.00693	C _{n_{δ_r}}	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.105	RAD	θ	=	.105	RAD	δ_e	=	-.015	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.043	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	22.18	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01104
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	146.4	Kw	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02270
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.459	μ_b	=	7.44	K _{XZ}	=	-.00100		
			μ_c	=	68.59					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.70599			
C _{Xu}	=	-.12611	C _{Zu}	=	-1.22331	C _{m_u}	=	.08679
C _{Xα}	=	.43044	C _{Zα}	=	-5.53991	C _{mα}	=	-1.01738
C _{Xq}	=	-.98418	C _{Zq}	=	-2.86722	C _{m_q}	=	-15.53645
C _{Xδ}	=	-.04195	C _{Zδ}	=	-.40056	C _{mδ}	=	-1.91849
C _{Xδ_f}	=	-.13660	C _{Zδ_f}	=	-1.50326	C _{mδ_f}	=	.41604
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06097	C _{nβ}	=	.01221
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.51828	C _{n_p}	=	-.10479
C _{Yr}	=	.36758	C _{l_r}	=	.15293	C _{n_r}	=	-.12787
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10765	C _{nδ_a}	=	.00744
C _{Yδ_r}	=	.11622	C _{lδ_r}	=	-.00176	C _{nδ_r}	=	-.08292

TABLE 4. (CONTINUED). CRUISE.
V = 55.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.062	RAD	θ	=	.062	RAD	δ_e	=	.010	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	29.28	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	193.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.383		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.02879	C_{Z_0}	=	-.46379		
C_{X_u}	=	-.05170	C_{Z_u}	=	-.78871	C_{m_u}	= .06910
C_{X_α}	=	.64156	C_{Z_α}	=	-5.53832	C_{m_α}	= -.83331
C_{X_q}	=	-.67490	C_{Z_q}	=	-2.95433	C_{m_q}	= -15.53794
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40000	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	-.02591	$C_{Z_{\delta f}}$	=	-1.45517	$C_{m_{\delta f}}$	= .41600
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10430	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.062	RAD	θ	=	.062	RAD	δ_e	=	.010	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

P _Z	=	29.28	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01098
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	193.2	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02277
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.383		μ_b	=	5.83		K _{XZ}	=	-.00049
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.46468			
C _{Xu}	=	-.10046	C _{Zu}	=	-.78400	C _{m_u}	=	.06909
C _{Xα}	=	.29696	C _{Zα}	=	-5.56742	C _{mα}	=	-.83376
C _{Xq}	=	-.85876	C _{Zq}	=	-2.90661	C _{m_q}	=	-15.53751
C _{Xδ}	=	-.02478	C _{Zδ}	=	-.39997	C _{mδ}	=	-1.91850
C _{Xδ_f}	=	-.11600	C _{Zδ_f}	=	-1.45077	C _{mδ_f}	=	.41577
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06143	C _{nβ}	=	.00957
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.51332	C _{n_p}	=	-.12694
C _{Yr}	=	.36759	C _{l_r}	=	.15878	C _{n_r}	=	-.12102
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10433	C _{nδ_a}	=	.00260
C _{Yδ_r}	=	.11617	C _{lδ_r}	=	.00180	C _{nδ_r}	=	-.08292

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.067	RAD	θ	=	.067	RAD	δ_e	=	.007	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				
Pz	=	27.88	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	186.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.390		μ_b	=	6.19		K _{XZ}	=	.00024	
				μ_c	=	57.04					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.03308	C _{Z0}	=	-.49219			
C _{Xu}	=	-.04644	C _{Zu}	=	-.84211	C _{mu}	=	.07079
C _{Xα}	=	.68801	C _{Zα}	=	-5.53150	C _{mα}	=	-.85515
C _{Xq}	=	-.67489	C _{Zq}	=	-2.95438	C _{mq}	=	-15.53781
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40006	C _{mδ}	=	-1.91857
C _{Xδ_f}	=	-.02025	C _{Zδ_f}	=	-1.46162	C _{mδ_f}	=	.41605
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00576
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{lp}	=	-.50450	C _{np}	=	-.15850
C _{Yr}	=	.36761	C _{lr}	=	.16950	C _{nr}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10472	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.067	RAD	θ	=	.067	RAD	δ_e	=	.007	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	27.88	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01098
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	186.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02276
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.390		μ_b	=	6.19		K _{XZ}	=	-.00055
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.49330		
C_{X_u}	=	-.10280	C_{Z_u}	=	-.83710	C_{m_u}	= .07079
C_{X_α}	=	.31526	C_{Z_α}	=	-5.56518	C_{m_α}	= -.85564
C_{X_q}	=	-.87379	C_{Z_q}	=	-2.90219	C_{m_q}	= -15.53739
C_{X_δ}	=	-.02683	C_{Z_δ}	=	-.40003	C_{m_δ}	= -1.91850
$C_{X_{\delta_f}}$	=	-.11821	$C_{Z_{\delta_f}}$	=	-1.45698	$C_{m_{\delta_f}}$	= .41580
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06138	C_{n_β}	= .00989
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51396	C_{n_p}	= -.12431
C_{Y_r}	=	.36759	C_{l_r}	=	.15811	C_{n_r}	= -.12185
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10474	$C_{n_{\delta_a}}$	= .00316
$C_{Y_{\delta_r}}$	=	.11618	$C_{l_{\delta_r}}$	=	.00138	$C_{n_{\delta_r}}$	= -.08293

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.073	RAD	θ	=	.073	RAD	δ_e	=	.004	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

P _Z	=	26.57	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	180.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.399		μ_b	=	6.58		K _{XZ}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.03802	C_{Z_0}	=	-.52274		
C_{X_u}	=	-.04037	C_{Z_u}	=	-.89933	C_{m_u}	= .07272
C_{X_α}	=	.73714	C_{Z_α}	=	-5.52356	C_{m_α}	= -.87867
C_{X_q}	=	-.67488	C_{Z_q}	=	-2.95442	C_{m_q}	= -15.53766
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40013	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	-.01416	$C_{Z_{\delta f}}$	=	-1.46856	$C_{m_{\delta f}}$	= .41610
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10517	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.073	RAD	θ	=	.073	RAD	δ_e	=	.004	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	26.57	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01099
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	180.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02276
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.399		μ_b	=	6.58		K _{XZ}	=	-.00062
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.52412			
C _{Xu}	=	-.10551	C _{Zu}	=	-.89403	C _{m_u}	=	.07272
C _{Xα}	=	.33420	C _{Zα}	=	-5.56248	C _{mα}	=	-.87917
C _{Xq}	=	-.88994	C _{Zq}	=	-2.89736	C _{m_q}	=	-15.53725
C _{Xδ}	=	-.02903	C _{Zδ}	=	-.40010	C _{mδ}	=	-1.91850
C _{Xδ_f}	=	-.12066	C _{Zδ_f}	=	-1.46367	C _{mδ_f}	=	.41583
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06133	C _{nβ}	=	.01023
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.51464	C _{n_p}	=	-.12148
C _{Yr}	=	.36759	C _{l_r}	=	.15737	C _{n_r}	=	-.12273
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10517	C _{nδ_a}	=	.00377
C _{Yδ_r}	=	.11619	C _{lδ_r}	=	.00092	C _{nδ_r}	=	-.08294

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.079	RAD	θ	=	.079	RAD	δ_e	=	.001	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

P _Z	=	25.34	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	174.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.408		μ_b	=	6.99		K _{XZ}	=	.00024
				μ_c	=	64.44				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.04373	C_{Z_0}	=	-.55564		
C_{X_u}	=	-.03339	C_{Z_u}	=	-.96067	C_{m_u}	= .07494
C_{X_α}	=	.78910	C_{Z_α}	=	-5.51430	C_{m_α}	= -.90399
C_{X_q}	=	-.67487	C_{Z_q}	=	-2.95448	C_{m_q}	= -15.53750
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40020	C_{m_δ}	= -1.91857
$C_{X_{\delta_f}}$	=	-.00760	$C_{Z_{\delta_f}}$	=	-1.47604	$C_{m_{\delta_f}}$	= .41616
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10566	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 8000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.079	RAD	θ	=	.079	RAD	δ_e	=	.001	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	8000.0	FT	r	=	.000	RAD/S				

Pz	=	25.34	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01100
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	174.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02275
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.408		μ_b	=	6.99		K _{XZ}	=	-.00069
				μ_c	=	64.44				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	0.00000	C_{Z_0}	=	-.55735		
C_{X_u}	=	-.10865	C_{Z_u}	=	-.95509	C_{m_u}	= .07494
C_{X_α}	=	.35374	C_{Z_α}	=	-5.55922	C_{m_α}	= -.90452
C_{X_q}	=	-.90731	C_{Z_q}	=	-2.89205	C_{m_q}	= -15.53711
C_{X_δ}	=	-.03140	C_{Z_δ}	=	-.40018	C_{m_δ}	= -1.91850
$C_{X_{\delta f}}$	=	-.12338	$C_{Z_{\delta f}}$	=	-1.47089	$C_{m_{\delta f}}$	= .41587
C_{Y_0}	=	0.00000					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06127	C_{n_β}	= .01059
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51534	C_{n_p}	= -.11843
C_{Y_r}	=	.36759	C_{l_r}	=	.15657	C_{n_r}	= -.12369
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10564	$C_{n_{\delta_a}}$	= .00443
$C_{Y_{\delta_r}}$	=	.11619	$C_{l_{\delta_r}}$	=	.00043	$C_{n_{\delta_r}}$	= -.08294

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 8000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.085	RAD	θ	=	.085	RAD	δ_e	=	-.003	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	0.000	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	0.00	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	24.20	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	169.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.419		μ_b	=	7.44		K _{XZ}	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.05032	C_{Z_0}	=	-.59109		
C_{X_u}	=	-.02536	C_{Z_u}	=	-1.02643	C_{m_u}	= .07750
C_{X_α}	=	.84399	C_{Z_α}	=	-5.50352	C_{m_α}	= -.93129
C_{X_q}	=	-.67485	C_{Z_q}	=	-2.95455	C_{m_q}	= -15.53733
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40029	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	-.00054	$C_{Z_{\delta f}}$	=	-1.48409	$C_{m_{\delta f}}$	= .41622
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10619	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00 M/S	φ	=	.000 RAD	δ_a	=	.009 RAD
α	=	.085 RAD	θ	=	.085 RAD	δ_e	=	-.003 RAD
β	=	-.012 RAD	ψ	=	.000 RAD	δ_r	=	-.042 RAD
γ	=	0.000 RAD	P	=	.000 RAD/S	δ_f	=	.000 RAD
C	=	0.00 M/S	q	=	.000 RAD/S			
h	=	10000.0 FT	r	=	.000 RAD/S			

P _Z	=	24.20 " HG	X _{cg}	=	.380 M.A.C.	K _{X2}	=	.01101
n	=	1800.0 RPM	Y _{cg}	=	.000 M.A.C.	K _{Y2}	=	1.20150
P	=	169.8 KW	Z _{cg}	=	-.560 M.A.C.	K _{Z2}	=	.02274
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.419	μ_b	=	7.44	K _{XZ}	=	-.00076
			μ_c	=	68.59			

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	0.00000	C _{Z0}	=	-.59323			
C _{Xu}	=	-.11233	C _{Zu}	=	-1.02058	C _{m_u}	=	.07750
C _{Xα}	=	.37382	C _{Zα}	=	-5.55528	C _{mα}	=	-.93184
C _{Xq}	=	-.92598	C _{Zq}	=	-2.88622	C _{m_q}	=	-15.53695
C _{Xδ}	=	-.03395	C _{Zδ}	=	-.40026	C _{mδ}	=	-1.91850
C _{Xδ_f}	=	-.12642	C _{Zδ_f}	=	-1.47870	C _{mδ_f}	=	.41591
C _{Y0}	=	0.00000						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06120	C _{nβ}	=	.01098
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.51609	C _{n_p}	=	-.11513
C _{Yr}	=	.36759	C _{l_r}	=	.15571	C _{n_r}	=	-.12471
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10613	C _{nδ_a}	=	.00515
C _{Yδ_r}	=	.11620	C _{lδ_r}	=	-.00010	C _{nδ_r}	=	-.08294

TABLE 4. (CONTINUED). CRUISE.
V = 60.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

DESCEND

Table 4. (continued)

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.106	RAD	θ	=	.057	RAD	δ_e	=	-.003	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.049	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.46	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.162		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.03817	C_{Z_0}	=	-.66805		
C_{X_u}	=	.04908	C_{Z_u}	=	-1.30091	C_{m_u}	= .01751
C_{X_α}	=	.99267	C_{Z_α}	=	-5.46122	C_{m_α}	= -1.02292
C_{X_q}	=	-.67470	C_{Z_q}	=	-2.95480	C_{m_q}	= -15.53689
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40031	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.02315	$C_{Z_{\delta f}}$	=	-1.51110	$C_{m_{\delta f}}$	= .41643
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10796	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.106	RAD	θ	=	.057	RAD	δ_e	=	-.003	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.049	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.46	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
P _Z	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01104	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02270	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.162		μ_b	=	5.83		K _{XZ}	=	-.00101	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.03295	C_{Z_0}	=	-.66833		
C_{X_u}	=	-.08928	C_{Z_u}	=	-1.29877	C_{m_u}	= .01751
C_{X_α}	=	.40711	C_{Z_α}	=	-5.53574	C_{m_α}	= -1.02355
C_{X_q}	=	-.98825	C_{Z_q}	=	-2.86584	C_{m_q}	= -15.53671
C_{X_δ}	=	-.04249	C_{Z_δ}	=	-.40028	C_{m_δ}	= -1.91851
$C_{X_{\delta f}}$	=	-.13737	$C_{Z_{\delta f}}$	=	-1.50502	$C_{m_{\delta f}}$	= .41605
C_{Y_0}	=	-.00040					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06093	C_{n_β}	= .01229
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51844	C_{n_p}	= -.10405
C_{Y_r}	=	.36759	C_{l_r}	=	.15297	C_{n_r}	= -.12810
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10776	$C_{n_{\delta a}}$	= .00761
$C_{Y_{\delta r}}$	=	.11620	$C_{l_{\delta r}}$	=	-.00183	$C_{n_{\delta r}}$	= -.08292

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.121	RAD	θ	=	.081	RAD	δ_e	=	-.014	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.040	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.98	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	46.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.227		μ_b	=	6.58		K _{XZ}	=	.00024	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.06118	C _{Z0}	=	-.75225			
C _{Xu}	=	.07038	C _{Zu}	=	-1.43899	C _{m_u}	=	.03259
C _{Xα}	=	1.10292	C _{Zα}	=	-5.42745	C _{mα}	=	-1.08456
C _{Xq}	=	-.67466	C _{Zq}	=	-2.95501	C _{m_q}	=	-15.53648
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40049	C _{mδ}	=	-1.91857
C _{Xδ_f}	=	.03908	C _{Zδ_f}	=	-1.52926	C _{mδ_f}	=	.41657
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00577
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10915	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.121	RAD	θ	=	.081	RAD	δ_e	=	-.014	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.040	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.58	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01108
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	46.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02267
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.227		μ_b	=	6.58		K _{XZ}	=	-.00118
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02987	C_{Z_0}	=	-.75414		
C_{X_u}	=	-.10346	C_{Z_u}	=	-1.43699	C_{m_u}	= .03259
C_{X_α}	=	.44084	C_{Z_α}	=	-5.52078	C_{m_α}	= -1.08521
C_{X_q}	=	-1.02993	C_{Z_q}	=	-2.85140	C_{m_q}	= -15.53635
C_{X_δ}	=	-.04824	C_{Z_δ}	=	-.40046	C_{m_δ}	= -1.91851
$C_{X_{\delta f}}$	=	-.14540	$C_{Z_{\delta f}}$	=	-1.52284	$C_{m_{\delta f}}$	= .41614
C_{Y_0}	=	-.00037					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06074	C_{n_β}	= .01317
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51988	C_{n_p}	= -.09658
C_{Y_r}	=	.36758	C_{l_r}	=	.15096	C_{n_r}	= -.13034
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10882	$C_{n_{\delta_a}}$	= .00930
$C_{Y_{\delta_r}}$	=	.11621	$C_{l_{\delta_r}}$	=	-.00307	$C_{n_{\delta_r}}$	= -.08288

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.138	RAD	θ	=	.106	RAD	δ_e	=	-.027	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.032	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.59	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

P _Z	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.302		μ_b	=	7.44		K _{XZ}	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.09019	C _{Z0}	=	-.84943			
C _{Xu}	=	.09777	C _{Zu}	=	-1.59821	C _{m_u}	=	.05013
C _{Xα}	=	1.22487	C _{Zα}	=	-5.38266	C _{mα}	=	-1.15646
C _{Xq}	=	-.67461	C _{Zq}	=	-2.95528	C _{m_q}	=	-15.53601
C _{Xδ}	=	.00000	C _{Zδ}	=	-.40074	C _{mδ}	=	-1.91856
C _{Xδ_f}	=	.05765	C _{Zδ_f}	=	-1.55044	C _{mδ_f}	=	.41674
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00578
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{l_p}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{l_r}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.11054	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	50.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.138	RAD	θ	=	.106	RAD	δ_e	=	-.027	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.032	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.59	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				
Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01112	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02263	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.302		μ_b	=	7.44		K _{xz}	=	-.00137	
				μ_c	=	68.59					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02713	C_{Z_0}	=	-.85382		
C_{X_u}	=	-.12227	C_{Z_u}	=	-1.59653	C_{m_u}	= .05013
C_{X_α}	=	.47498	C_{Z_α}	=	-5.49976	C_{m_α}	= -1.15714
C_{X_q}	=	-1.07825	C_{Z_q}	=	-2.83383	C_{m_q}	= -15.53591
C_{X_δ}	=	-.05494	C_{Z_δ}	=	-.40070	C_{m_δ}	= -1.91851
$C_{X_{\delta f}}$	=	-.15546	$C_{Z_{\delta f}}$	=	-1.54370	$C_{m_{\delta f}}$	= .41625
C_{Y_0}	=	-.00036					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06052	C_{n_β}	= .01419
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52142	C_{n_p}	= -.08783
C_{Y_r}	=	.36758	C_{l_r}	=	.14855	C_{n_r}	= -.13292
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11002	$C_{n_{\delta_a}}$	= .01132
$C_{Y_{\delta_r}}$	=	.11623	$C_{l_{\delta_r}}$	=	-.00446	$C_{n_{\delta_r}}$	= -.08282

TABLE 4. (CONTINUED). DESCEND.
V = 50.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.085	RAD	θ	=	.030	RAD	δ_e	=	.009	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.055	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-3.03	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.143		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.01652	C_{Z_0}	=	-.55276		
C_{X_u}	=	.01279	C_{Z_u}	=	-1.07909	C_{m_u}	= .01316
C_{X_α}	=	.82182	C_{Z_α}	=	-5.50344	C_{m_α}	= -.93152
C_{X_q}	=	-.67478	C_{Z_q}	=	-2.95455	C_{m_q}	= -15.53746
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40009	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	-.00049	$C_{Z_{\delta f}}$	=	-1.48415	$C_{m_{\delta f}}$	= .41622
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10620	$C_{n_{\delta a}}$	= -.00387
$C_{Y_{\delta r}}$	=	.11580	$C_{l_{\delta r}}$	=	.00693	$C_{n_{\delta r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.085	RAD	θ	=	.030	RAD	δ_e	=	.009	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.055	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-3.03	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01101
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02274
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.143		μ_b	=	5.83		K _{XZ}	=	-.00076
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	-.03046	C _{Z0}	=	-.55217			
C _{Xu}	=	-.07883	C _{Zu}	=	-1.07628	C _{m_u}	=	.01316
C _{Xα}	=	.35152	C _{Zα}	=	-5.55333	C _{mα}	=	-.93209
C _{Xq}	=	-.92605	C _{Zq}	=	-2.38619	C _{m$\dot{q}$}	=	-15.53719
C _{Xδ}	=	-.03395	C _{Zδ}	=	-.40006	C _{mδ}	=	-1.91851
C _{Xδ_f}	=	-.12644	C _{Zδ_f}	=	-1.47875	C _{mδ_f}	=	.41592
C _{Y0}	=	-.00035						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06118	C _{nβ}	=	.01098
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l$\dot{p}$}	=	-.51610	C _{n$\dot{p}$}	=	-.11511
C _{Yr}	=	.36759	C _{l$\dot{r}$}	=	.15587	C _{n$\dot{r}$}	=	-.12472
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10613	C _{nδ_a}	=	.00515
C _{Yδ_r}	=	.11618	C _{lδ_r}	=	-.00011	C _{nδ_r}	=	-.08294

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.097	RAD	θ	=	.053	RAD	δ_e	=	.000	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.044	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.44	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				
Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	46.8	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.192		μ_b	=	6.58		K _{XZ}	=	.00024	
				μ_c	=	60.60					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.03281	C_{Z_0}	=	-.62288		
C_{X_u}	=	.02736	C_{Z_u}	=	-1.19655	C_{m_u}	= .02449
C_{X_α}	=	.92124	C_{Z_α}	=	-5.48094	C_{m_α}	= -.98274
C_{X_q}	=	-.67475	C_{Z_q}	=	-2.95468	C_{m_q}	= -15.53712
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40022	C_{m_δ}	= -1.91857
$C_{X_{\delta_f}}$	=	.01276	$C_{Z_{\delta_f}}$	=	-1.49926	$C_{m_{\delta_f}}$	= .41634
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10719	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.097	RAD	θ	=	.053	RAD	δ_e	=	.000	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.044	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.44	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01103
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	46.8	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02272
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.192		μ_b	=	6.58		K _{xz}	=	-.00090
				μ_c	=	60.60				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02764	C_{Z_0}	=	-.62313		
C_{X_u}	=	-.08859	C_{Z_u}	=	-1.19358	C_{m_u}	= .02449
C_{X_α}	=	.38608	C_{Z_α}	=	-5.54437	C_{m_α}	= -.98335
C_{X_q}	=	-.96098	C_{Z_q}	=	-2.87494	C_{m_q}	= -15.53690
C_{X_δ}	=	-.03874	C_{Z_δ}	=	-.40019	C_{m_δ}	= -1.91851
$C_{X_{\delta f}}$	=	-.13242	$C_{Z_{\delta f}}$	=	-1.49345	$C_{m_{\delta f}}$	= .41599
C_{Y_0}	=	-.00033					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06104	C_{n_β}	= .01171
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51744	C_{n_p}	= -.10892
C_{Y_r}	=	.36759	C_{l_r}	=	.15425	C_{n_r}	= -.12662
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10705	$C_{n_{\delta a}}$	= .00652
$C_{Y_{\delta r}}$	=	.11619	$C_{l_{\delta r}}$	=	-.00110	$C_{n_{\delta r}}$	= -.08293

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.111	RAD	θ	=	.076	RAD	δ_e	=	-.009	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.035	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.94	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

P _Z	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.248		μ_b	=	7.44		K _{XZ}	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.05336	C_{Z_0}	=	-.70398		
C_{X_u}	=	.04662	C_{Z_u}	=	-1.33226	C_{m_u}	= .03766
C_{X_α}	=	1.03223	C_{Z_α}	=	-5.45100	C_{m_α}	= -1.04240
C_{X_q}	=	-.67471	C_{Z_q}	=	-2.95486	C_{m_q}	= -15.53673
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40039	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.02819	$C_{Z_{\delta f}}$	=	-1.51685	$C_{m_{\delta f}}$	= .41648
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00577
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10834	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, h = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	55.00	M/S	φ	=	.000	RAD	δ_a	=	.010	RAD
α	=	.111	RAD	θ	=	.076	RAD	δ_e	=	-.009	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.035	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-1.94	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01105
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02269
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.248		μ_b	=	7.44		K _{XZ}	=	-.00107
				μ_c	=	68.59				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	-.02488	C _{Z0}	=	-.70556			
C _{Xu}	=	-.10111	C _{Zu}	=	-1.32924	C _{m_u}	=	.03766
C _{Xα}	=	.42233	C _{Zα}	=	-5.53176	C _{mα}	=	-1.04304
C _{Xq}	=	-1.00147	C _{Zq}	=	-2.86134	C _{m$\dot{q}$}	=	-15.53655
C _{Xδ}	=	-.04431	C _{Zδ}	=	-.40036	C _{mδ}	=	-1.91851
C _{Xδ_f}	=	-.13985	C _{Zδ_f}	=	-1.51065	C _{mδ_f}	=	.41608
C _{Y0}	=	-.00031						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06087	C _{nβ}	=	.01257
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l$\dot{p}$}	=	-.51890	C _{n$\dot{p}$}	=	-.10169
C _{Yr}	=	.36759	C _{l$\dot{r}$}	=	.15232	C _{n$\dot{r}$}	=	-.12881
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10810	C _{nδ_a}	=	.00814
C _{Yδ_r}	=	.11621	C _{lδ_r}	=	-.00226	C _{nδ_r}	=	-.08291

TABLE 4. (CONTINUED). DESCEND.
V = 55.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.069	RAD	θ	=	.006	RAD	δ_e	=	.017	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.063	RAD	p	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-3.77	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.130		μ_b	=	5.83		K _{xz}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{X0}	=	.00277	C _{Z0}	=	-.46467			
C _{Xu}	=	-.00993	C _{Zu}	=	-.90898	C _{m_u}	=	.01013
C _{Xα}	=	.68293	C _{Zα}	=	-5.52917	C _{mα}	=	-.86227
C _{Xq}	=	-.67482	C _{Zq}	=	-2.95439	C _{m_q}	=	-15.53788
C _{Xδ}	=	.00000	C _{Zδ}	=	-.39993	C _{mδ}	=	-1.91857
C _{Xδ_f}	=	-.01842	C _{Zδ_f}	=	-1.46371	C _{mδ_f}	=	.41606
C _{Y0}	=	.00000						
C _{Yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00576
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12400	C _{lp}	=	-.50450	C _{n_p}	=	-.15850
C _{Yr}	=	.36761	C _{lr}	=	.16950	C _{n_r}	=	-.11072
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10486	C _{nδ_a}	=	-.00387
C _{Yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.069	RAD	θ	=	.006	RAD	δ_e	=	.017	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.063	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-3.77	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01098
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	28.3	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02276
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.130		μ_b	=	5.83		K _{xz}	=	-.00057
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	-.02916	C _{Z0}	=	-.46377			
C _{Xu}	=	-.07236	C _{Zu}	=	-.90615	C _{m_u}	=	.01013
C _{Xα}	=	.30116	C _{Zα}	=	-5.56303	C _{mα}	=	-.86276
C _{Xq}	=	-.87860	C _{Zq}	=	-2.90076	C _{m_q}	=	-15.53755
C _{Xδ}	=	-.02748	C _{Zδ}	=	-.39990	C _{mδ}	=	-1.91851
C _{Xδ_f}	=	-.11894	C _{Zδ_f}	=	-1.45899	C _{mδ_f}	=	.41581
C _{Y0}	=	-.00032						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06135	C _{nβ}	=	.00999
C _{Y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{l_p}	=	-.51417	C _{n_p}	=	-.12346
C _{Yr}	=	.36759	C _{l_r}	=	.15802	C _{n_r}	=	-.12211
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10487	C _{nδ_a}	=	.00334
C _{Yδ_r}	=	.11617	C _{lδ_r}	=	.00124	C _{nδ_r}	=	-.08293

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.079	RAD	θ	=	.028	RAD	δ_e	=	.011	RAD
β	=	-.011	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.051	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-3.06	M/S	q	=	.000	RAD/S				
h	=	6000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	46.8	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.168		μ_b	=	6.58		K _{xz}	=	.00024
				μ_c	=	60.60				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.01460	C_{Z_0}	=	-.52391		
C_{X_u}	=	.00011	C_{Z_u}	=	-1.00992	C_{m_u}	= .01886
C_{X_α}	=	.77201	C_{Z_α}	=	-5.51372	C_{m_α}	= -.90555
C_{X_q}	=	-.67480	C_{Z_q}	=	-2.95448	C_{m_q}	= -15.53760
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40003	C_{m_δ}	= -1.91857
$C_{X_{\delta_f}}$	=	-.00721	$C_{Z_{\delta_f}}$	=	-1.47649	$C_{m_{\delta_f}}$	= .41616
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10569	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 6000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00 M/S	ϕ	=	.000 RAD	δ_a	=	.009 RAD
α	=	.079 RAD	θ	=	.028 RAD	δ_e	=	.011 RAD
β	=	-.011 RAD	ψ	=	.000 RAD	δ_r	=	-.039 RAD
γ	=	-.051 RAD	P	=	.000 RAD/S	δ_f	=	.000 RAD
C	=	-3.06 M/S	q	=	.000 RAD/S			
h	=	6000.0 FT	r	=	.000 RAD/S			
Pz	=	15.00 "HG	x _{cg}	=	.380 M.A.C.	K _{X2}	=	.01100
n	=	1800.0 RPM	y _{cg}	=	.000 M.A.C.	K _{Y2}	=	1.20150
P	=	46.8 KW	z _{cg}	=	-.560 M.A.C.	K _{Z2}	=	.02275
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.168	μ_b	=	6.58	K _{XZ}	=	-.00069
			μ_c	=	60.60			

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C _{X0}	=	-.02674	C _{Z0}	=	-.52344			
C _{Xu}	=	-.07948	C _{Zu}	=	-1.00679	C _{m_u}	=	.01886
C _{Xα}	=	.33479	C _{Zα}	=	-5.55742	C _{mα}	=	-.90609
C _{Xq}	=	-.90830	C _{Zq}	=	-2.89174	C _{m_q}	=	-15.53731
C _{Xδ}	=	-.03153	C _{Zδ}	=	-.40001	C _{mδ}	=	-1.91851
C _{Xδ_f}	=	-.12355	C _{Zδ_f}	=	-1.47133	C _{mδ_f}	=	.41588
C _{Y0}	=	-.00030						
C _{Yβ}	=	-.76775	C _{lβ}	=	-.06125	C _{nβ}	=	.01061
C _{Y$\dot{\beta}$}	=	-.07999	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{Yp}	=	-.12399	C _{lp}	=	-.51539	C _{n_p}	=	-.11825
C _{Yr}	=	.36759	C _{lr}	=	.15668	C _{n_r}	=	-.12374
C _{Yδ_a}	=	-.02956	C _{lδ_a}	=	-.10566	C _{nδ_a}	=	.00447
C _{Yδ_r}	=	.11618	C _{lδ_r}	=	.00040	C _{nδ_r}	=	-.08294

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 6000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.091	RAD	θ	=	.050	RAD	δ_e	=	.003	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.041	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.45	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.211		μ_b	=	7.44		K _{XZ}	=	.00024
				μ_c	=	68.59				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.02954	C_{Z_0}	=	-.59250		
C_{X_u}	=	.01370	C_{Z_u}	=	-1.12669	C_{m_u}	= .02901
C_{X_α}	=	.87213	C_{Z_α}	=	-5.49309	C_{m_α}	= -.95593
C_{X_q}	=	-.67477	C_{Z_q}	=	-2.95461	C_{m_q}	= -15.53728
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40016	C_{m_δ}	= -1.91857
$C_{X_{\delta_f}}$	=	.00582	$C_{Z_{\delta_f}}$	=	-1.49135	$C_{m_{\delta_f}}$	= .41628
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10667	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 10000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	60.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.091	RAD	θ	=	.050	RAD	δ_e	=	.003	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.039	RAD
γ	=	-.041	RAD	P	=	.000	RAD/S	δ_f	=	.000	RAD
C	=	-2.45	M/S	q	=	.000	RAD/S				
h	=	10000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01101
n	=	1800.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	63.5	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02273
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.211		μ_b	=	7.44		K _{XZ}	=	-.00083
				μ_c	=	68.59				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02424	C_{Z_0}	=	-.59274		
C_{X_u}	=	-.08838	C_{Z_u}	=	-1.12330	C_{m_u}	= .02901
C_{X_α}	=	.37084	C_{Z_α}	=	-5.54949	C_{m_α}	= -.95651
C_{X_q}	=	-.94272	C_{Z_q}	=	-2.88088	C_{m_q}	= -15.53702
C_{X_δ}	=	-.03624	C_{Z_δ}	=	-.40014	C_{m_δ}	= -1.91851
$C_{X_{\delta f}}$	=	-.12924	$C_{Z_{\delta f}}$	=	-1.48575	$C_{m_{\delta f}}$	= .41595
C_{Y_0}	=	-.00028					
C_{Y_β}	=	-.76775	C_{l_β}	=	-.06112	C_{n_β}	= .01133
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51675	C_{n_p}	= -.11216
C_{Y_r}	=	.36759	C_{l_r}	=	.15509	C_{n_r}	= -.12562
$C_{Y_{\delta a}}$	=	-.02956	$C_{l_{\delta a}}$	=	-.10657	$C_{n_{\delta a}}$	= .00580
$C_{Y_{\delta r}}$	=	.11619	$C_{l_{\delta r}}$	=	-.00058	$C_{n_{\delta r}}$	= -.08294

TABLE 4. (CONTINUED). DESCEND.
V = 60.0 M/S, H = 10000.0 FT.
STABILITY REFERENCE SYSTEM.

APPROACH

Table 4. (continued)

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.138	RAD	θ	=	.084	RAD	δ_e	=	.027	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.054	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.88	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

P _Z	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	32.0	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.320		μ_b	=	5.50		K _{XZ}	=	.00024
				μ_c	=	50.65				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.09308	C_{Z_0}	=	-.87653		
C_{X_u}	=	.09591	C_{Z_u}	=	-1.64365	C_{m_u}	= .05444
C_{X_α}	=	1.51917	C_{Z_α}	=	-5.71182	C_{m_α}	= -1.15573
C_{X_q}	=	-.67461	C_{Z_q}	=	-2.95328	C_{m_q}	= -15.53600
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40076	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	.05813	$C_{Z_{\delta f}}$	=	-1.55098	$C_{m_{\delta f}}$	= .41674
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76730	C_{l_β}	=	-.06180	C_{n_β}	= .00578
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11058	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 0.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.138	RAD	θ	=	.084	RAD	δ_e	=	.027	RAD
β	=	-.013	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.054	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.88	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				
Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01112	
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150	
P	=	32.0	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02263	
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.320		μ_b	=	5.50		K _{xz}	=	-.00138	
				μ_c	=	50.65					

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02835	C_{Z_0}	=	-.88100		
C_{X_u}	=	-.13104	C_{Z_u}	=	-1.64122	C_{m_u}	= .05444
C_{X_α}	=	.71383	C_{Z_α}	=	-5.86648	C_{m_α}	= -1.15627
C_{X_q}	=	-1.07950	C_{Z_q}	=	-2.83114	C_{m_q}	= -15.53590
C_{X_δ}	=	-.05511	C_{Z_δ}	=	-.40073	C_{m_δ}	= -1.91851
$C_{X_{\delta_f}}$	=	-.15572	$C_{Z_{\delta_f}}$	=	-1.54424	$C_{m_{\delta_f}}$	= .41625
C_{Y_0}	=	-.00037					
C_{Y_β}	=	-.76771	C_{l_β}	=	-.06051	C_{n_β}	= .01421
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52146	C_{n_p}	= -.08761
C_{Y_r}	=	.36758	C_{l_r}	=	.14848	C_{n_r}	= -.13299
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11005	$C_{n_{\delta_a}}$	= .01137
$C_{Y_{\delta_r}}$	=	.11623	$C_{l_{\delta_r}}$	=	-.00450	$C_{n_{\delta_r}}$	= -.08282

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 0.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.150	RAD	θ	=	.101	RAD	δ_e	=	.016	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	-.049	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.71	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				
Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096	
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150	
P	=	41.1	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278	
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.404		μ_b	=	5.83		K _{XZ}	=	.00024	
				μ_c	=	53.73					

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.11904	C_{Z_0}	=	-.94839		
C_{X_u}	=	.11079	C_{Z_u}	=	-1.74798	C_{m_u}	= .07405
C_{X_α}	=	1.60404	C_{Z_α}	=	-5.67687	C_{m_α}	= -1.20612
C_{X_q}	=	-.67458	C_{Z_q}	=	-2.95349	C_{m_q}	= -15.53566
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40103	C_{m_δ}	= -1.91856
$C_{X_{\delta f}}$	=	.07114	$C_{Z_{\delta f}}$	=	-1.56581	$C_{m_{\delta f}}$	= .41686
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00578
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11155	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	ϕ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.150	RAD	θ	=	.101	RAD	δ_e	=	.016	RAD
β	=	-.014	RAD	ψ	=	.000	RAD	δ_r	=	-.042	RAD
γ	=	-.049	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.71	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

P _Z	=	15.00	"HG	X _{cg}	=	.380	M.A.C.	K _{X2}	=	.01115
n	=	2000.0	RPM	Y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	41.1	KW	Z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02259
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.404		μ_b	=	5.83		K _{XZ}	=	-.00151
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.02377	C_{Z_0}	=	-.95554		
C_{X_u}	=	-.15119	C_{Z_u}	=	-1.74495	C_{m_u}	= .07405
C_{X_α}	=	.73887	C_{Z_α}	=	-5.85263	C_{m_α}	= -1.20666
C_{X_q}	=	-1.11317	C_{Z_q}	=	-2.81833	C_{m_q}	= -15.53553
C_{X_δ}	=	-.05982	C_{Z_δ}	=	-.40100	C_{m_δ}	= -1.91850
$C_{X_{\delta_f}}$	=	-.16323	$C_{Z_{\delta_f}}$	=	-1.55891	$C_{m_{\delta_f}}$	= .41632
C_{Y_0}	=	-.00033					
C_{Y_β}	=	-.76771	C_{l_β}	=	-.06035	C_{n_β}	= .01493
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52245	C_{n_p}	= -.08147
C_{Y_r}	=	.36758	C_{l_r}	=	.14670	C_{n_r}	= -.13478
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11087	$C_{n_{\delta_a}}$	= .01281
$C_{Y_{\delta_r}}$	=	.11625	$C_{l_{\delta_r}}$	=	-.00547	$C_{n_{\delta_r}}$	= -.08276

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.162	RAD	θ	=	.118	RAD	δ_e	=	.004	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.044	RAD
γ	=	-.044	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.55	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	49.7	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.495		μ_b	=	6.19		K _{xz}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.14853	C_{Z_0}	=	-1.02516		
C_{X_u}	=	.12645	C_{Z_u}	=	-1.85910	C_{m_u}	= .09515
C_{X_α}	=	1.69293	C_{Z_α}	=	-5.63604	C_{m_α}	= -1.26041
C_{X_q}	=	-.67455	C_{Z_q}	=	-2.95374	C_{m_q}	= -15.53528
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40144	C_{m_δ}	= -1.91856
$C_{X_{\delta_f}}$	=	.08515	$C_{Z_{\delta_f}}$	=	-1.58179	$C_{m_{\delta_f}}$	= .41698
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00580
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11260	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	35.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.162	RAD	θ	=	.118	RAD	δ_e	=	.004	RAD
β	=	-.015	RAD	ψ	=	.000	RAD	δ_r	=	-.044	RAD
γ	=	-.044	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.55	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01119
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	49.7	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02255
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.495		μ_b	=	6.19		K _{XZ}	=	-.00166
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.01918	C_{Z_0}	=	-1.03568		
C_{X_u}	=	-.17579	C_{Z_u}	=	-1.85508	C_{m_u}	= .09515
C_{X_α}	=	.75895	C_{Z_α}	=	-5.83561	C_{m_α}	= -1.26093
C_{X_q}	=	-1.14927	C_{Z_q}	=	-2.80410	C_{m_q}	= -15.53506
C_{X_δ}	=	-.06490	C_{Z_δ}	=	-.40140	C_{m_δ}	= -1.91848
$C_{X_{\delta_f}}$	=	-.17171	$C_{Z_{\delta_f}}$	=	-1.57474	$C_{m_{\delta_f}}$	= .41639
C_{Y_0}	=	-.00028					
C_{Y_β}	=	-.76771	C_{l_β}	=	-.06018	C_{n_β}	= .01570
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.52343	C_{n_p}	= -.07484
C_{Y_r}	=	.36757	C_{l_r}	=	.14469	C_{n_r}	= -.13668
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.11173	$C_{n_{\delta_a}}$	= .01438
$C_{Y_{\delta_r}}$	=	.11628	$C_{l_{\delta_r}}$	=	-.00652	$C_{n_{\delta_r}}$	= -.08268

TABLE 4. (CONTINUED). APPROACH.
V = 35.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.086	RAD	θ	=	.030	RAD	δ_e	=	.059	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.056	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-2.24	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

P _Z	=	15.00	" HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	32.0	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.243		μ_b	=	5.50		K _{XZ}	=	.00024
				μ_c	=	50.65				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.02968	C_{Z_0}	=	-.59596		
C_{X_u}	=	.00206	C_{Z_u}	=	-1.11863	C_{m_u}	= .03647
C_{X_α}	=	1.12813	C_{Z_α}	=	-5.83149	C_{m_α}	= -.93459
C_{X_q}	=	-.67480	C_{Z_q}	=	-2.95255	C_{m_q}	= -15.53739
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40013	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.00097	$C_{Z_{\delta f}}$	=	-1.48582	$C_{m_{\delta f}}$	= .41623
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10631	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 0.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.086	RAD	θ	=	.030	RAD	δ_e	=	.059	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.056	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-2.24	M/S	q	=	.000	RAD/S				
h	=	0.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	Xcg	=	.380	M.A.C.	Kx2	=	.01101
n	=	2000.0	RPM	Ycg	=	.000	M.A.C.	Ky2	=	1.20150
P	=	32.0	KW	Zcg	=	-.560	M.A.C.	Kz2	=	.02274
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.243		μ_b	=	5.50		Kxz	=	-.00078
				μ_c	=	50.65				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

Cx0	=	-.02179	Cz0	=	-.59630			
Cxu	=	-.09436	Czu	=	-1.11465	Cmu	=	.03647
Cx α	=	.62105	Cz α	=	-5.90702	Cm α	=	-.93507
Cxq	=	-.92994	Czq	=	-2.88282	Cmq	=	-15.53711
Cx δ	=	-.03448	Cz δ	=	-.40011	Cm δ	=	-1.91851
Cx δ_f	=	-.12708	Cz δ_f	=	-1.48037	Cm δ_f	=	.41592
Cy0	=	-.00025						
Cy β	=	-.76773	C1 β	=	-.06117	Cn β	=	.01106
Cy $\dot{\beta}$	=	-.07999	C1 $\dot{\beta}$	=	.00000	Cn $\dot{\beta}$	=	.00000
Cyp	=	-.12399	C1p	=	-.51625	Cnp	=	-.11443
Cyr	=	.36759	C1r	=	.15567	Cnr	=	-.12493
Cy δ_a	=	-.02956	C1 δ_a	=	-.10624	Cn δ_a	=	.00530
Cy δ_r	=	.11619	C1 δ_r	=	-.00022	Cn δ_r	=	-.08294

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 0.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.095	RAD	θ	=	.045	RAD	δ_e	=	.052	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.050	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-2.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01096
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	41.1	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.300		μ_b	=	5.83		K _{XZ}	=	.00024
				μ_c	=	53.73				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C_{X_0}	=	.04462	C_{Z_0}	=	-.65215		
C_{X_u}	=	.00990	C_{Z_u}	=	-1.20461	C_{m_u}	= .04961
C_{X_α}	=	1.20662	C_{Z_α}	=	-5.81434	C_{m_α}	= -.97365
C_{X_q}	=	-.67478	C_{Z_q}	=	-2.95266	C_{m_q}	= -15.53712
C_{X_δ}	=	.00000	C_{Z_δ}	=	-.40027	C_{m_δ}	= -1.91857
$C_{X_{\delta f}}$	=	.01108	$C_{Z_{\delta f}}$	=	-1.49734	$C_{m_{\delta f}}$	= .41632
C_{Y_0}	=	.00000					
C_{Y_β}	=	-.76780	C_{l_β}	=	-.06180	C_{n_β}	= .00576
$C_{Y_{\dot{\beta}}}$	=	-.08000	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12400	C_{l_p}	=	-.50450	C_{n_p}	= -.15850
C_{Y_r}	=	.36761	C_{l_r}	=	.16950	C_{n_r}	= -.11072
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10706	$C_{n_{\delta_a}}$	= -.00387
$C_{Y_{\delta_r}}$	=	.11580	$C_{l_{\delta_r}}$	=	.00693	$C_{n_{\delta_r}}$	= -.08265

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 2000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.095	RAD	θ	=	.045	RAD	δ_e	=	.052	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.040	RAD
γ	=	-.050	RAD	p	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-2.00	M/S	q	=	.000	RAD/S				
h	=	2000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{X2}	=	.01102
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{Y2}	=	1.20150
P	=	41.1	KW	z _{cg}	=	-.560	M.A.C.	K _{Z2}	=	.02272
$\frac{\Delta P_t}{\frac{1}{2}\rho V^2}$	=	.300		μ_b	=	5.83		K _{XZ}	=	-.00089
				μ_c	=	53.73				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

C_{X_0}	=	-.01772	C_{Z_0}	=	-.65343		
C_{X_u}	=	-.10492	C_{Z_u}	=	-1.20007	C_{m_u}	= .04960
C_{X_α}	=	.64680	C_{Z_α}	=	-5.90286	C_{m_α}	= -.97415
C_{X_q}	=	-.95658	C_{Z_q}	=	-2.87422	C_{m_q}	= -15.53686
C_{X_δ}	=	-.03814	C_{Z_δ}	=	-.40024	C_{m_δ}	= -1.91850
$C_{X_{\delta_f}}$	=	-.13164	$C_{Z_{\delta_f}}$	=	-1.49158	$C_{m_{\delta_f}}$	= .41598
C_{Y_0}	=	-.00021					
C_{Y_β}	=	-.76772	C_{l_β}	=	-.06107	C_{n_β}	= .01162
$C_{Y_{\dot{\beta}}}$	=	-.07999	$C_{l_{\dot{\beta}}}$	=	.00000	$C_{n_{\dot{\beta}}}$	= .00000
C_{Y_p}	=	-.12399	C_{l_p}	=	-.51727	C_{n_p}	= -.10971
C_{Y_r}	=	.36759	C_{l_r}	=	.15440	C_{n_r}	= -.12638
$C_{Y_{\delta_a}}$	=	-.02956	$C_{l_{\delta_a}}$	=	-.10694	$C_{n_{\delta_a}}$	= .00635
$C_{Y_{\delta_r}}$	=	.11620	$C_{l_{\delta_r}}$	=	-.00097	$C_{n_{\delta_r}}$	= -.08293

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 2000.0 FT.
STABILITY REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.105	RAD	θ	=	.061	RAD	δ_e	=	.045	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	-.045	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.78	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT.	r	=	.000	RAD/S				

Pz	=	15.00	"HG	x _{cg}	=	.380	M.A.C.	K _{x2}	=	.01096
n	=	2000.0	RPM	y _{cg}	=	.000	M.A.C.	K _{y2}	=	1.20150
P	=	49.7	KW	z _{cg}	=	-.560	M.A.C.	K _{z2}	=	.02278
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.360		μ_b	=	6.19		K _{xz}	=	.00024
				μ_c	=	57.04				

STABILITY DERIVATIVES IN BODY-FIXED REFERENCE SYSTEM.

C _{x0}	=	.06163	C _{z0}	=	-.71232			
C _{xu}	=	.01909	C _{zu}	=	-1.29654	C _{mu}	=	.06374
C _{xα}	=	1.28919	C _{zα}	=	-5.79397	C _{mα}	=	-1.01568
C _{xq}	=	-.67476	C _{zq}	=	-2.95278	C _{mq}	=	-15.53684
C _{xδ}	=	.00000	C _{zδ}	=	-.40043	C _{mδ}	=	-1.91857
C _{xδ_f}	=	.02195	C _{zδ_f}	=	-1.50973	C _{mδ_f}	=	.41642
C _{y0}	=	.00000						
C _{yβ}	=	-.76780	C _{lβ}	=	-.06180	C _{nβ}	=	.00577
C _{y$\dot{\beta}$}	=	-.08000	C _{l$\dot{\beta}$}	=	.00000	C _{n$\dot{\beta}$}	=	.00000
C _{yp}	=	-.12400	C _{lp}	=	-.50450	C _{np}	=	-.15850
C _{yr}	=	.36761	C _{lr}	=	.16950	C _{nr}	=	-.11072
C _{yδ_a}	=	-.02956	C _{lδ_a}	=	-.10787	C _{nδ_a}	=	-.00387
C _{yδ_r}	=	.11580	C _{lδ_r}	=	.00693	C _{nδ_r}	=	-.08265

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 4000.0 FT.
BODY-FIXED REFERENCE SYSTEM.

NOMINAL FLIGHT CONDITION :

V	=	40.00	M/S	φ	=	.000	RAD	δ_a	=	.009	RAD
α	=	.105	RAD	θ	=	.061	RAD	δ_e	=	.045	RAD
β	=	-.012	RAD	ψ	=	.000	RAD	δ_r	=	-.041	RAD
γ	=	-.045	RAD	P	=	.000	RAD/S	δ_f	=	.262	RAD
C	=	-1.78	M/S	q	=	.000	RAD/S				
h	=	4000.0	FT	r	=	.000	RAD/S				

Pz	=	15.00	"HG	Xcg	=	.380	M.A.C.	Kx2	=	.01104
n	=	2000.0	RPM	Ycg	=	.000	M.A.C.	Ky2	=	1.20150
P	=	49.7	KW	Zcg	=	-.560	M.A.C.	Kz2	=	.02270
$\frac{\Delta P_t}{\frac{1}{2} \rho V^2}$	=	.360		μ_b	=	6.19		Kxz	=	-.00100
				μ_c	=	57.04				

STABILITY DERIVATIVES IN STABILITY REFERENCE SYSTEM.

Cx0	=	-.01355	Cz0	=	-.71485		
Cxu	=	-.11723	Czu	=	-1.29137	Cm _u	= .06374
Cx α	=	.67299	Cz α	=	-5.89735	Cm α	= -1.01620
Cxq	=	-.98516	Czq	=	-2.86471	Cmq	= -15.53658
Cx δ	=	-.04207	Cz δ	=	-.40040	Cm δ	= -1.91850
Cx δ_f	=	-.13679	Cz δ_f	=	-1.50368	Cm δ_f	= .41604
Cy0	=	-.00017					
Cy β	=	-.76772	C _l β	=	-.06095	Cn β	= .01222
Cy $\dot{\beta}$	=	-.07999	C _l $\dot{\beta}$	=	.00000	Cn $\dot{\beta}$	= .00000
Cyp	=	-.12399	C _l p	=	-.51832	Cnp	= -.10462
Cyr	=	.36758	C _l r	=	.15301	Cnr	= -.12793
Cy δ_a	=	-.02956	C _l δ_a	=	-.10768	Cn δ_a	= .00748
Cy δ_r	=	.11621	C _l δ_r	=	-.00179	Cn δ_r	= -.08292

TABLE 4. (CONTINUED). APPROACH.
V = 40.0 M/S, H = 4000.0 FT.
STABILITY REFERENCE SYSTEM.

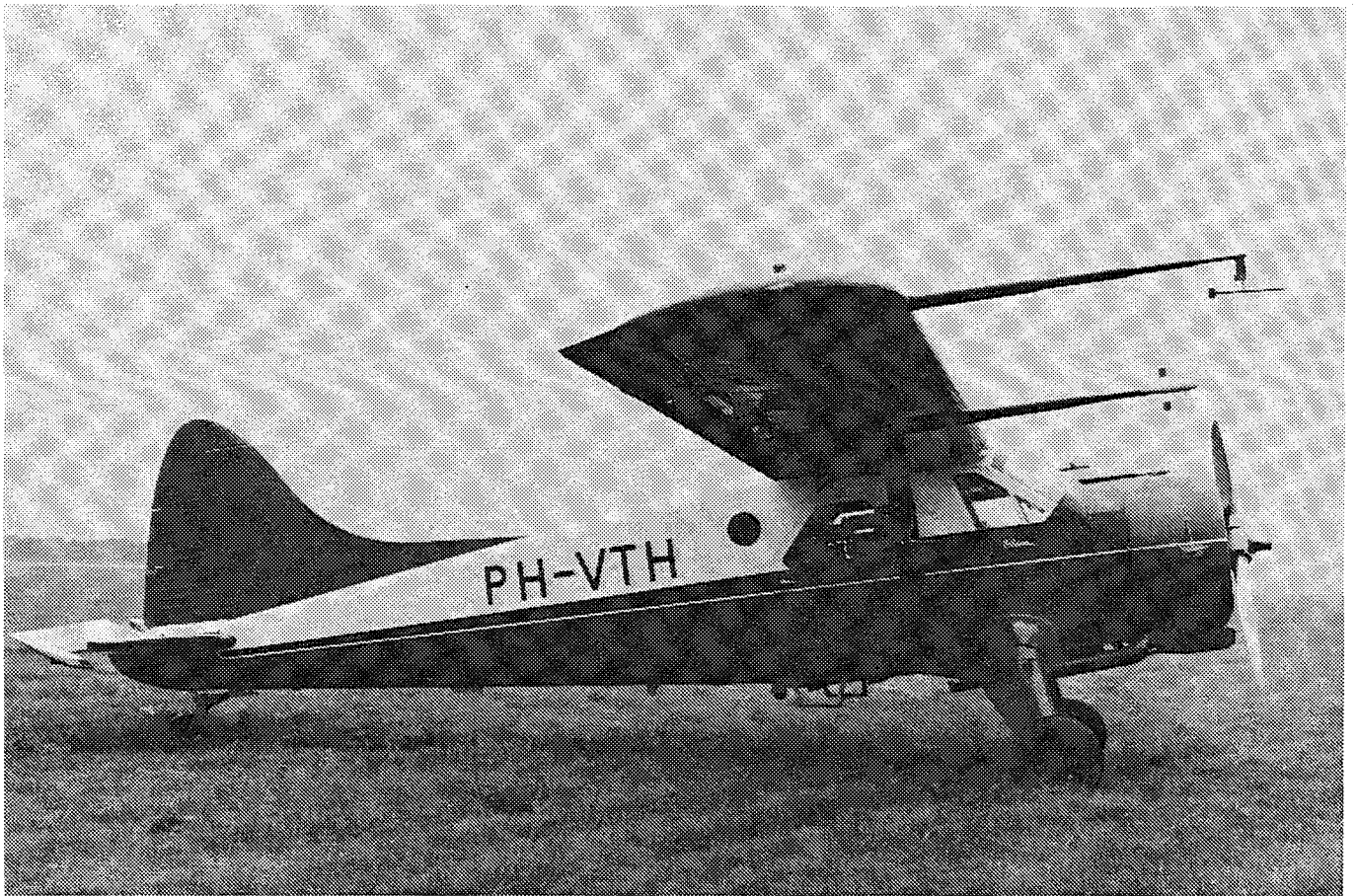


Figure 1. The De Havilland DHC-2 'Beaver' experimental aircraft of Delft University of Technology.

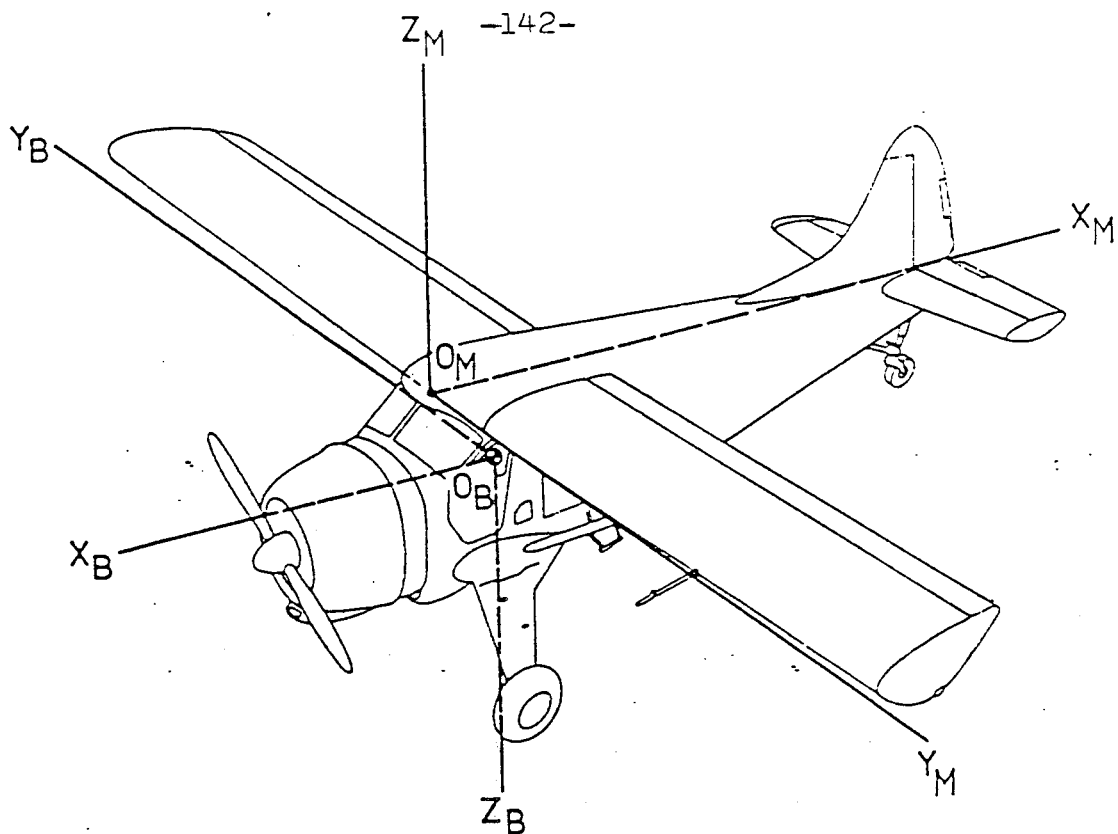


Figure 2. The measurement reference frame F_M and the body-fixed reference frame F_B .

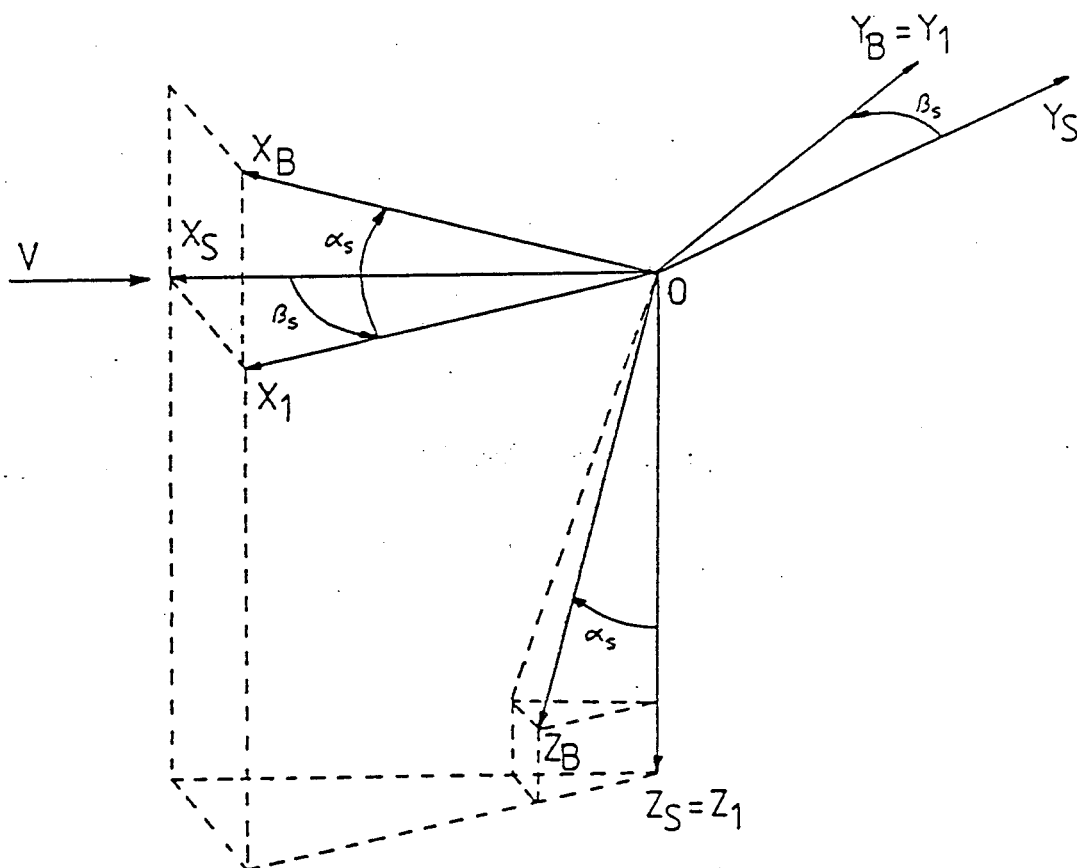


Figure 3. The body-fixed reference frame F_B and the stability reference frame F_S .

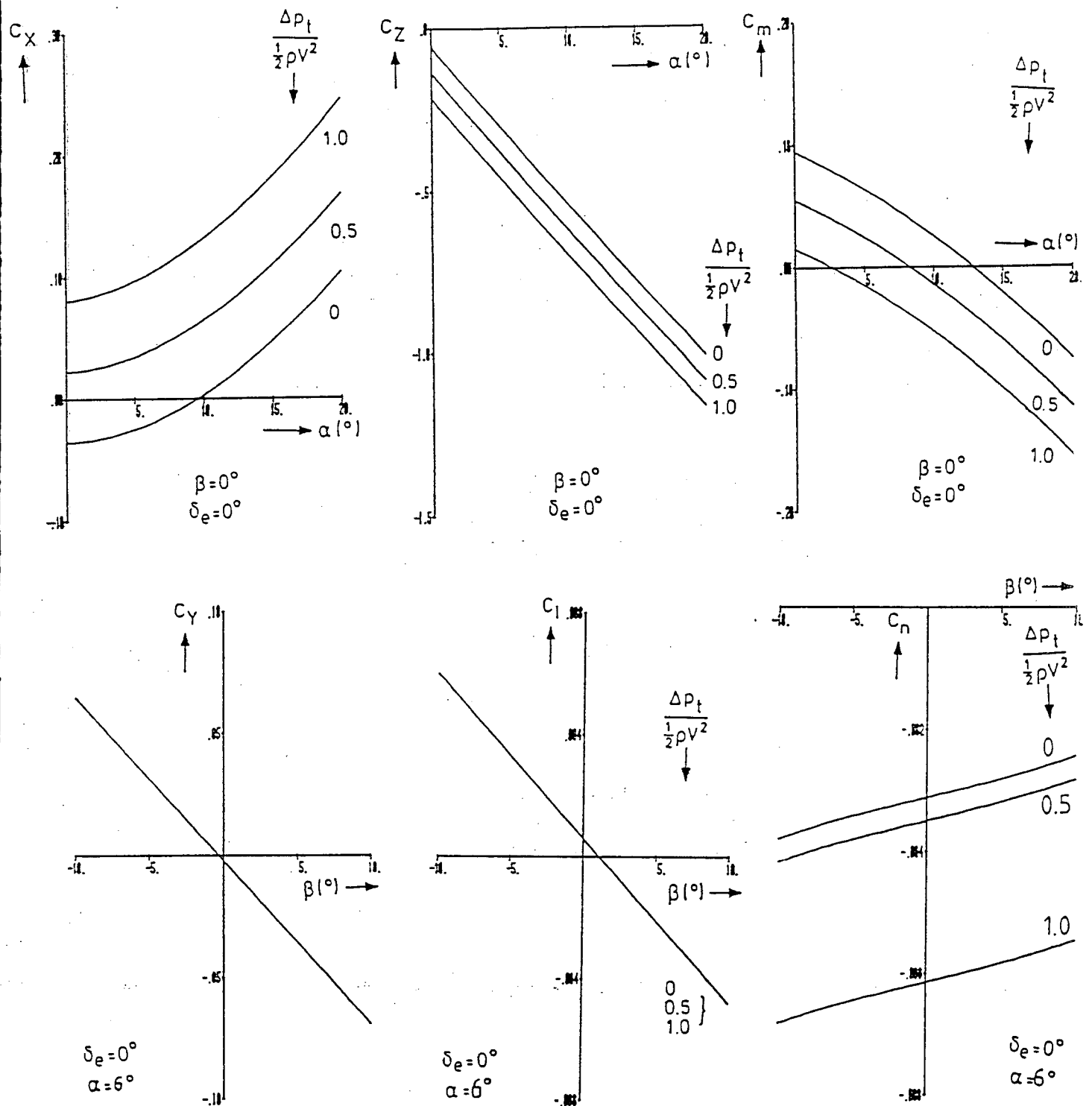


Figure 4. The dimensionless aerodynamic forces and moments of the nonlinear 'Beaver'-aircraft model as a function of angle of attack α and side slip angle β .

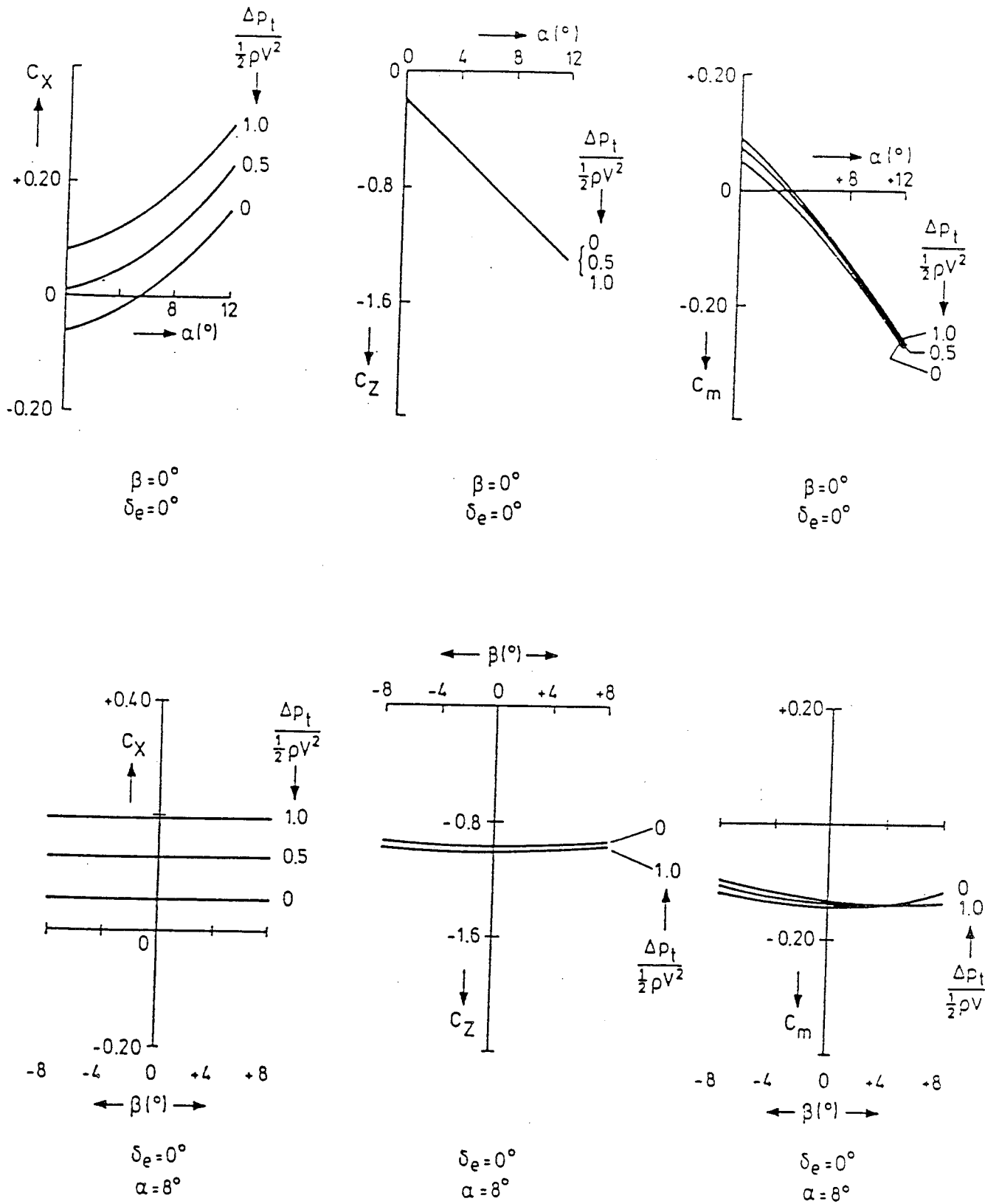


Fig. 5-a : Longitudinal aerodynamic force and moment coefficients of the DHC-2 'Beaver' aircraft as a function of angle of attack α and side slip angle β for three different values of $\Delta p_t / \frac{1}{2} \rho V^2$, as measured at a Reynolds number of 0.47×10^6 on a 1:11 scale model in the wind-tunnel.

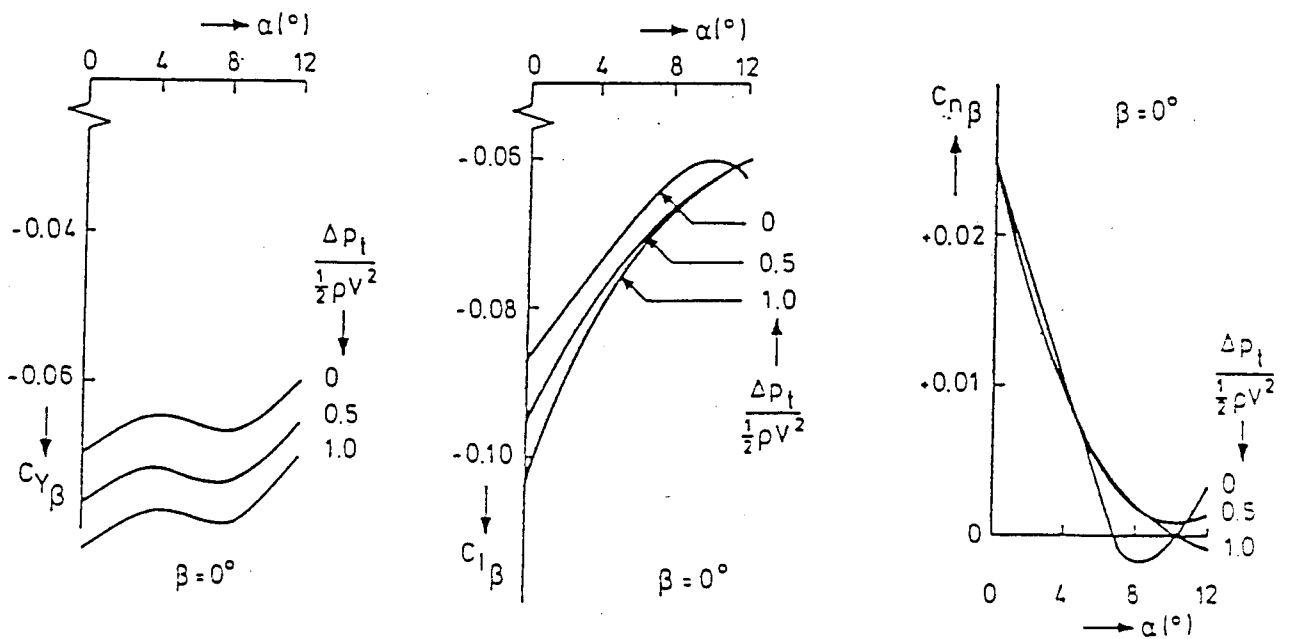
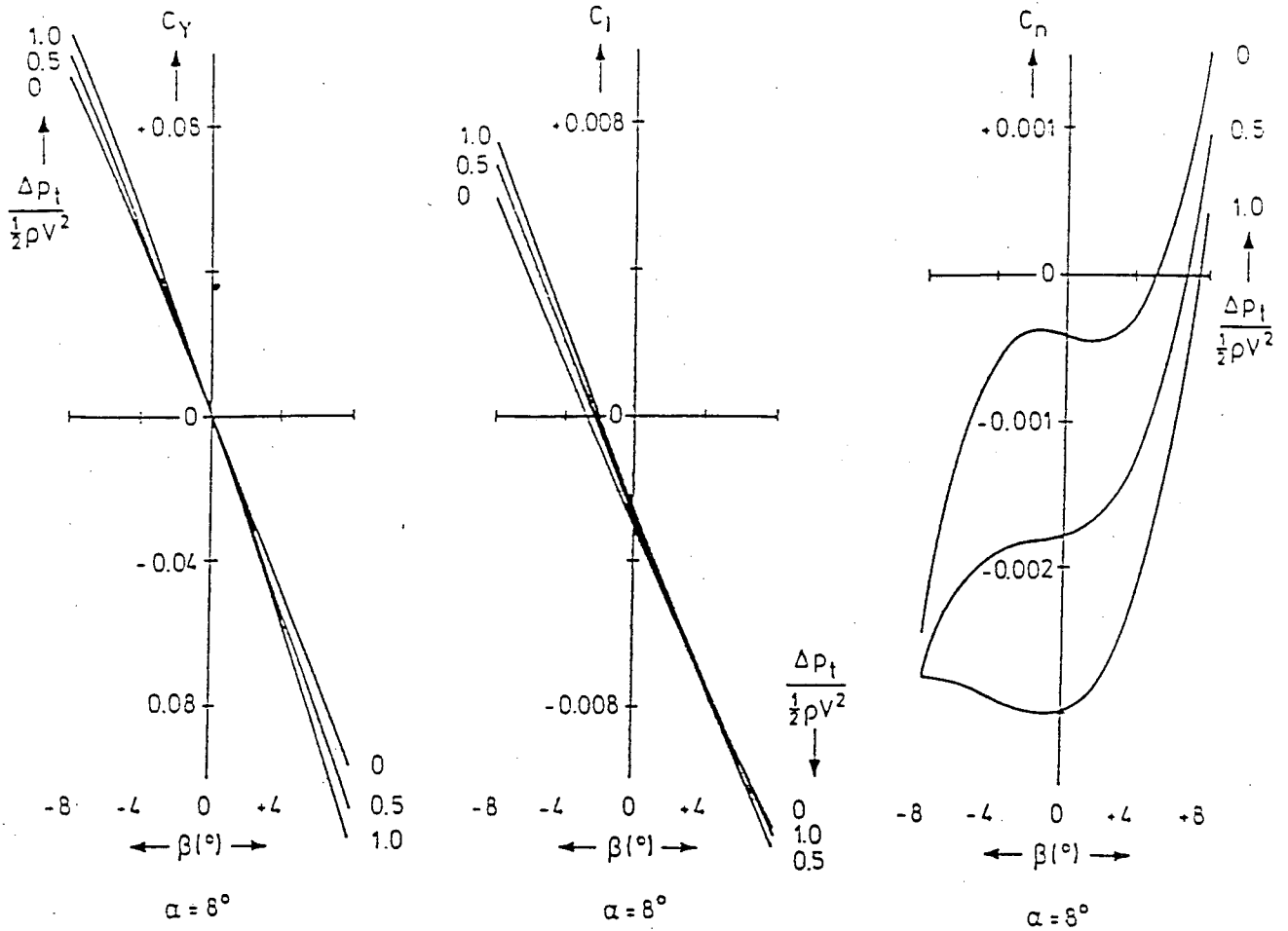


Fig. 5-b: Lateral aerodynamic force and moment coefficients as a function of sideslip angle β , for three different values of $\frac{\Delta p_t}{\frac{1}{2}\rho V^2}$.

